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Project and Professionalism (6CS007)

Project Report
Aurum (AI Powered Wealth and Lifestyle Management Hub)

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Declaration Sheet

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(Presented in partial fulfillment of the assessment requirements for the above award.)

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Abstract

With an integrated, AI powered platform, Aurum, the AI-powered wealth and lifestyle management hub, seeks in solving the increasing complexity of managing various financial portfolios and lifestyle demands of users. The initiative provides users with forecasts, investments suggestions, as well as personalized financial insights through fusion of cutting edge machine learning algorithms with user centric design. It's a web application developed with React.js and Django, which ensures smooth interaction while empowering the users to make informed decisions, automate lifestyle choices and reach financial goals. Despite obstacles like protecting user privacy, system scalability and regulatory compliance, Aurum is made to offer a complete, integrated solution to optimize finances and lifestyles. In addition to this, it provides a roadmap for future improvements to satisfy changing user needs. This report examines the project's goals, design, techniques, and competitive advantages.

Integrating of artificial intelligence, revolves around the precisely personalized investment strategies in the financial portfolio management. This report has application in dynamic markets, in user preference collection, risk management and performance monitoring is highlighted. The report analyzes the practical and theoretic parallels, whereby it is determined which of these methods are applicable to build adaptive, user target platforms, like Aurum. In addition, key metrics and models for subject of personalization, scalability and computational efficiency are evaluated in this study.

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1. Introduction

The Aurum (AI Powered Wealth and Lifestyle Management Hub) is a tool that is AI powered designed to maximize management of wealth and lifestyle for people with varied as well as intricate financial portfolios. The project aims to improve lifestyle choices, offer personalized financial suggestions, provide with real time forecasts by incorporating AI driven insights. The platform is for the users to make well-informed decisions based on their financial information as well as personal preferences through utilizing machine learning algorithms. Aurum provides a user friendly and smooth experience to connect financial planning and lifestyle automation by allowing users to track their assets, set financial objectives, and automate their lifestyle management effectively. Aurum is presented as a complete solution for intelligently and unified manage wealth and lifestyle preferences with an emphasis on user customization and need. The project address need for a single website to track the financial part of people and help them with intelligent investments and give appropriate suggestions. It is made with the focus of fulfilling most of the financial part of very person while integrating finance with their lifestyle.

The increasing complexities of financial portfolios and demands of lifestyle needs integrated solutions. Current systems have a gap in delivering such actions through insights and recommendations for both lifestyle and wealth (Srikanth, 2023). In recent years, financial portfolio management has developed from classic methods to complex AI based systems that are suited to the intricacies of the current markets and many different investors' needs. To address these challenges demanding tailored, adaptive and scalable Aurum, the AI powered wealth and lifestyle management hub, provides these with credible, tailored and adapted solutions.

1.1 AI Implementation

After learning about a person's finances and goals, the Nepal Investment Recommendation System recommends investments for them. The aim is to make financial planning available to Nepali retail investors by utilizing data in an organized way and with better algorithms.

Supervised Learning

This project will focus on ML which relates to supervised learning. The technique calculates the percentages for stocks, bonds, crypto, real estate and commodities by using previous recommendations that were tagged as such.

1.1.1 Training and Detailing the Model with Mathematics

Understanding AI requires an understanding of the mathematics involved. Prediction models seek to decrease the error using the traditional regression loss functions.

MSE helps in understanding, on average, how far off the predicted allocation had been from the true value.

R^2 Score: Determines how well the independent factors describe the differences in how the target allocations are formed.

Also, the project uses FeatureScaler and various domain-based engineering techniques such as Interactions, Square Transformations and specific rules.

1.1.2 Models Implemented

Two types of machine learning models are being employed:

Random Forest Regressor (RF): Uses many decision trees to help reduce variance and prevent your model from learning too much from the training data.

GB uses boosting by building each weak learner one after another and correcting any oversights from the previous learner.

For PicklableEnsemblePredictor, two models are combined using a weighted average.

Prediction = 0.6 × RF Prediction + 0.4 × GB Prediction

The prediction is 60% of RF prediction and 40% of GB prediction.

All combination ensembles performed better than the single models, reaching extremely high accuracy (R^2 more than 0.90) for each asset class.

1.1.3 Providing Information on the Agent

Nepal Investment Recommender is the main class in the system architecture that performs the AI agent’s tasks. Scale and transform the input features using a trained model before moving further. Forecasting the allocation of assets using various types of models. Advising users on individual investment products that work best for them. Following financial rules created for each domain to provide useful and practical advice. The agent is fully capable of being pickled and is used in both the backend code and the Django API endpoints.

1.2 Project Title

Aurum (AI Powered Wealth and Lifestyle Management Hub)

1.3 Aims

To address complex need of people with financial portfolios with an AI powered platform which integrates wealth management and automation of personalized lifestyle.

1.4 Objectives

1. To develop AI models for the purpose of personal investment recommendations, optimizing lifestyles and financial forecasts.

2. To give users high decision making power through data driven reports, personal recommendations and categorizing transactions.

3. To have the system which can provide diverse financial portfolio, offer action taking solutions and can report comprehensively such that user can have informed financial and lifestyle planning.

1.5 Artefact

1.5.1 FDD of the Whole Project

The complete FDD model as shown in this figure; which contained the core functions of the project. To track progress effectively, I was able to decompose the system into manageable features. The development of each feature follows a cycle with a planning, designing and implementation stage. This structured approach helped in developing the application efficiently, with all the functional requirements addressed at the end. The FDD acted as step by step guideline that I could follow to get the project done systematically.

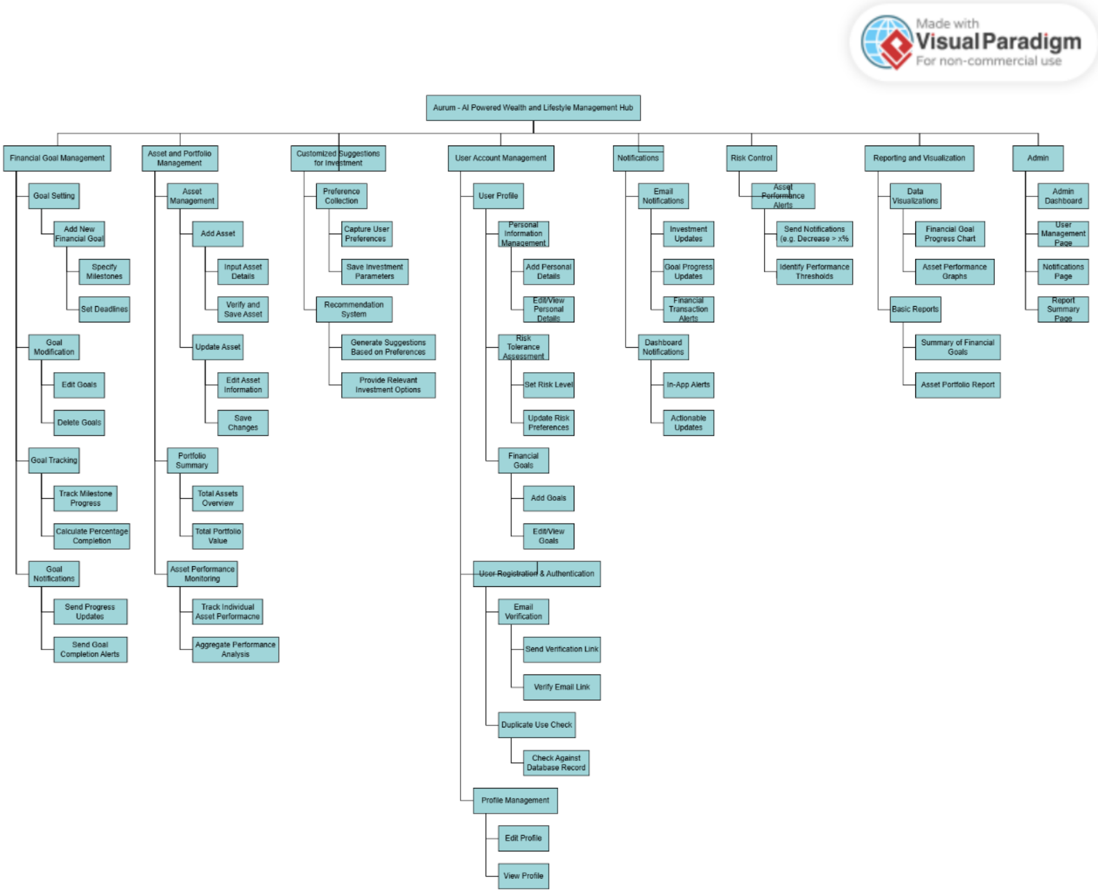


Figure 1 FDD of the whole project

1.5.2 User Account Management

User Registration & Authentication: User should be verified user, either through OTP or email so there is no duplicate user.

User Profiles (Personal info, financial goals, risk tolerance)

Create a management system for user profiles (view/edit)

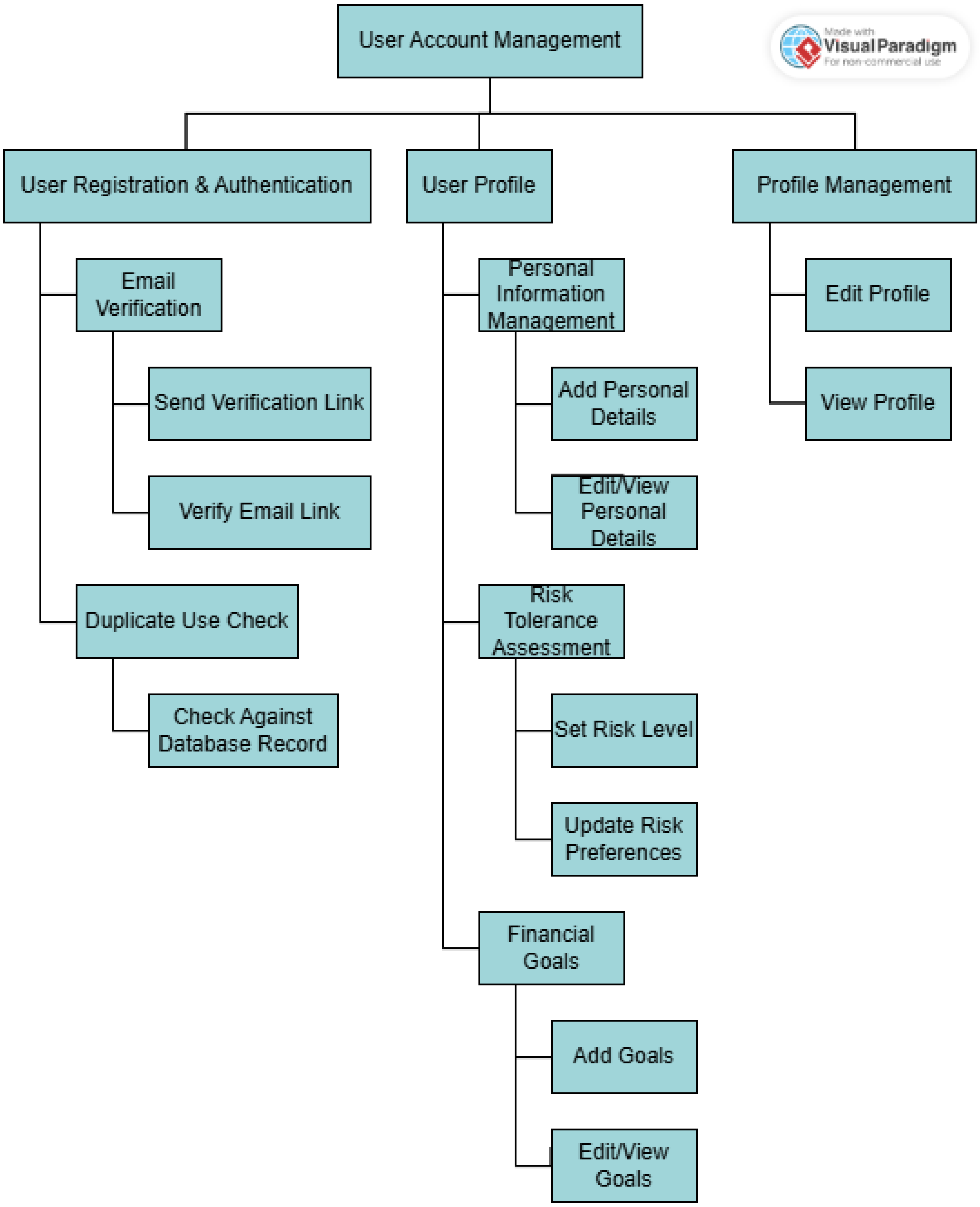


Figure 2 FDD User Account Management

1.5.3 Asset and Portfolio Management

Provide tools for adding and updating assets

Put the portfolio summary view (total assets, total value) into practice.

Establish a system to monitor asset performance

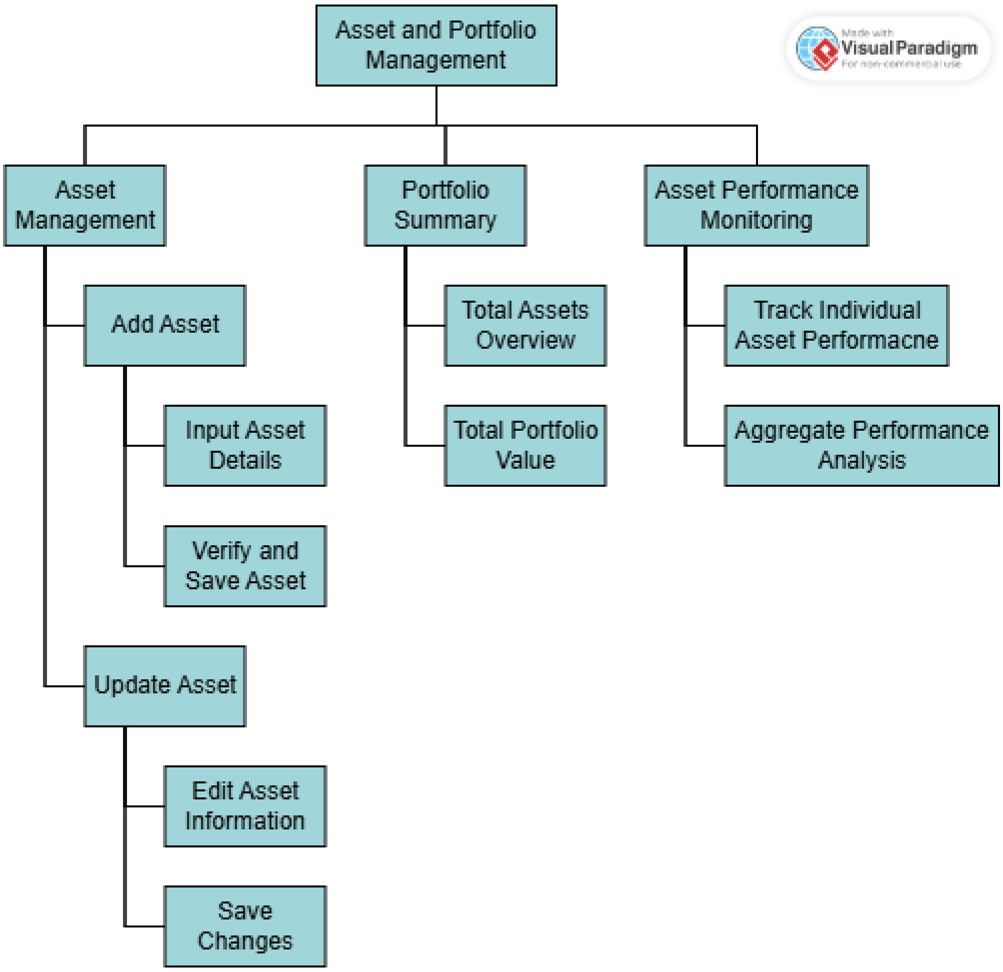


Figure 3 FDD Asset and Portfolio Management

1.5.4 Financial Goal Management

Give consumers the ability to set and modify financial objectives

Establishing goal tracking (accomplishing milestones, progress percentage)

Creating notifications pertaining to goals

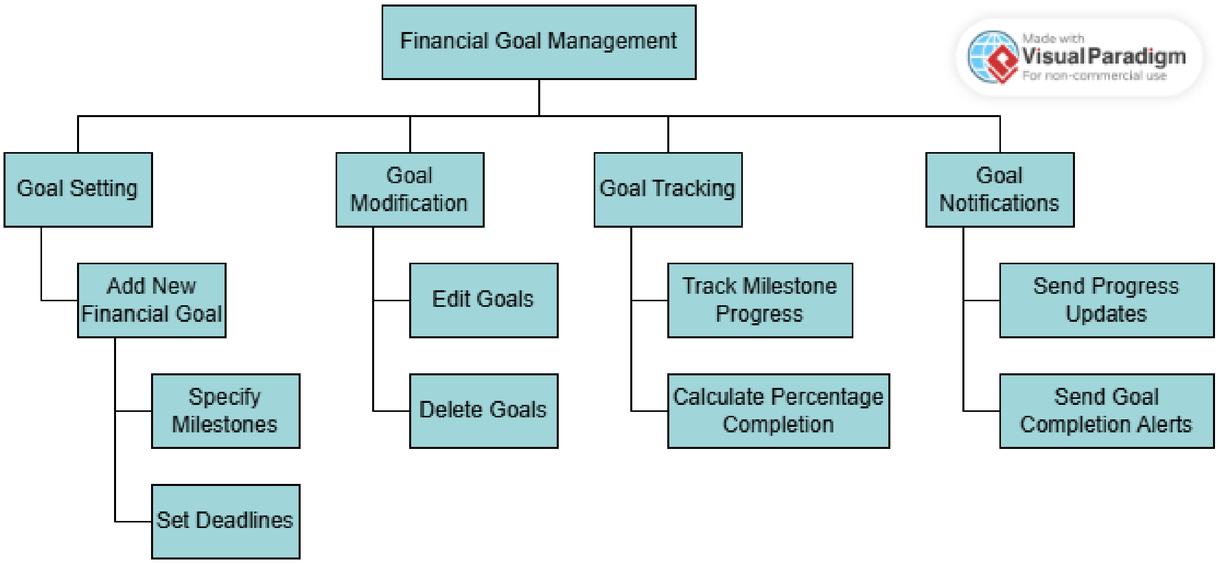


Figure 4 FDD Financial Goal Management

1.5.5 Customized Suggestion for Investment

Create a basic recommendation system which takes user preferences into account

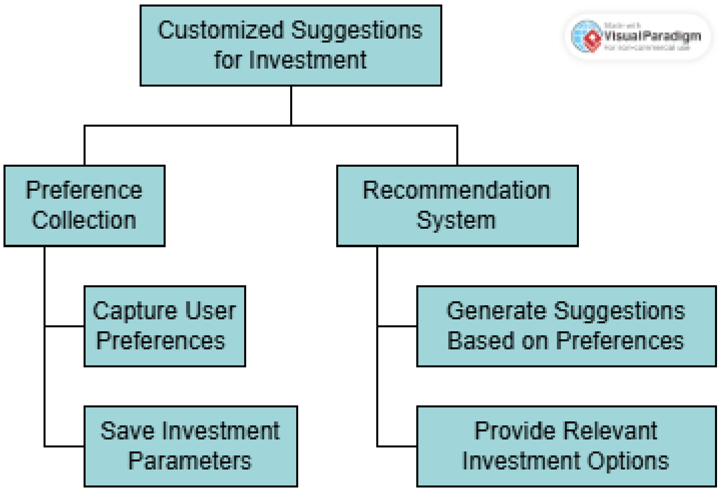


Figure 5 FDD Customized Suggestions for Investment

1.5.6 Notifications

Email notification: Updates on investments, objective progress, and financial transactions. Dashboard notification: In- app alerts and changes

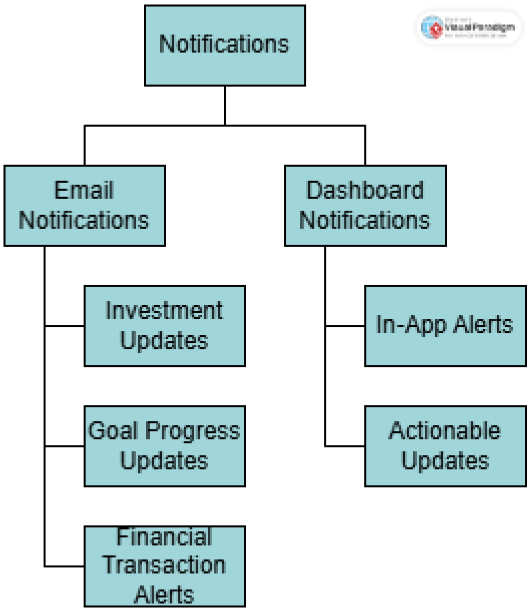


Figure 6 FDD Notifications

1.5.7 Risk Control

Simple asset performance notifications (i.e. decrease by certain percentage)

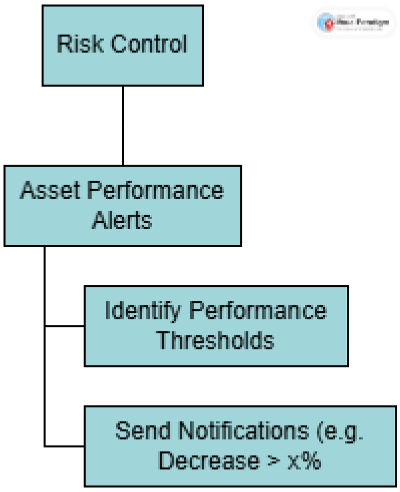


Figure 7 FDD Risk Control

1.5.8 Reporting and Visualization

Basic data visualizations for advancement of financial goal as well as asset performance Basic financial reports

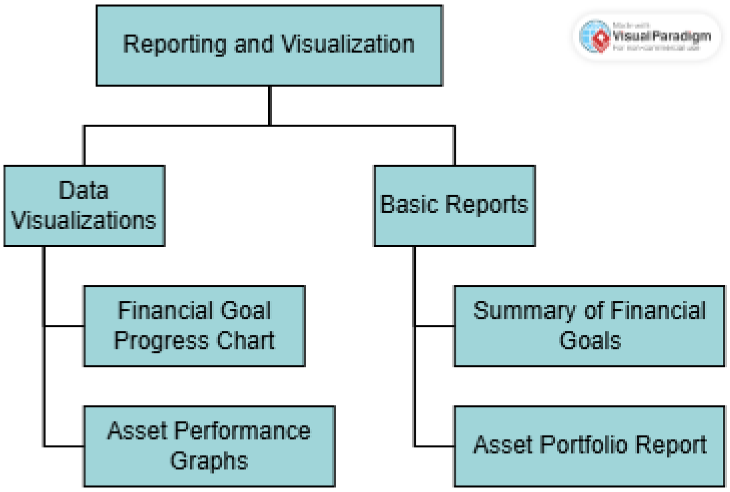


Figure 8 FDD Reporting and Visualization

1.6 Academic Question

How can wealth of a person and their personal lifestyle be optimized and managed such that it is able to address complex needs of individuals financially who have diverse and significant financial portfolios with the aid of AI-driven platform integrated?

1.7 Problem Statement

As there isn't a single platform which combines financial management and lifestyle automation, people frequently struggle tracking and managing a variety of financial portfolios while also meeting their lifestyle demands resulting in a lost opportunities and decreased productivity.

1.8 Project as a Solution

This is an AI powered platform which integrates financial and lifestyle aspects of the users which in turn offers them with a comprehensive solution to their individual financial portfolio and personalized lifestyle needs. The platform is designed such that the user can optimized their financial decisions, automate lifestyle choices, achieve personalized financial and lifestyle goals and create a unified user experience. Thus, this project has real time adaptability, personalization, comprehensive management, and is efficient.

1.9 Limitations

Protecting confidential data about people and their finances is challenging sometimes. If information is dealt with wrongly, it could lead to major issues regarding both the law and the company's reputation. The system is limited by how complex integration and customization can get. Doing the job or designing the scene becomes more complex. Merging the systems would make it harder for the population to find better applicability. As people use the platform more, managing scalability and real time data processing can challenge the platform. Users find it challenging to use big data effectively to get custom insights. Their impact can be seen in how accurate and fast the answers are. A lack of user confidence and meeting various country rules may pose a problem. Things may have to change regularly due to evolving laws.

2. Literature Review

In recent years, financial portfolio management has developed from classic methods to complex AI based systems that are suited to the intricacies of the current markets and many different investors' needs. To address these challenges demanding tailored, adaptive and scalable Aurum, the AI powered wealth and lifestyle management hub, provides these with credible, tailored and adapted solutions. Secure account management, personalized risk assessment, financial goal tracking, and dynamic asset and portfolio management, are the core functionalities of Aurum. With AI driven insights, the platform offers personalized investment recommendations, keeps a check on your risks, and produces very detailed financial reports. Market sentiment analysis, as well as dynamic asset allocation, enhance user decision making. In this literature review we consider several AI methodologies like Modern Portfolio Theory (MPT), Collaborative Filtering (CF), Reinforcement Learning (RL) and Genetic Algorithms (GA). In the review, how their work is related to investment optimization, personalized recommendations, and risk control, and the opportunities and limitations of using advanced machine learning in the Aurum framework is discussed.

2.1 Active Preference Learning for Personalized Portfolio Construction

In the literature on constructing individual portfolios, there is a consensus that one of the major concerns of portfolio management is the process of attaining a number of goals at the same time, including targeted return, risk, liquidity, and volatility, considering the preferences of the user. Tee et al. (2017) to overcome this problem, proposed to incorporate an active preference learning module in Bayesian optimization to acquire the user's sense of differences between financial portfolios (Tee, 2017). This can be more closely associated with the race in building a sustainable financial systems for learning and application of asset and portfolio management personalization process and optimization as seen on your project artifacts.

One of the themes of the research is the application of Bayesian optimization for rational decision making without excessive resource consumption. In this context, the study employs a questionnaire to seek users' opinions towards specifying probabilistic model to order the portfolios according to their levels of variability. All of the abovementioned characteristics are relevant and crucial for navigating elements like risk tolerance assessment and performance monitoring within your project as this methodological approach is built with the understanding of user preference integration into asset management systems in mind (Tee, 2017). For instance, the investment recommendations that are generated based on user input in your system are a nod to the call for personalized portfolio liquidity as elaborated in the study.

These findings of Tee et al. are most relevant to the domain of personalized financial systems. This is a perfect example of AI implementation at its best in that investors can create diverse portfolio investments consisting of the best performing assets and the users' defined criteria. Their work demonstrates many things including how active learning can increase system usability, reduce variations positively and all this without having to compromise on the portfolio. These form the basis of the need for preference collection and recommendations systems, both core components of your suggested investment for module.

However, the reporting and visualization artifacts highlighted in the study, including Pareto frontiers and performance trade-offs, correspond to those of the final year project. It confirms that goal optimization based on adaptive learning aligns with the project's goal-setting and tracking features that deal with progress of milestones and alerts when some of them are achieved. Moreover, the similarly nature of the portfolio selection through iterative queries of Aurum's system aligns with the user-oriented design of your goal construction and tracking features. There are useful insights for the project from the combination of active learning and adaptive systems in portfolio management, especially risk control or the ability to set financial goals. The research also illustrates how AI systems can be used to attain both diversification and performance enhancement.

2.2 Applications and Challenges of AI Technology in Financial Portfolio Management

One major idea that can be identified in the literature on financial portfolio management is the application of artificial intelligence (AI) and machine learning to develop idealized and realistic investment strategies. The document entitled Application As Well As The Challenges Of Artificial Intelligence In Financial Portfolio Management states that it is critical to harness the power of data analytics as the basis for intelligible decision-making at the investment arena and because of the variation and increasing complexity of investors' needs, particularly in today's world, it is crucial to tailor an improvised and diversified approach to satisfy these needs.

The positions the research in the wider spectrum of AI processes then presents a framework of intelligent portfolio management for taking risk and return efficiencies using quantitative analysis and an understanding of machine learning. Cores piece of evidence indicate that enterprise risk management and intelligent systems powered by artificial intelligence led to the optimization of portfolio performance through risk assessment, trend analysis and resource allocation (Wang, 2024). These findings exemplify that AI is a critical component to contemporary digitization of financial decision making processes.

Going further, the study provides descriptions of constructing the intelligent portfolio management platforms with deep coverage of the key processes, including data cleaning, entity recognition, and event extraction. Many of these techniques include; supervised and unsupervised learning enable the systems to analyze historical and real time data to find central processing information. Further, proliferation structures like the Graph Neural Networks substantiate the relationships between the financial entities to support better decision making (Wang, 2024). This methodological precision allows and enables platforms with providing the investors relevant tools to adjust for and navigate through various market context.

These findings are therefore important for their ability to revolutionize financial markets. In the conventional risk assessment and market analysis, AI driven system performs much better than human because there is less likelihood of making an error.

Readily available and finely tuned to reflect the highly-specialized risk tolerance of volatile financial landscapes, these systems are now critical components for asset managers seeking to address complex financial challenges.

When compared and contrasted to the AI-Powered Wealth and Lifestyle Management System the similarities are clear. Components like the asset performance, risk management and tailored investment advice reflect on the document's call for a rationality and risk assessment. Hence, the elements such as financial goal tracking and data visualization reflect on the above described AI applications, which aligns with the overall goal of providing user-oriented financial management tools. Consequently, the alignment demonstrated here can illustrate the combination between theoretical academic work and application of AI in financial systems.

2.3 Evaluation of Collaborative Filtering for Recommender Systems

This paper makes the following key observations: A common problem in recommender systems research is the accurate and efficient development of methods that can scale well in terms of user experience. To understand this, the study on Evaluation of Collaborative Filtering for Recommender Systems compares one variant of the CF algorithms against another to show and tell how well they perform in terms of recommendation accuracy and computational cost.

Again, the research situates CF as the most prevalent approach to recommend systems where users are expected to select items personalized based on past activity and user/ item similarities. Two primary categories are discussed: To be specific, this paper distinguishes memory-based and model-based CF. Memory-based methods that employ the direct approach together with K-Nearest Neighbor (KNN) formulas utilize user-item rating matrices to calculate recommendations based on similarity metrics, such as the cosine similarity (Al-Ghamdi, 2021). On the other hand, model-based methods for example, Slope One, co-clustering, and Non-negative Matrix Factorization (NMF) solve problems of sparsity and scalability that affect memory-based methods.

Exploring the detail of the outcomes, the work points out that item-based KNN is most effective in terms of accuracy with the lowest MAE and RMSE among all the applied algorithms. But it says that co-clustering is more effective in terms of coverage of the given dataset as it accurately predicted all of the data points in the dataset while the others were only around 57% (Al-Ghamdi, 2021). This paper also provides insights into hyper parameter optimization and cross-validation to choose the best model performance with less computational cost and high accuracy.

The importance of these discoveries is reflected in their ability to be implemented in actual recommendation systems, including those used on e-commerce websites. The research thus defines a conceptual framework for the creation of future systems capable of delivering accurate context-sensitive recommendations with respect to key issues identifying certain types of social networks, such as sparsity and scalability. Feedback-oriented performance measures like Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Fraction of Concordant Pairs (FCP) contribute workable reference points to measuring and increasing the system's efficiency.

These concepts are well correlated with the work of AI-Powered Wealth and Lifestyle Management System project. CF methodologies enhance such features as personalized investment recommendations and user preference capturing that the system can use to recommend suitable investments based on the user's pattern of use. Infrastructural models like the NMF and approaches based on similarity could complement the existing modules like the asset performance monitoring and the money financial goal tracking by making the results more accurate due to mathematical modeling. The acquisition of these superior evaluation metrics and optimization methodologies is the integral support of the system that begins to provide far more accurate, tailor-made, and efficient solutions in the encompassing financial territory for the identified purpose of the system.

2.4 Large-scale Recommendation for Portfolio Optimization

This integrating advanced algorithms to personalize the investment decision recommendation is in fact a prominent theme in financial portfolio management domain. We move from traditional brokers to online platforms; they require solutions able to provide scaled out personalized advisory solutions. Such research is evident when MPT is combined with Collaborative Filtering (CF) to optimally satisfy both the goals of maximizing returns and aligning with user preferences.

Swezey and Charron point out that the combination of MPT and CF is effective for improving investment recommendations. MPT is concerned about optimizing portfolios between Risk and Return, while CF utilizes user's behavior data to better refine the recommendations based on Familiarity and Affinity. Through domain expert validation and statistical evaluations on the provided dataset, we show that this hybrid approach of personalization and scalability outperforms standalone methods.

Modern Portfolio Theory gives us, the mathematical foundation, which enables us to optimize asset allocation by maximizing expected returns compared to risk (variance). It utilizes utility functions adapted for risk aversion levels, presented using Efficient Frontiers. This is complemented by an approach, known as Collaborative Filtering, which utilizes dense item-item matrices to model user familiarity with assets using personalized suggestions (Swezey, 2018). Through such integration, these techniques not only increase the computational efficiency in handling large scale datasets, but also lower the complexity level of them. The methodology validated against baselines such as the Random Selection and Pure Selection of CF show superior performance in expert evaluation especially in retail transaction context.

This capability to scale and yet remain personal for online wealth management platforms addresses critical gaps in the current approach to online wealth management. This method fills this vacancy by accurately estimating user risk aversion while making individual asset recommendations that optimizes both automated systems and human advisors. The versatility of it in applying across a wide spectrum of asset classes and its ability to adapt for various forecasting models highlights why it is very widely applicable, which matches the need for autonomous, data driven financial management tools (Swezey, 2018).

The 'Customized Suggestion for Investment' and 'Risk Control' artifacts of the AI powered Wealth and Lifestyle Management System were inspired by the core concepts of MPT and CF. By matching project goal setting and tracking to MPT's structured risk-return tradeoff, and CF's focus on user familiarity with matching project user preference collection and recommendation, the system is consistent with MPT's stakeholder preference collection. The visualization techniques used in Efficient Frontiers and item-based recommendations also apply to the "Reporting and Visualization" artifacts. User specific risk aversion γ captures the tradeoff between return and risk, which is optimized by utility function, $U(w) = E(rw) - \gamma \times Var(rw)$. The Efficient Frontier is portfolios that represent an optimal combination for any given level of risk.

2.5 Portfolio Optimization Using Evolutionary Algorithms

A theme that recurs frequently in financial optimization is its use of advanced computational techniques to augment portfolio management strategies. Evolutionary algorithms (EAs) have recently become popular as tools to solve the complex portfolio selection and optimization problems, inspired by natural selection processes. The methods in this thesis handle challenges from large data sets and dynamic financial markets.

Yu, Wang, and Lai propose a two stage genetic algorithm for portfolio optimization in their study. It takes asset ranking and optimal asset allocation together to produce high performing portfolios. It overcomes the limitations of conventional methods such as Markowitz's mean variance model, which prescribes fixed asset quality, and the information asymmetry and variable market environment (Yu, 2008).

First, based on financial indicators such as Return on Capital Employed (ROCE), Price to Earnings ratio (P/E ratio), Earnings per Share (EPS), and Liquidity Ratio, given a set of assets, genetic algorithms (GAs) select good quality assets. These indicators are stored as such on chromosomes, and fitness functions are used to evaluate the quality of chromosomes by minimizing Root Mean Square Error Root Mean Square Error (RMSE) RMSE). In the second stage, we optimize asset allocation using Markowitz's mean-variance theory with the use of GAs. A fitness function balancing risk minimization with return maximization is a fitness function, and chromosomes embody portfolio weights (Yu, 2008). There are experimental results of superior performance of portfolios produced from this method in comparison with traditional equally weighted portfolios.

By showing the ability to combine evolutionary algorithms with financial theories, the study highlights how such approaches help overcome the shortfalls of conventional models. The approach selects high quality assets and optimizes allocation to better result in a volatile market. The latter makes it a valuable tool for investors aspiring to efficient decision making in the face of a wide gamut of financial scenarios and big data sets.

Genetic Algorithms (GAs): Indirection recruitment for selection, crossover, and mutation to improve selection and allocation of assets mimicking biological evolution.

Markowitz's Mean-Variance Theory: Balances expected return and risk, mathematically represented as: where is the asset weight, the expected return, γ the risk aversion parameter, and σ_{nm} the covariance.

Fitness Function Design: It chooses portfolios where the risk and return are at the best trade-off balance.

The concepts overlay so closely with the "Customized Suggestions for Investment" and "Risk Control" parts of the AI powered Wealth and Lifestyle Management System that it is virtually one in the same. The first stage ranking is consistent with the system's ability to complete preference collection and risk tolerance assessment, and the second stage optimization aids in the asset performance monitoring and provides investment recommendations. These include financial goal progress charts that integrate outputs from the evolutionary optimization process to enhance user experience through reporting and visualization features.

2.6 Portfolio Optimization using Machine Learning Techniques

Portfolio optimization with machine learning techniques is a radical transformation for the bread industry. Traditional methods, like the mean-variance model, lack the flexibility to accommodate our complex financial markets. Through machine learning, we can employ diverse models like regression, clustering, reinforcement learning, neural networks to create data driven portfolio analysis which are quicker, cheaper and more adroit.

Recent studies show the application of advanced machine learning methods to overcome the shortcomings of conventional portfolio optimization techniques. Deep reinforcement learning is shown to balance portfolios dynamically by Sugadev et al. Like Ma et al., we also demonstrated the integration of deep learning with classical methods to increase risk adjusted returns. Singh et al. looked into the advantages of applying recurrent neural networks (RNNs) for time dependent adjustments in portfolio; while Yoo et al. utilized clustering approaches to functionalize such investments in the portfolio a la diversification by grouping assets under identical characteristics.

Many aspects of portfolio management have been improved significantly by machine learning methods. Asset return prediction use regression models, and asset risk is assessed with classification models (e.g., support vector machines). Diversification of assets is customarily done by clustering techniques like K-Means or hierarchical clustering. We use reinforcement learning algorithms optimizing dynamic portfolio balancing: DQNs (Deep Q Networks) adjust to market conditions. Finances model are introduced to bring in relation to deep learning approaches that enable non-linear relationship to be exposed, and hidden pattern to be unveiled (Sugadev, 2023). Their results improve decision making through data driven, precise and adaptive insights.

Machine learning is adopted to solve the problems introduced by the dynamic market conditions and different types of assets. This gives investors the chance to utilize tools that help with better risk management and higher returns to optimally cope with the craftiness of financial climate. Robust model training and synthetic data generation, and increased portfolio resilience is provided by techniques such as ensemble learning and GANs (Sugadev, 2023).

The discussed concepts are intertwined with the 'Customized Suggestions for Investment' and 'Asset Performance Monitoring' artifacts in the Environment for the AI powered Wealth and Lifestyle Management System. Both "Risk Control" and "Portfolio Summary" have straightforward techniques such as clustering and RNNs that directly apply, while "Asset Management" can be modeled with reinforcement learning which matches the changing asset allocation. Outputs from machine learning models can add capabilities like progress charts and performance graphs for reporting or visualization viewers to help build user insights.

Regression Models: Used for predicting asset returns, represented as

$y = \beta_0 + \beta_1 X + \varepsilon$.

Clustering Algorithms: Optimize diversification by minimizing within-cluster variance:

$$J = \Sigma^{i}_{k} \Sigma_{x \in c_i} ||x - \mu_i||^2$$

Reinforcement Learning: Employs Q-learning to optimize portfolio allocation dynamically.

$$Q(s,a) = r + \gamma \max\_a' Q(s',a')$$

Neural Networks: RNNs can model complex relationships (e.g. time dependent patterns) and CNNs identify historical price data trends.

The coupling of machine learning techniques with financial theories is the basis of creation of adaptive, precise and efficient portfolio management systems.

2.7 Real Time Portfolio Management System Using Machine Learning Techniques

Recent financial research has consistently exposed the transformative virtues of machine learning in the management of portfolio, namely the selection, optimization, and real time management of assets. Foundational are traditional approaches such as the mean variance model of Markowitz, but they do not address the complexity of modern, dynamic markets. As a result, machine learning, with its adaptive and predictive properties, is filling this gap and becomes a key tool in providing assistance to portfolio management strategy.

In his research findings, advances such as clustering algorithms like K-Means for asset selection and metaheuristic algorithms like Genetic Algorithms (GA) for optimization are mentioned. Studies show how clustering techniques can help diversified portfolios, by grouping similar asset characteristics together to reduce sector bias and volatility in portfolios. Genetic Algorithms, fertilized by Sliding Window techniques, dynamically optimize asset allocations adapting to changing market dynamics in the meantime (Aithal, 2023).

Elaboration is made on these findings using an exposure to application of advanced optimization models. Metaheuristic methods, like Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO) and Artificial Bee Colony (ABC) algorithms are used for optimization of the GMV and Maximum Sharpe Ratio (MSR) portfolios. Both returns and risk adjusted ways to achieve these techniques outperform traditional methods. Additionally, we back test these optimized portfolios using real time data and they are consistently better than benchmarks like the Nifty-100 index (Aithal, 2023).

The relevance of these results is in the application to dynamic portfolio management. Pipelines which combine machine learning and financial heuristics result in portfolios of higher expected returns but with higher stability. Robustness of these approaches is underscored by the solid key metric like the rolling Sharpe ratio and beta values.

In asset management, performance monitoring, and customized investment suggestions, these core concepts resonate in the project’s artifacts. K-Means clustering for diversification is a technique that compliments your asset categorization and tracking capabilities. Likewise risk control mechanisms in your project should include metrics such as rolling beta and Sharpe ratios for real time performance alerts. Aurum project's focus on leading financial planning through adaptive and robust financial planning—from the mathematical core including optimization equations and risk metrics, directly supports user centric features like financial goal tracking and financial portfolio recommendations.

2.8 Reinforcement-Learning based Portfolio Management with Augmented Asset Movement Prediction States

Portfolio management research has frequently pointed out that traditional methods based on the Markowitz framework have limitations and a lot can be done by leveraging RL to address those. These classical models also have the propensity of being evaluated under the transitory environment of static market behaviors that these models typically miss out. With modern advancements in RL, we have a very robust framework in which to use RL to optimize asset allocation by dynamically adjusting strategies based on new data inputs.

The results of research confirm the feasibility of using augmented state modeling for portfolio management in RL. In the state augmented RL (SARL) framework, we introduce diverse data, including historical prices and financial news, to improve the predictions of asset movement. We perform experiments on cryptocurrency and stock datasets and find large gains in terms of portfolio value and Sharpe Ratio with SARL compared to standard RL methods, of 140.9% on Bitcoin datasets and 15.7% on High-tech datasets. External information such as news embedding, help prediction with noisy or sparse datasets while keeping prediction accuracy high (Ye, 2020).

The findings extend further the adaptability of the SARL in dynamic situations or heterogeneous environments. SARL addresses data heterogeneity and environmental uncertainty by integrating into the framework an augmented state. As an example of application areas, the accuracy of movement prediction is improved by the use of LSTM networks and HANs for asset reallocation based on informed decisions. The existence of this adaptability guarantees higher profitability and lower risk than that of static models.

Furthermore, given the practical gains for investors, these findings have a special significance. SARL is robust to market dynamics, as emissions such as portfolio value and Sharpe Ratio underscore, and is immune to overfitting problems frequently experienced in conventional RL (Ye, 2020).

The SARL framework compares favorably to features such as customized investment suggestions, risk control and asset performance monitoring when compared to these concepts. For instance, supplementing state modeling with augmented state modeling using financial data and user specific inputs would be in parallel to your design's recommendation systems and asset alerts. Similar to aurum system, the focus of SARL on dynamic reward optimization and action-driven reallocation of assets respects the same the adaptive tracking and goal setting. This point of view of applying RL to wealth management solutions brings something new.

2.9 Implementing AI

2.9.1 Modern Portfolio Theory Utility Function:

$U(w) = \sum(i=1 \text{ to } n)\ wi \cdot \mu i - (\gamma/2) \cdot \sum(i=1 \text{ to } n)\sum(j=1 \text{ to } n)\ wi \cdot wj \cdot \sigma ij$ where:

- wi = weight of asset i
- μi = expected return of asset i

- σ_{ij} = covariance between assets i and j
- γ = risk aversion coefficient

2.9.10 Similarity and Recommendation Metrics

Cosine Similarity:

$$Sim(A,B) = \sum_{i=1}^{n} A_i \cdot B_i / \sqrt{\sum_{i=1}^{n} A_i^2} \cdot \sqrt{\sum_{i=1}^{n} B_i^2}$$

Matrix Factorization:

$R \approx P \cdot Q^T$ where:

- R = user-item interaction matrix
- P,Q = latent factor matrices

2.9.11 Reinforcement Learning Formula

Q-Learning Update Rule:

$Q(s,a) = r + \gamma \cdot \max[Q(s',a')]$ where:

- s = current state
- a = action
- r = reward
- γ = discount factor
- s' = next state
- a' = next action

2.9.12 Performance Metrics

Sharpe Ratio:

$SR = (E(r) - rf) / \sigma$

where:

- $E(r)$ = expected return
- r_f = risk-free rate

- σ = standard deviation of returns

2.9.13 Key Algorithms

Portfolio Optimization:

Algorithm: Two-Stage Genetic Algorithm
Input: Asset universe, constraints, risk parameters
Output: Optimal portfolio weights
Stage 1 - Asset Selection:

1. Initialize population of binary chromosomes

2. Evaluate fitness using MPT utility function

3. Apply crossover and mutation

4. Select best asset combination

Stage 2 - Weight Optimization:

1. Initialize weight distribution

2. Apply weight constraints

3. Optimize using utility function

4. Return optimal weights

Collaborative Filtering:

Algorithm: Memory-based CF
Input: User-item interaction matrix
Output: Predicted ratings

1. Calculate similarity matrix using cosine similarity

2. For each user-item pair:

a. Find k nearest neighbors

b. Compute weighted average of ratings

3. Return prediction matrix

2.9.14 Finding of Analysis

Active Preference Learning and Bayesian Optimization

By combining active learning with user preference integration in a Bayesian optimization framework, proposed by Tee, Jones, and Boyd (2017), Bayesian Optimization helps personalized portfolio construction. This approach serves efficiently for risk tolerance assessment and investment recommendation and adheres to its goal setting, adaptation system.

Collaborative Filtering (CF)

By taking advantage of user-item interactions data, CF techniques including memory based KNN, model based NMF can further improve recommendation system performance. By way of comparative analysis, it is established that item based KNN demonstrate much better accuracy (low MAE and RMSE) while co clustering is better at data coverage. As directly applicable as these insights are for Aurum's customized investment suggestions.

Modern Portfolio Theory (MPT) and Evolutionary Algorithms

One of the key aspects of optimizing risk return tradeoffs using utility functions and efficient frontiers is what is provided by MPT. GA make this more evolutionary process by treating dynamic market environment and big data. Significant performance gains are shown with a two stage GA approach of combining asset ranking with optimal allocation.

Augmented States and Reinforcement Learning (RL)

The SARL framework coupled with reinforcement learning improves portfolio management by dynamically adapting the strategies, based on market or exogenous data. Altering financial news and historical data into augmented states results in dramatically increased prediction accuracy and stability in Aurum features such as risk control and performance monitoring.

Diversification and Dynamic Adjustments using Machine Learning

Portfolio diversification and asset reallocation are optimized using clustering algorithms (e.g., k-Means), and deep reinforcement learning. RNNs provide a way to make time dependent adjustments, as required by Aurum's reporting and visualization.

2.9.15 Understanding the Intelligence AI Tries to Achieve

The system attempts to reproduce how a human financial advisor thinks by going over organized investment past. The intelligence covered in the command is called:

Tailored decisions: Changes the way money is allocated using an investor's age, investment ambition and risk affinity.

Adaptive learning: Analyzes the investing strategies of thousands of real individuals to determine the top assets to include in your portfolio.

Using domain rules: If someone chooses a conservative strategy, more bonds will be invested. Easy for users to understand why a specific asset is being suggested.

Essentially, the aim is to make smart, relevant and justifiable financial recommendations that reflect the skills of a financial professional, able to work on a large scale.

2.9.16 Comparison between the performance of algorithm (Mathematics and Discussion)

RF and GB were the two regression models employed in the analysis. Performance was improved by developing an ensemble model.

Key Observations:

Model	R^2 Score (Avg)	RMSE (Avg)	MAE (Avg)	Strength
RF	0.90–0.96	Low	Low	Stable, interpretable
GB	0.91–0.97	Very Low	Very Low	Better with non-linearity
Ensemble	**0.91–0.97**	**Lowest**	**Lowest**	Combines best of both models

2.9.17 Mathematical function of ensemble

$$y^{ensemble} = \alpha \cdot fRF(X) + (1 - \alpha) \cdot fGB(X), \alpha = 0.\hat{6}\{y\}_{\{ensemble\}}$$

$$= \alpha \cdot f_{\{RF\}(X)} + (1 - \alpha) \cdot f_{\{GB\}(X)}, \backslash quad\ \alpha = 0.6\boldsymbol{y^{ensemble}}$$

$$= \boldsymbol{\alpha \cdot fRF(X) + (1 - \alpha) \cdot fGB(X), \alpha = 0.6}$$

This approach reduces bias and variance, improving generalization.

The ensemble approach delivered the best trade-off between accuracy and robustness, making it ideal for the recommendation system deployed.

3. Project Methodology

3.1 Software Development Methodology

Agile technique and iterative SDLC will be used such that the project will be completed timely and successfully. The approach encourages iterative development and flexibility making it a perfect feature for this project as it requires multiple integrations and requirements. The breakdown is as:

3.1.1 Preparation and Requirements Gathering

In this phase the objective, scope and specific requirements of the projects are clearly mentioned whereby identifying key features of above mentioned artifacts such as user authentication, asset management, financial goals and AI connections is required. Likewise, identifying technology stack and dividing project into components and creating sprints are required.

3.1.2 Design Goal

Creating user interfaces and visualizing the system architecture task is completed in this one whereby flowcharts, and diagrams to understand and comprehend system components and user interactions is done. Likewise, database schema holding preferences, financial information and specifying data flow and API structure for front end, back end, and AI model communications is carried out. Planning of UI/UX is also in this phase.

3.1.3 Iterative Development

Creating the system in iterative sprints, providing a functional feature at last of each sprint is required. Whereby different core features are divided into different sprints and done accordingly. The main objective is to complete the task in well-organized manner and on time thus with minimal or none delay. The division makes clear sprints such that every factors of the project is considered and given timeline thus better integration can be achieved.

3.1.4 Testing and Quality Assurance

It is to guarantee the usability, functionality and stability of the system and project and test whether the project has been concluded with accordance to requirements. UI/UX should live up to user expectations and the product is integrated and properly tested. It is also necessary to verify the correctness and functionality of AI models and to check the flaws in the system and fix it.

3.1.5 Observation and Maintenance

Tracking system performances and managing enhancements to make sure the system runs well by continuously monitoring is an essential step. User input can be taken to enhance UI and functionalities and regular maintenance for AI models can make the accuracy go higher. Based on real world usages problems and flaws can be fixed.

3.1.6 Iteration and Continuous Improvement

Constantly enhancing system in response to client input and evolving needs can be considered future goals of the project and can be regarded as its scope. Improving AI models by analyzing user fresh data as well as user behavior to improve the system and satisfy the user needs can be done. Including extra models in the future to iterate based on user feedback can be a good aim in future.

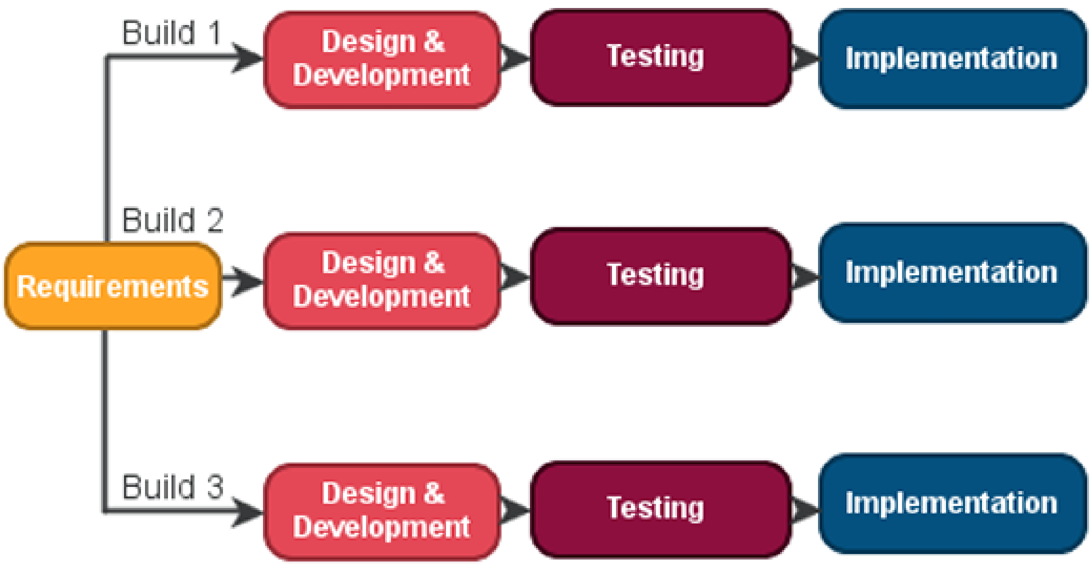


Figure 9 SDLC - Iterative (professionalqa.com, 2019)

3.2 WBS (Work Breakdown Structure)

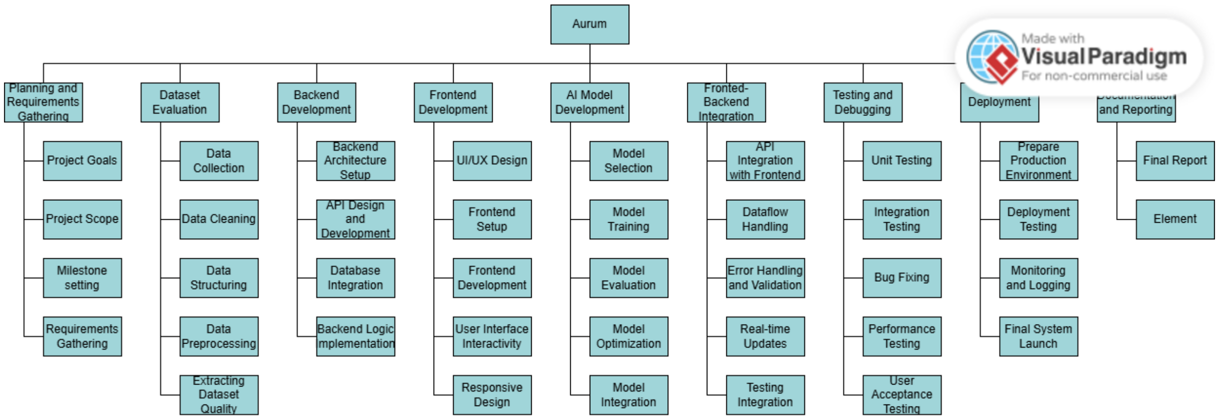


Figure 10 WBS (Work Breakdown Structure)

3.3 Gantt Chart

Figure 11 Gantt Chart

4. Tools and Technologies Used

4.1 Frontend Development

React.js: It’s an UI library that is used to build components.

HTML/CSS/JavaScript: Used for structure, styling and scripting respectively.

Bootstrap/Material-UI: UI frameworks that is responsible for responsive design.

4.2 Backend Development

Django: It’s a python web framework.

SQLite: It’s a built in database in django.

Django REST Framework: To build RESTful APIs.

4.3 AI/ML Tools

4.3.1 Supervised Machine Learning

Algorithms used:

RandomForestRegressor (from sklearn.ensemble)

GradientBoostingRegressor (from sklearn.ensemble)

Purpose: Predict asset allocation percentages based on user profile data.

Ensemble Learning**:**

A custom PicklableEnsemblePredictor combines predictions from both models

4.3.2 Feature Engineering

Implemented through a custom transformer class FinancialFeatureTransformer.

Techniques used:

One-hot encoding

Interaction terms

Non-linear transformations (e.g., age^2, horizon^2)

Rule-based logic to enrich inputs for modeling.

4.3.3 Data Preprocessing

Libraries used:

pandas, numpy, scikit-learn for data handling and transformation

Techniques:

Scaling with StandardScaler

Handling categorical data (e.g., age ranges, investment goals)

4.3.4 Model Evaluation & Metrics

Metrics used:

R^2 Score

Root Mean Square Error (RMSE)

Mean Absolute Error (MAE)

Validation Technique:

KFold Cross-Validation (5-fold)

4.3.5 Explainable AI (XAI) Concepts

Feature importance via:

model.feature_importances_ (for Random Forest)

permutation_importance (from sklearn.inspection)

Rule-based post-processing to apply domain logic (e.g., age > 50 ⇒ prefer bonds/real estate)

4.3.6 Agent Architecture

AI agent: NepalInvestmentRecommender

Handles preprocessing, prediction, normalization, and logic application

Deployed using Django API for real-time usage

Returns allocation + specific asset suggestions with reasoning

4.4 Integration Tools

Deployment-ready model serialization: pickle

API Integration: Example Django views for:

Quick recommendation

Comprehensive personalized financial planning

4.5 Development Tools

Git: Version control

GitHub/GitLab: Hosting repositories

Postman: API testing

VS Code: Code editors

4.6 Security

Django Security Features: There are built in protection.

4.7 Project Management

Click Up: To track the project timeline and list out the tasks to be completed.

4.8 Client Information

Mr. Nikunja Lamsal, supervisor to this project will be acting as a client to this project.

5. Artefact Design

5.1 Artefact – User Account Management

5.1.2 FDD of User Account Management

Specifically for user account management module, this figure shows the FDD model. It shows the core aspects one needs for resolving user's authentication, registration, and profile management stuff. The development process made by smaller feature sets enabled us to clearly define the development process so that the important functionalities were developed correctly. FDD model allowed me to structure the user account management in same development. Thus, this diagram also assisted in discovering the dependencies between various user-related features. The detailed breakdown of the implementation and testing processes streamlined my work, but I'm sure lacking the large numbers the other guys were aiming for slowed down their work a fair bit.

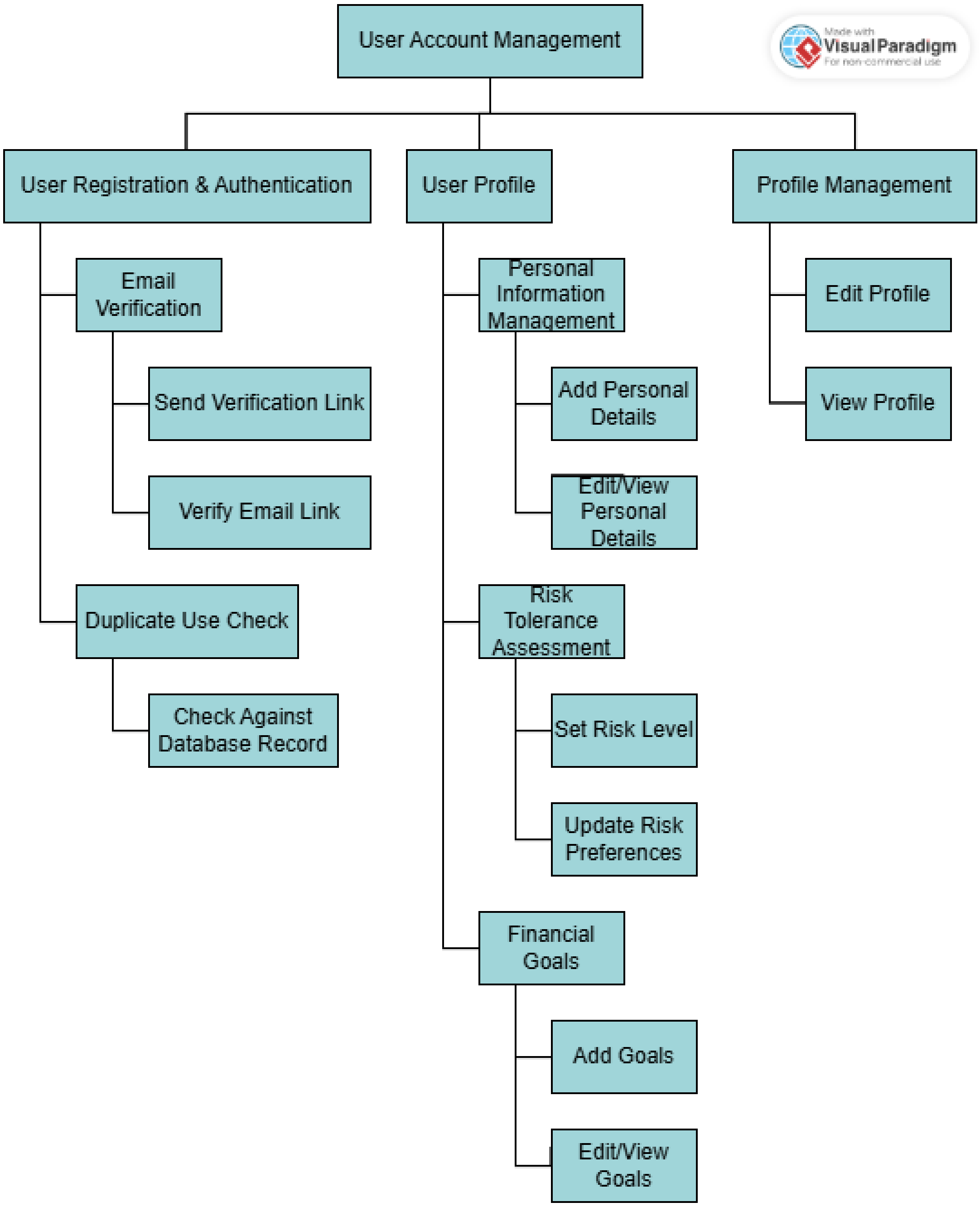


Figure 12 FDD of user account management

5.1.3 Product Backlog of User Account Management

This is a figure about the product backlog for the user account management module, showing all tasks and features that are to be developed. These are also some features of the backlog: user registration, login, authentication, update of the profile. The backlog has been prioritized with importance and complexity of each item to have a smooth development process. Throughout the project, I always refined and updated the backlog. It was a reference point to monitor progress and henceforth pending tasks. The rationale behind this backlog was key for planning my sprint activities.

Epic 1: User Account Management

ID	User Story	Priority	Acceptance Criteria
UAM-1	As a user, I want to register with my email and password so that I can create an account.	High	User can create an account, and duplicate emails are blocked.
UAM-2	As a user, I want to receive an email verification link so that I can confirm my account.	High	Email verification link is sent, and the account is verified after the user clicks it.
UAM-3	As a user, I want to update my personal details in my profile so that my account stays current.	Medium	User can edit and save personal details (name, email, etc.)
UAM-4	As a user, I want to set my risk tolerance level so that the system can provide tailored advice.	High	Risk tolerance is saved and updated in the user's profile.
UAM-5	As a user, I want to add financial goals so that I can track my progress.	High	Goals are saved, editable, and viewable in the user profile.

Figure 13 Product backlog for user account management

5.1.4 Sprint Planning for User Account Management

This is the sprint backlog for specifically user account management module, listing down what will be done in the sprint cycle. It is a collection of the user stories and tasks, deduced from the product backlog. It was the sprint backlog that got me to work on more important tasks than others while keeping my time regulated. I then allocated estimated time for each task as this helped me keep a watch on my progress. I updated this backlog along the way, completing tasks and refining other tasks to keep in the backlog. As a result, development process was structured and oriented on a goal.

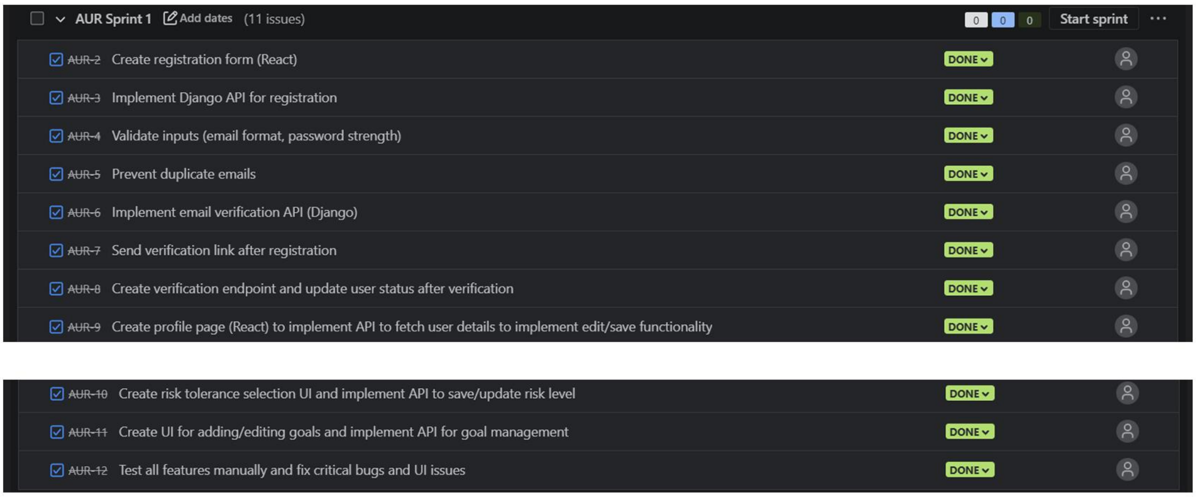


Figure 14 Sprint backlog for user account management

5.1.5 Activity Diagram for User Account Management

The sequence of actions performed by users and the system is represented in this figure, which is the activity diagram of the user account management module. It contains basic user's registration, login, authentication and profile updates. It allowed me to visualize how the user will interact with the system, and can engage in a smooth workflow. We designed each of the activity transitions to avoid conflicts and have them smooth. As a result, this diagram had a very important role in enforcing logical process flows within the system. It also helped to detect bottlenecks in the workflow.

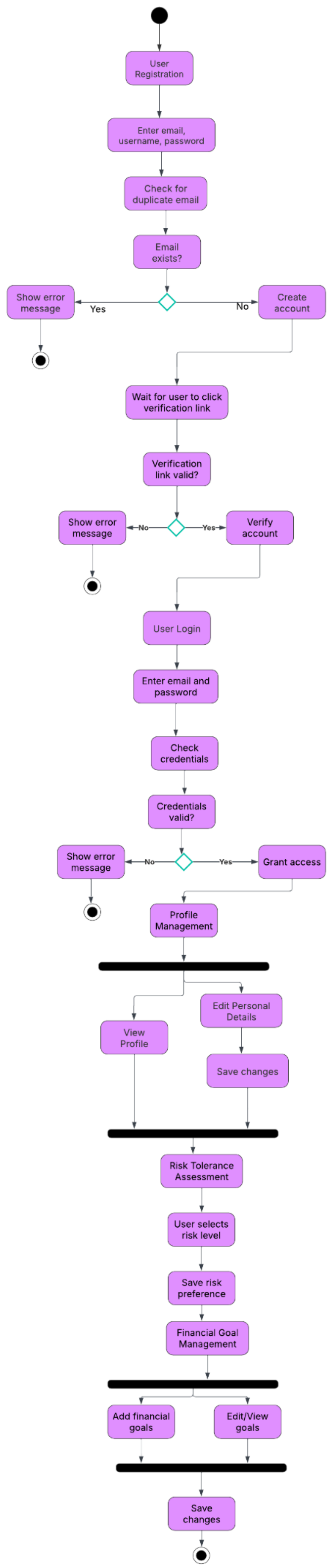

User Registration
Enter email, username, password
Check for duplicate email
Email exists?
Show error message
Yes
No
Create account
Wait for user to click verification link
Verification link valid?
Show error message
No
Yes
Verify account
User Login
Enter email and password
Check credentials
Credentials valid?
Show error message
No
Yes
Grant access
Profile Management
View Profile
Edit Personal Details
Save changes
Risk Tolerance Assessment
User selects risk level
Save risk preference
Financial Goal Management
Add financial goals
Edit/View goals
Save changes

Figure 15 Activity diagram for user account management

5.1.6 Use Case for User Account Management

Here is this figure which is a use case diagram for interactions of the user and the user account management system. It defines several user roles (registered users and admins) and what permissions you have on each. This diagram had helped me to be sure that all the functional requirements are covered. Each use case is a particular user's action he has to perform, for instance, the 'sign up', the 'login' and the 'update profile' steps. With this user interaction mapped clear, I had the opportunity to create a less user confusing experience. In case of user authentication security, this diagram helped determine how to implement security measures.

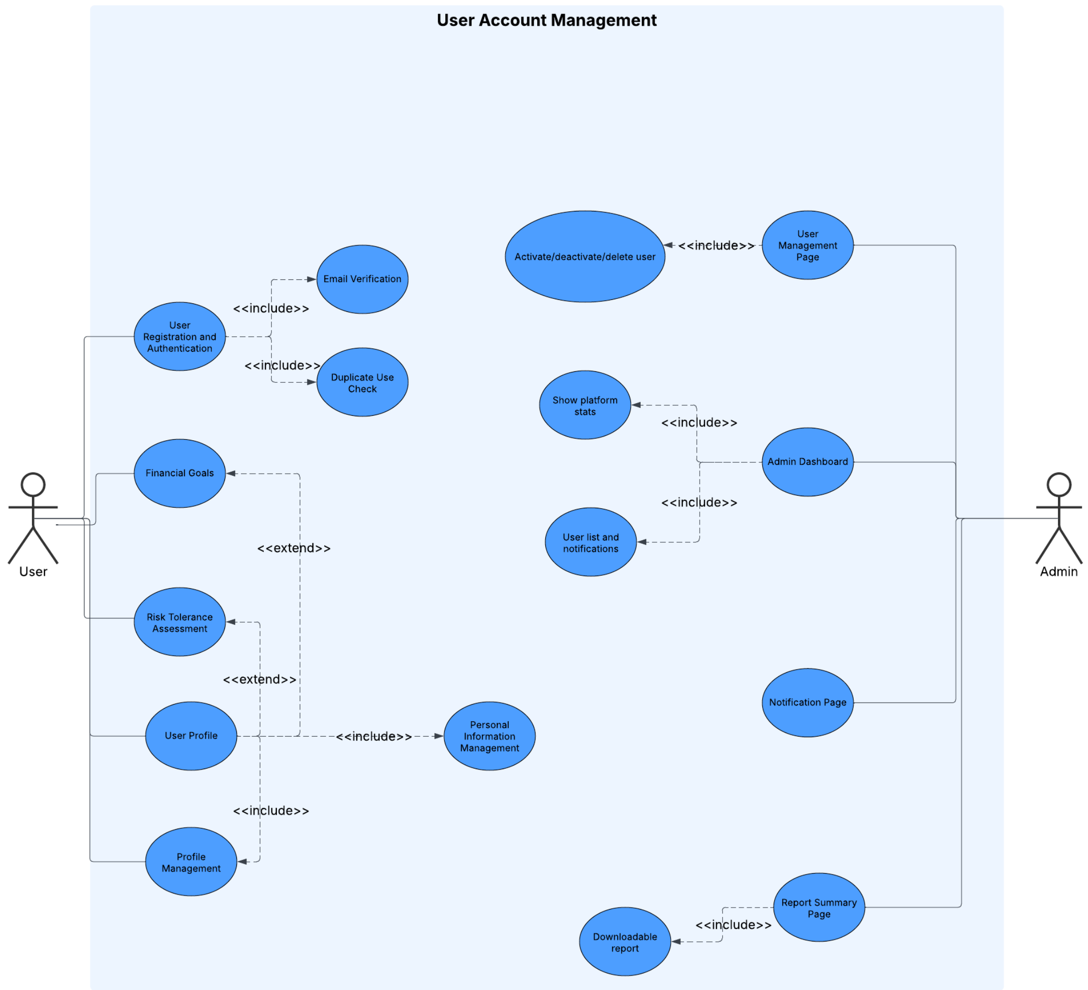


Figure 16 Use Case for user account management

5.1.7 Wireframe – Home

The wireframe of the home page is presented in this figure and can be considered as a blueprint of the website's first interface. It contains navigation elements, key sections, and spot for content. This is important, as I could use the wireframe to structure the layout before beginning with the actual UI design. I mapped out the most important interface element to create a clear and user friendly design. The wire frame was used as the basis for frontend work to maintain consistency in page structure. It also gave early feedback on the design before becoming fully implemented.

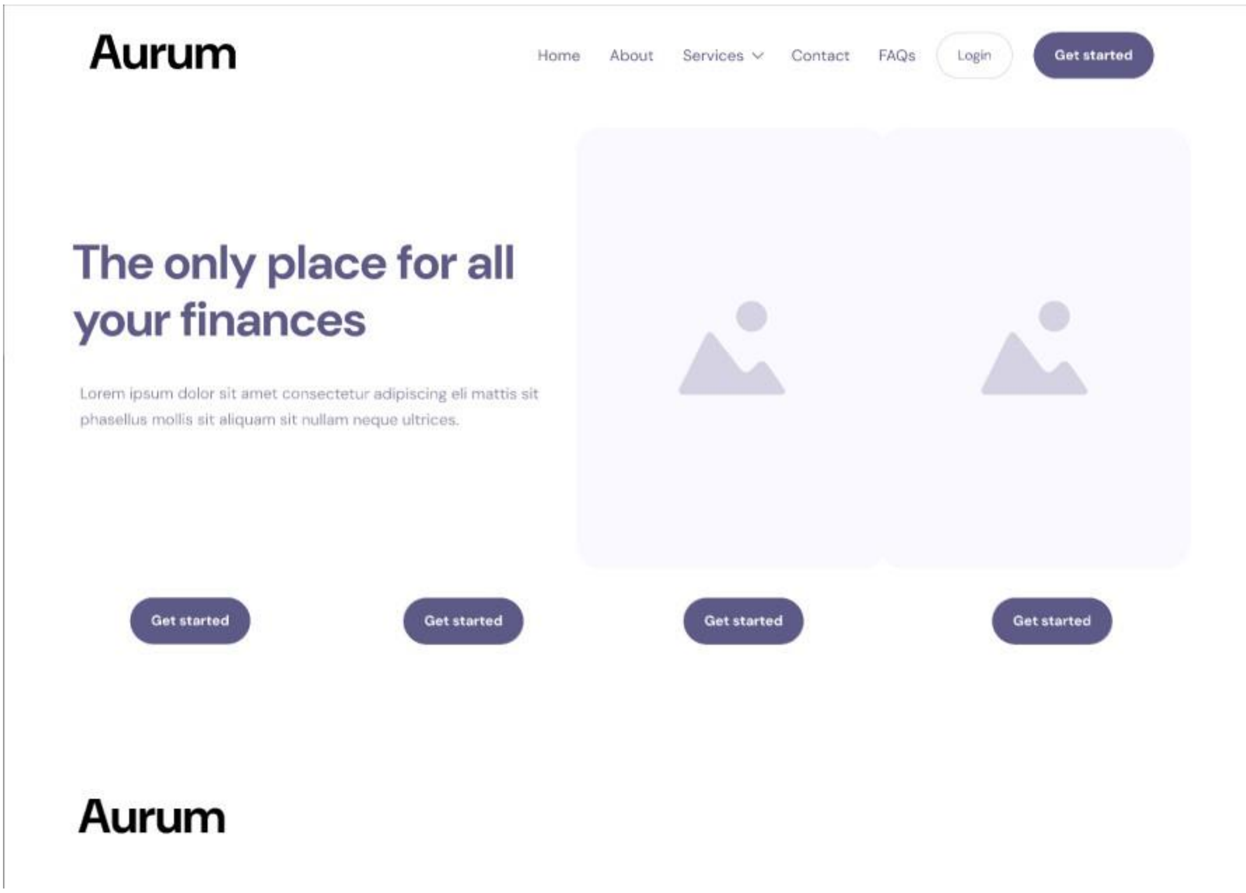


Figure 17 Wireframe - Home

5.1.8 Wireframe – Sign Up

The wireframe of the user sign up page is given in this figure which includes the input fields and the buttons necessary for user sign up. It has username field, email, password and the final confirm password so we have all information. The registration process was simple and minimal friction on purpose. This wireframe helped the development of the actual sign up page. This structured wireframe allowed me to create a good User Experience. On the one hand, the design was at some point subjected to validation feedback and error handling.

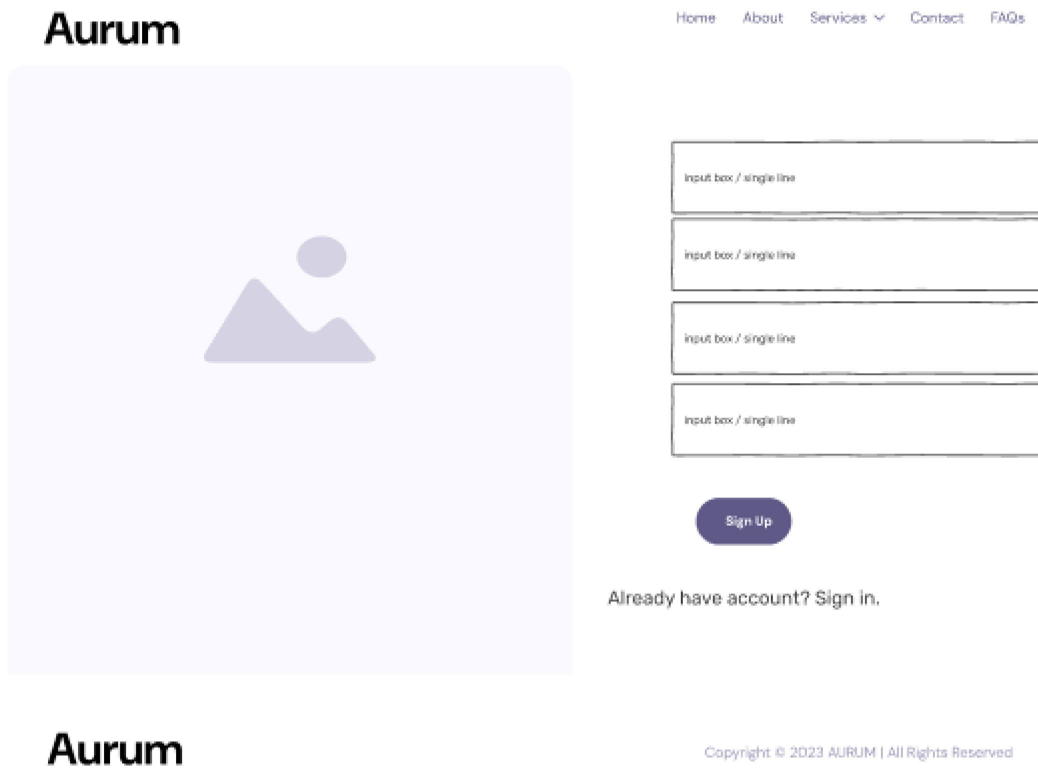


Figure 18 Wireframe - sign up

5.1.9 Wireframe – Login

The structure for user account makes sense in this kind of situation, and this is what this figure shows — the login page wireframe. The part includes your email and password field input so that you can click on the login button and if you can't remember your password you can use the 'forgot password' link. In designing such layout with planning the wire frame was helpful in figuring out what the users will be easy to get their accounts. I focused in on usability to make the authentication process simple. The frontend implementation of the login feature was carried out according to this wireframe. It also enabled us in planning the security feature like error handling while providing wrong credentials.

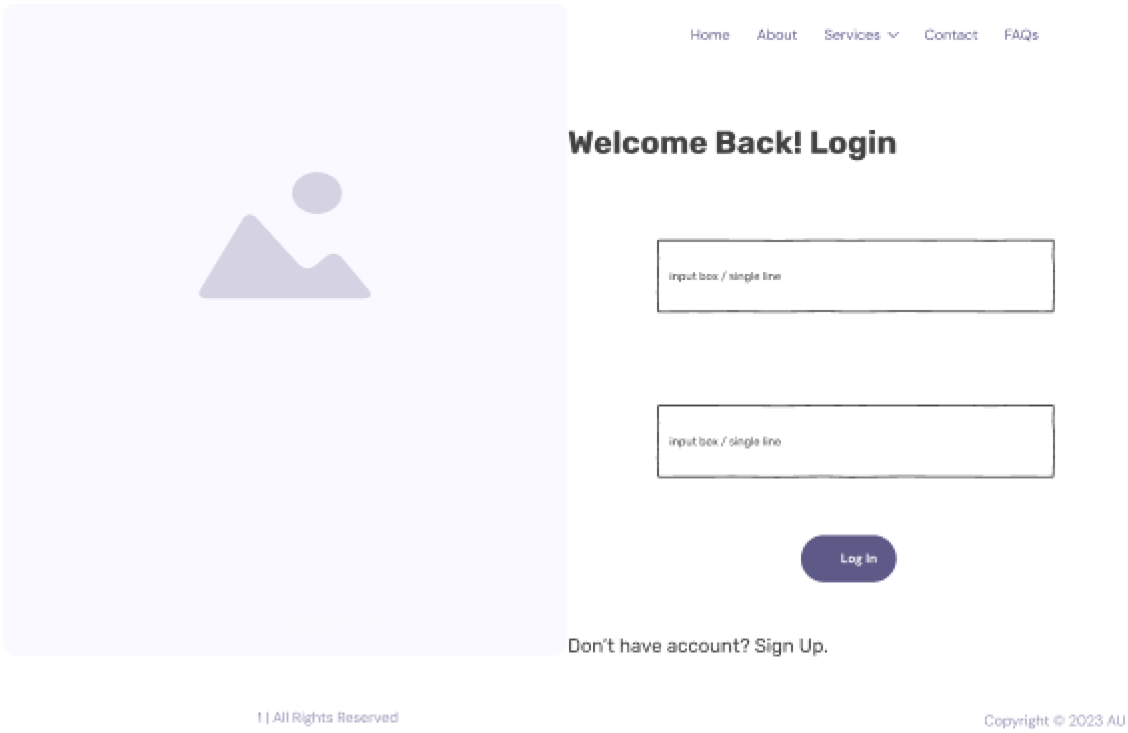


Figure 19 Wireframe – login

5.1.10 Wireframe – Personal Information

The wireframe for the personal information page where users may edit and manage the profile details has been presented in this figure. It includes name field and other personal user information such as email, phone number and so on. It is designed in such a way that makes the changes structured so that it becomes easy for the users to make the changes. By building on this wireframe, I was able to implement the actual personal information page accordingly in order for its user to have an intuitive experience. Additionally, data accuracy validation checks were also planned using it. I made the layout consistent across the platform by structuring the page layout beforehand.

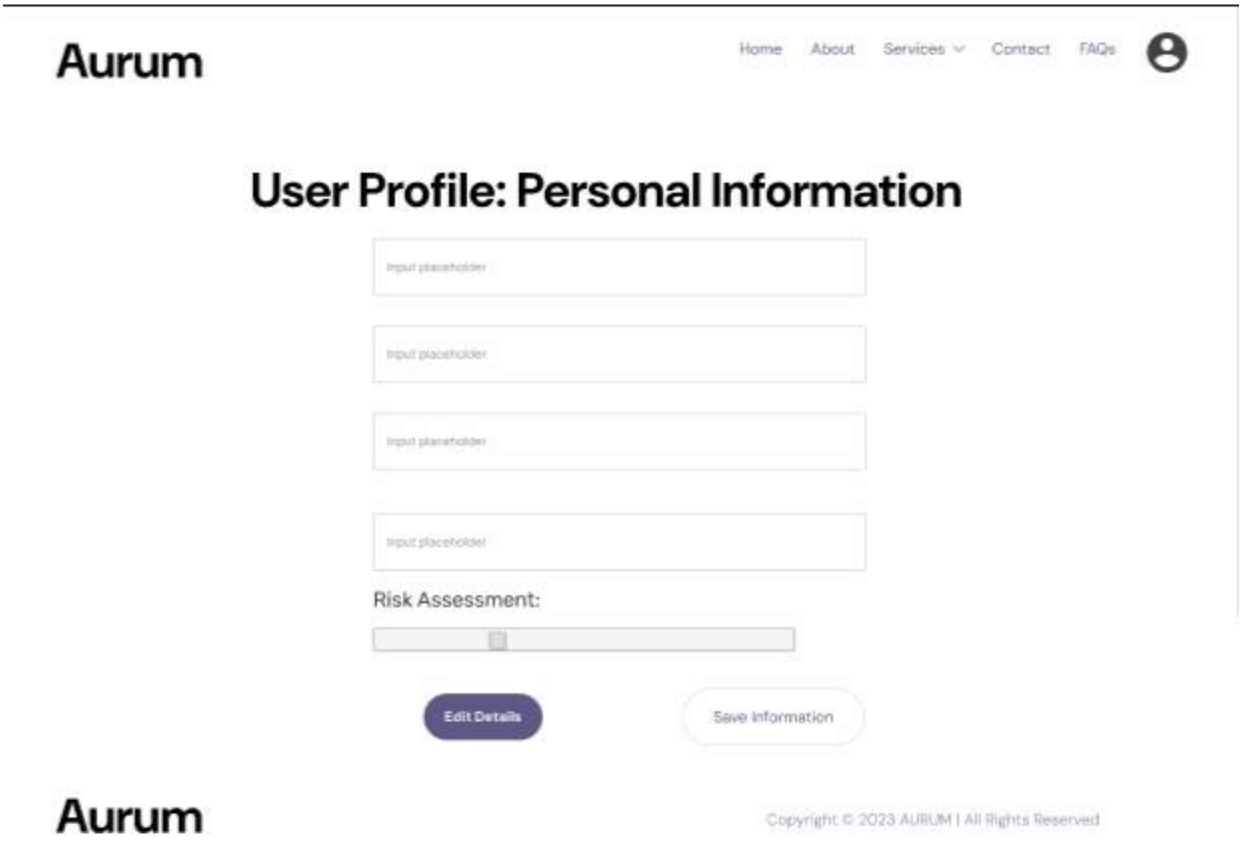


Figure 20 Wireframe - personal information

5.1.11 Wireframe – Admin Login

This is the wireframe of the admin login page which is a page dedicated to the interface for the administrators to access to the system. The admin credentials are included as input fields on the page as well as a secure authentication mechanism. Finally, the layout is minimized and focused on security. This wireframe helped me to define that administrators should have a seamless and secure login. Role based access control is also considered to limit unauthorized access. The actual implementation of the login functionality of the admin was guided by this wireframe.

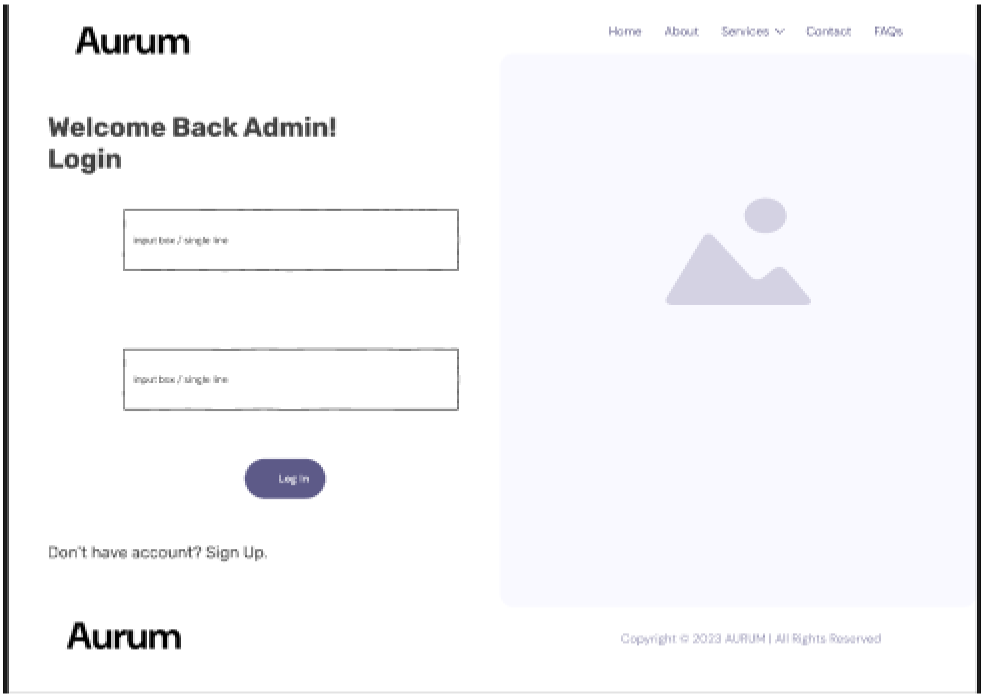


Figure 21 Wireframe - admin login

5.1.12 Wireframe – Admin Dashboard

This is wireframe admin dashboard, as central hub of managing users account system setting. The dashboard contains key system metrics overviews, users management, and other administrative controls. Layout was made in such a way that gives an ease of access on necessary functionalities. This wireframe was structured in such a way, which would help me streamline administration activities. This planning helped me on the development of an intuitive and well organized interface, with sections taking care of their own concerns. It also dictated frontend development of the dashboard.

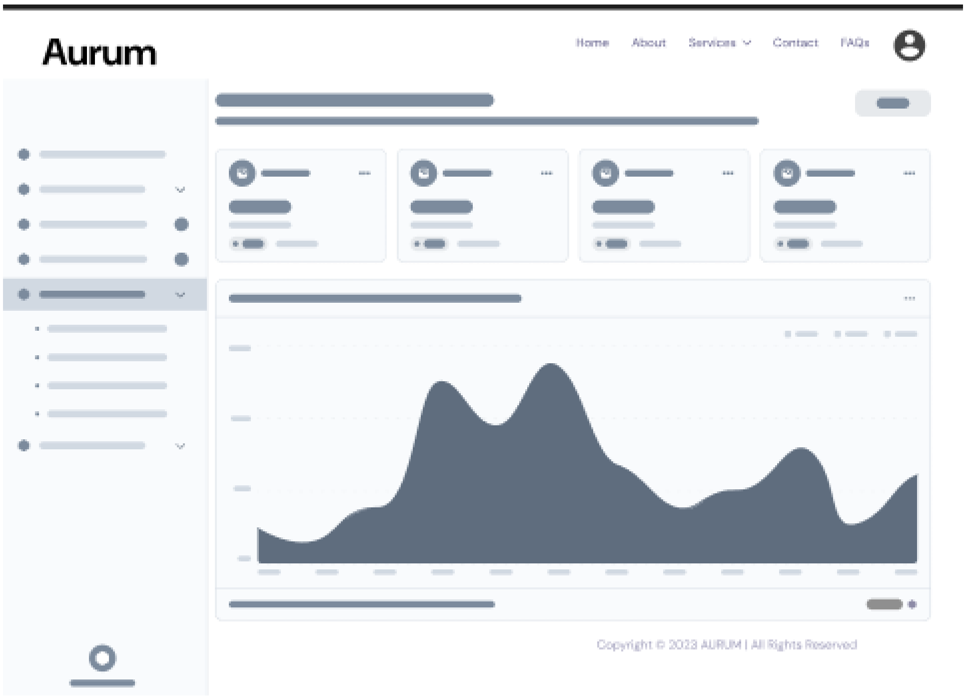


Figure 22 Wireframe - admin dashboard

5.1.13 ERD of User Account Management

This figure showcases the Entity-Relationship Diagram (ERD) for the user account management module. It shows how a database entity such as users connects to another like roles or Authentication records. I defined entity relationships so that the database schema would be well structured. The ERD like this helped designing the database tables and optimization of queries. It also was involved in ensuring data integrity and security. It gave a clear understanding of how data of users is stored and handled.

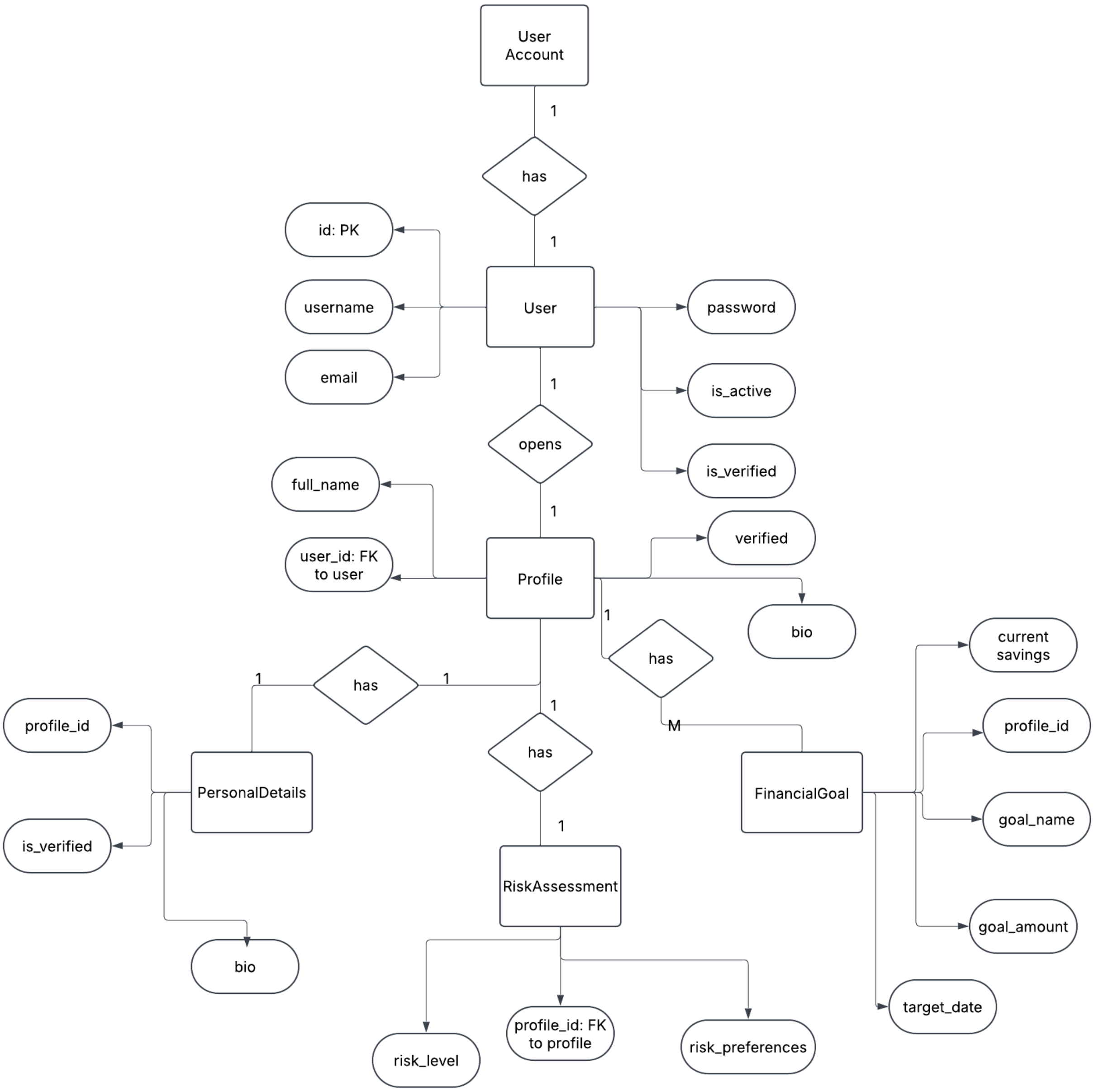


Figure 23 ERD of user account management

5.1.14 Class Diagram for User Account Management

The class diagram of the user account management module is defined in this figure and the description of the structure and the relationship among classes in different classes. It has classes for users prefixing all classes with Users, authentication classes and profile management. I mapped out class relationships so that class structure is organized and modular. It assisted for the design of object oriented components efficiently. The code reusability and maintainability that this approach offered was due to being structured. It was used as a reference during the implementation phase.

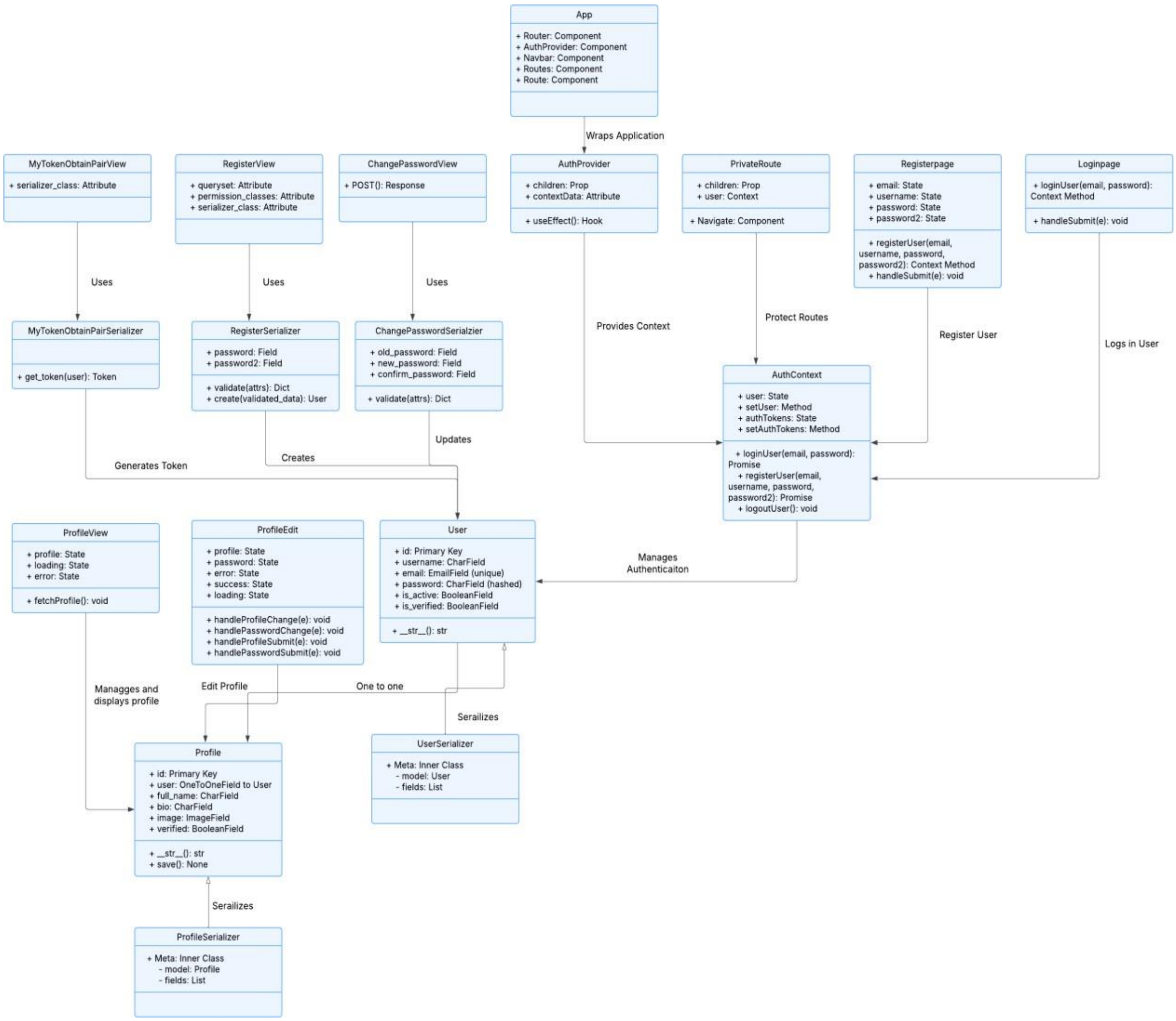


Figure 24 Class diagram of user account management

5.1.15 Sequence Diagram for User Account Management

The following figure shows process flows on how system should interact with users on the key processes such as login, registration, and profile update. It describes the interaction of frontend and backend flow of requests and response. Designing this sequence diagram makes sure all interactions are in following the logical order. It helped in revealing the possibility of delays in data flow and inefficiency in the flow path. However, it acted as reference while implementing workflow. It improved the system's responsiveness and accuracy, through this structured approach.

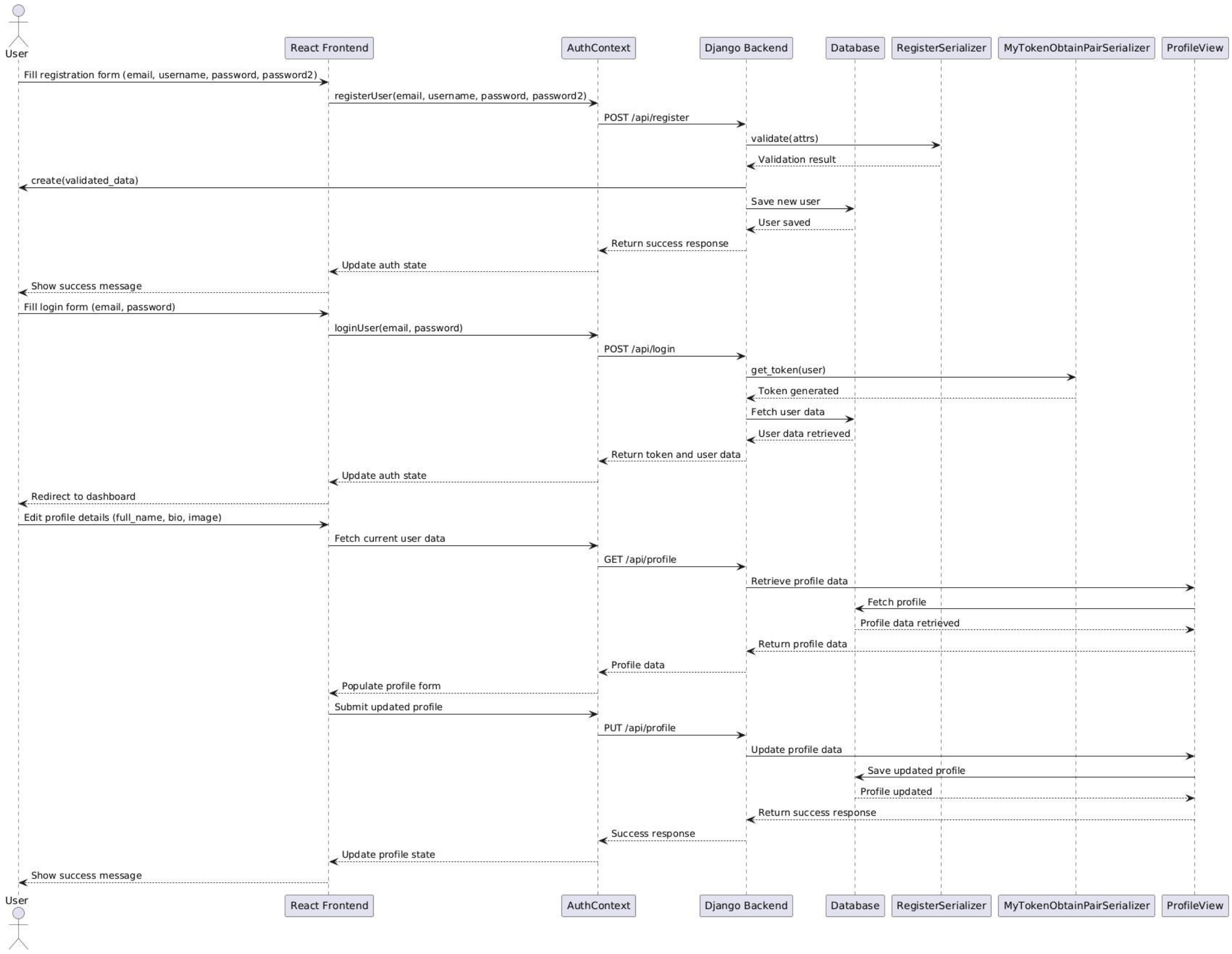


Figure 25 Sequence diagram of user account management

5.1.16 Testing

The testing process of the user account management module is documented by this figure, which also details test cases and their results. It is functional tests to registration, login, and profile updates. The test process covered all the features in the project requirements and tested that all features worked under their intended use. I started by trying one component at a time, systematically testing it and fixing any problems I found as I went along. The validation tests for user input and error handling is also part of this figure. The reliability of the module as a whole grew with the use of this structured testing technique.

User Story ID	Test Case	Priority	Steps	Expected Outcome	Status	Comments
UAM-1	Successful Registration	High	1. Navigate to the registration page. 2. Enter a unique email, username, and password. 3. Submit the form.	The account is created successfully. A confirmation message is displayed. Email verification link sent.	Passed	Account created successfully.
UAM-1	Duplicate Email Check	High	1. Attempt to register with an email that already exists in the database. 2. Submit the registration form.	The system displays an error message: "Email already exists." The account is not created.	Passed	Duplicate email blocked.
UAM-2	Email Verification Link Sent	High	1. Register a new account. 2. Check the inbox of the provided email address.	An email with a verification link is received.	Passed	Verification email received.
UAM-2	Account Verification	High	1. Click the verification link in the email.	The account is marked as verified in the database. A success message is displayed: "Your email has been verified."	Passed	Account verified successfully.
UAM-2	Invalid Verification Link	High	1. Modify the verification link URL (e.g., change the token). 2. Open the modified link.	The system displays an error message: "Invalid or expired link."	Passed	Error message displayed.
UAM-3	Update Personal Details	Medium	1. Log in to the application. 2. Navigate to the profile settings page. 3. Edit personal details. 4. Save.	The updated details are saved in the database. A success message is displayed: "Profile updated successfully."	Passed	Profile updated successfully.
UAM-3	Duplicate Email Check During Update	Medium	1. Attempt to update the email to one that already exists in the database. 2. Save the changes.	The system displays an error message: "Email already exists." The changes are not saved.	Passed	Duplicate email blocked.
UAM-4	Set Risk Tolerance	High	1. Log in to the application. 2. Navigate to the risk tolerance settings page. 3. Select a level. 4. Save.	The selected risk tolerance level is saved in the database. A success message is displayed: "Risk tolerance updated successfully."	Passed	Risk tolerance saved.
UAM-4	View Risk Tolerance	High	1. Navigate to the profile page.	The saved risk tolerance level is displayed correctly.	Passed	Risk tolerance displayed.
UAM-5	Add Financial Goal	High	1. Log in to the application. 2. Navigate to the financial goals page. 3. Add a new goal. 4. Save.	The goal is saved in the database. A success message is displayed: "Goal added successfully."	Passed	Goal added successfully.
UAM-5	Edit Financial Goal	High	1. Navigate to the financial goals page. 2. Select an existing goal and edit its details. 3. Save.	The updated goal is saved in the database. A success message is displayed: "Goal updated successfully."	Passed	Goal updated successfully.
UAM-5	Delete Financial Goal	High	1. Navigate to the financial goals page. 2. Select an existing goal and delete it.	The goal is removed from the database. A success message is displayed: "Goal deleted successfully."	Passed	Goal deleted successfully.
UAM-5	View Financial Goals	High	1. Navigate to the financial goals page.	All saved goals are displayed correctly with their details.	Passed	Goals displayed correctly.

Figure 26 Testing user account management artefact

5.1.17 Example Test Cases

When user tries to make an account with same email.

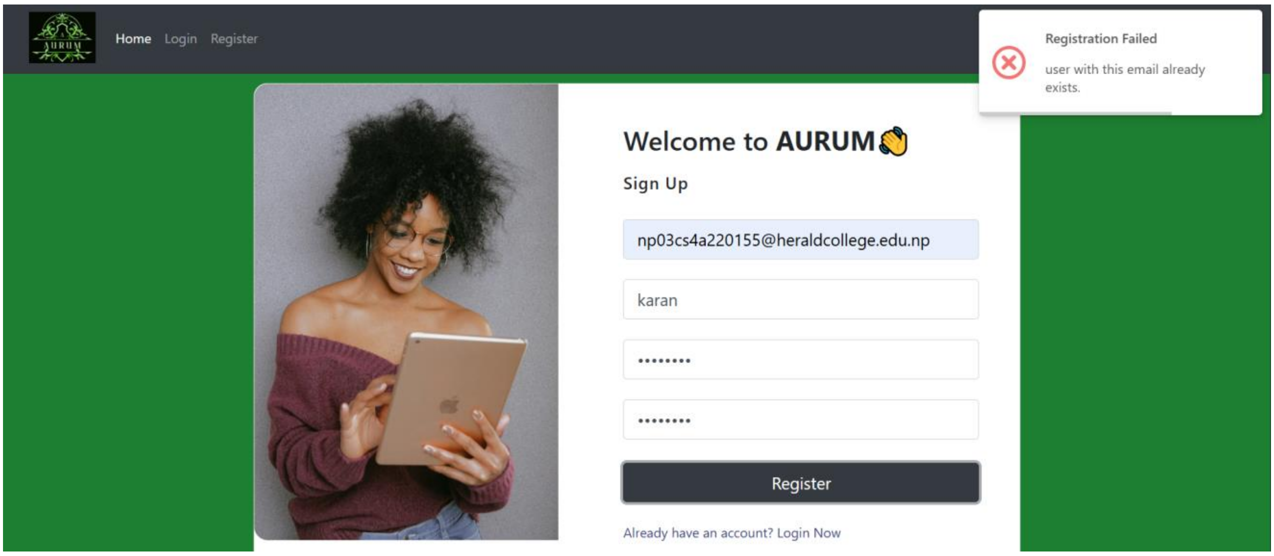


Figure 27 Example test case

Risk threshold added successfully:

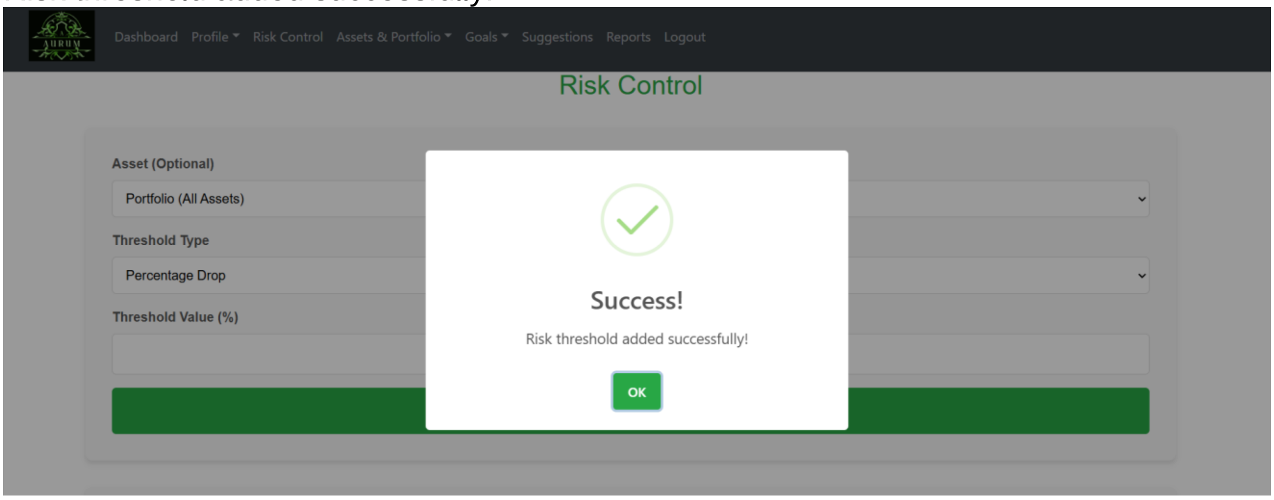


Figure 28 Risk threshold added successfully:

5.2 Artefact – Asset and Portfolio Management

5.2.1 FDD for Asset and Portfolio Management

The FDD model to asset and portfolio management is as is this figure. Its parts dissolve essential functions like adding assets, being a performance monitor and portfolio manager. The structured development approach made prioritization of tasks easy. This is because by defining each feature separately, I had ensured that the module satisfied all the functional requirements. It also was a roadmap that tracked progress for the FDD model. It increased the efficiency of development in general.

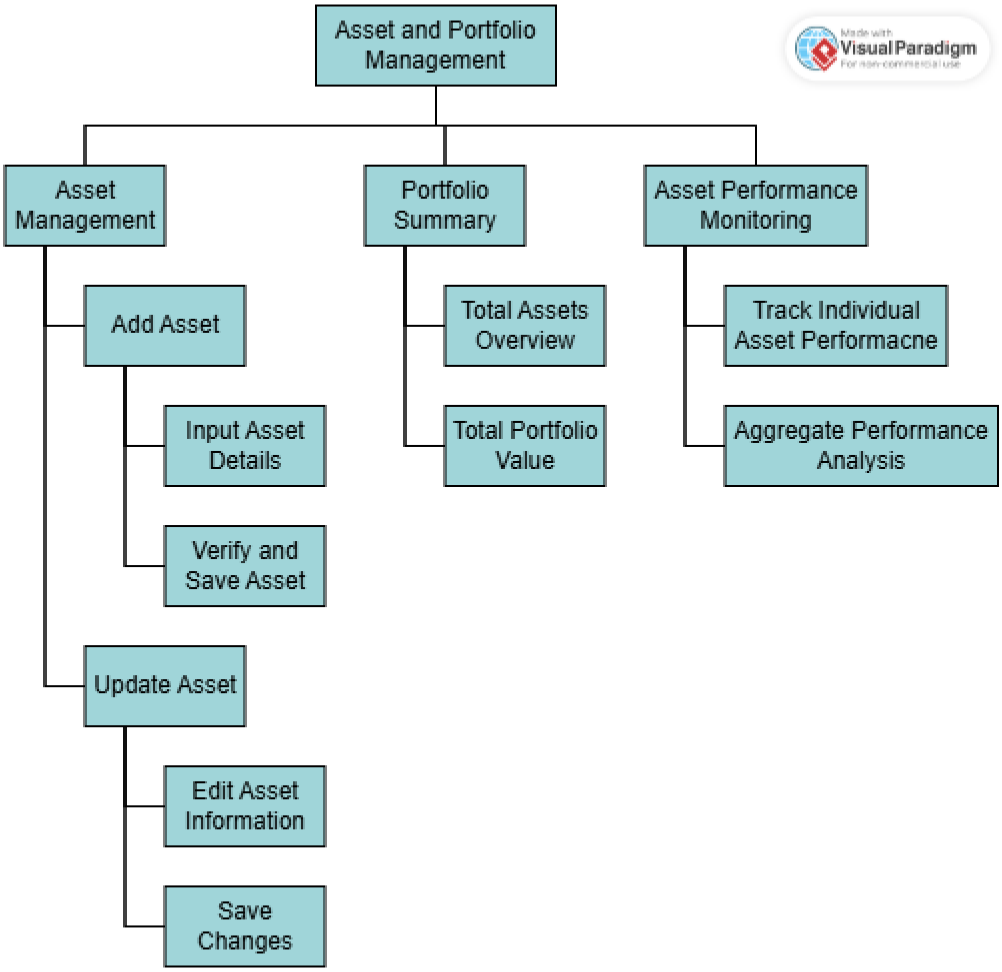


Figure 29 FDD of asset and portfolio management

5.2.2 Product Backlog of Asset and Portfolio Management

The product backlog for the asset and portfolio management module can be found in this figure which outlines what features and functionalities are to be added. This includes the essential tasks of adding assets, editing assets, tracking the performance of assets and generating the portfolio summaries. Starting from the bottom (least important to most important and then on to the least impact to the greatest user experience), each backlog item was prioritized based on importance and its impact on the user experience. Maintaining a well-structured backlog enforced that form of development was on track for achieving the project goals. It continually refined and updated this backlog to be based on testing feedback. Additionally, it was a guide to feature development and sprint planning.

Epic 2: Asset and Portfolio Management

ID	User Story	Priority	Acceptance Criteria
APM-1	As a user, I want to add assets with details so that I can manage my portfolio.	High	Assets are added with name, type, and value and saved in the database.
APM-2	As a user, I want to view a summary of my portfolio so that I can see my total assets and value	High	Portfolio summary displays total asset count and total portfolio value.
APM-3	As a user, I want to update asset information so that I can keep it accurate.	Medium	Changes to asset details are reflected in the portfolio.
APM-4	As a user, I want to monitor the performance of my assets so that I can track growth over time.	High	Graphs display asset performance trends.
APM-5	As a user, I want to see aggregate performance analysis so that I can evaluate my portfolio easily.	Medium	Portfolio performance graphs and summaries are available.

Figure 30 Backlog of Asset and Portfolio Management

5.2.3 Sprint Planning for Asset and Portfolio Management

This is a figure that shows the sprint backlog for the asset and portfolio management module and for what tasks were selected to be developed in the particular sprint cycle. The information it includes is also user stories and technical tasks about tasking assets, monitoring them, and visualizing financial data. I did this by establishing parameters for sprint focused objectives. The time it was thought that each task would take, gave me a good idea of progress. As the tasks were completed and new refinements were introduced, the backlog was updated again and again. But it also imposed a structure upon the development of the module, such that its development was smooth and incremental.

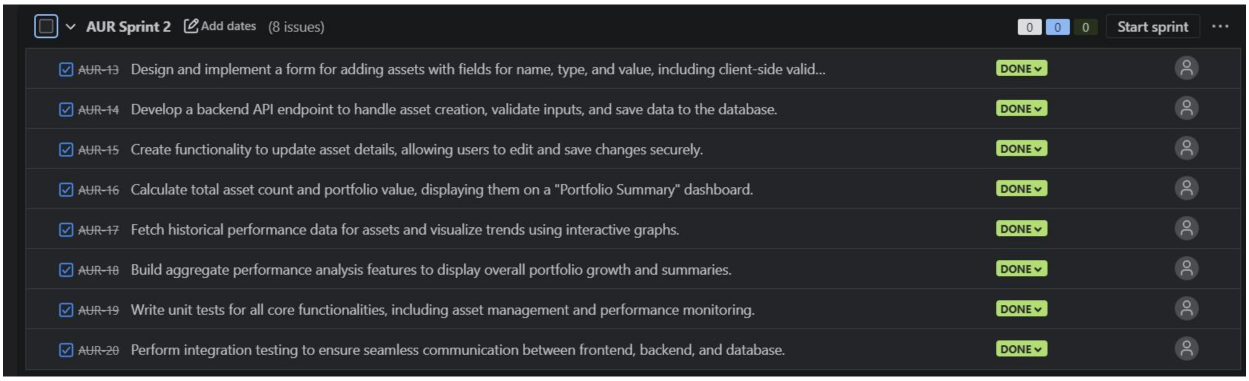


Figure 31 Sprint planning for Asset and Portfolio Management

5.2.4 Activity Diagram for Asset and Portfolio Management

The activity diagram of the asset and portfolio management module depicts the strokes performed by the users from starting to ending in the asset and portfolio management module. This includes adding, editing, analyzing assets etc. It helped in visualizing how users use the system. I mapped out how activities transitioned from stage to stage so it would be a smooth, easy to follow process. This also helped in identifying potential bottlenecks in the system as well. It was essential for designing efficient user interaction.

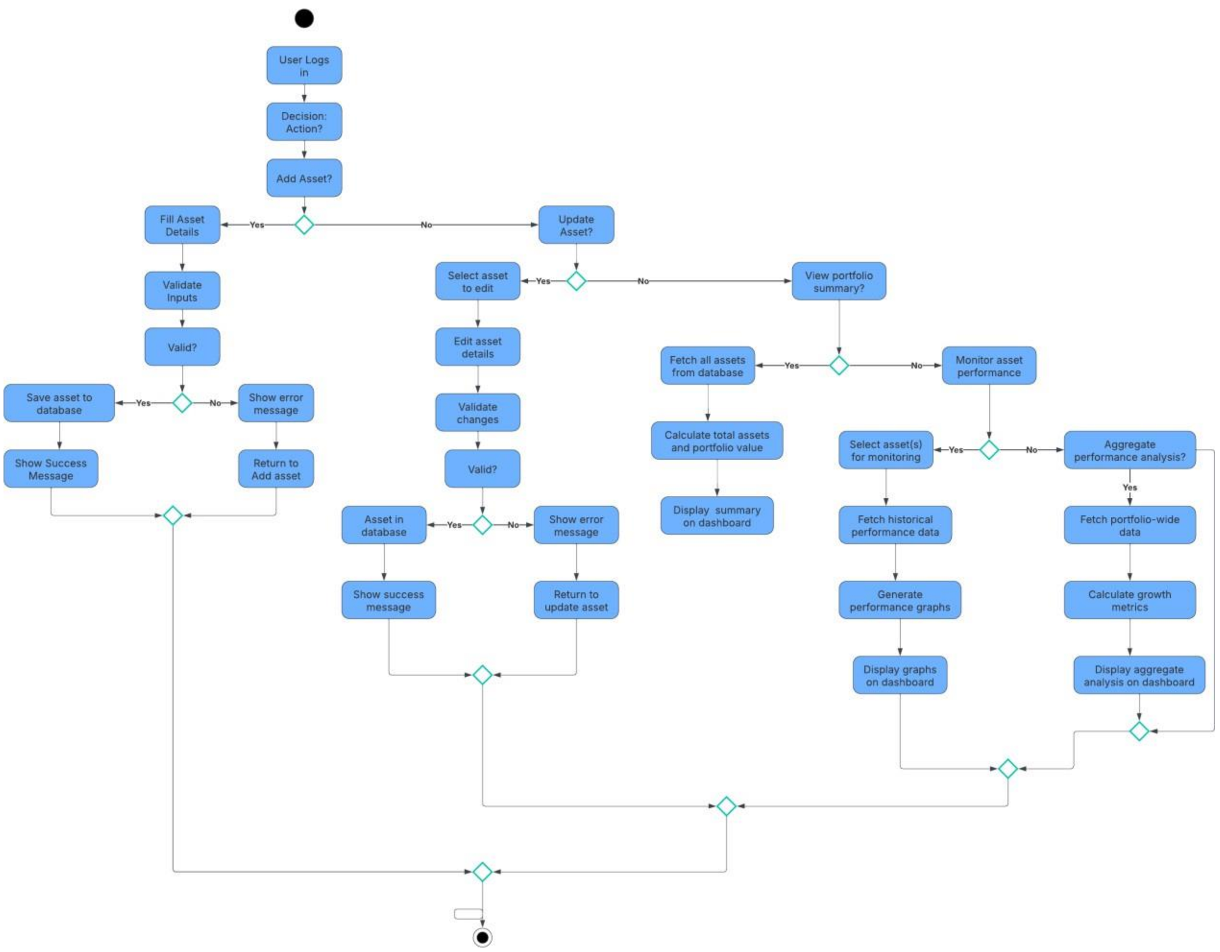


Figure 32 Activity diagram for asset and portfolio management

5.2.5 Use Case Diagram for Asset and Portfolio Management

This is where the use case diagram of the asset and portfolio management module shows the definition regarding different user roles and the relationship of this system with them. It includes actions like adding the assets, updating the portfolios and viewing the asset performance. The diagram was sure to complete all the functional requirements. Clearly defining user interaction meant an improvement in the usability of the module. Defining Role based access control was helped by a structure to this. Implementation of various user actions was based on the use case diagram.

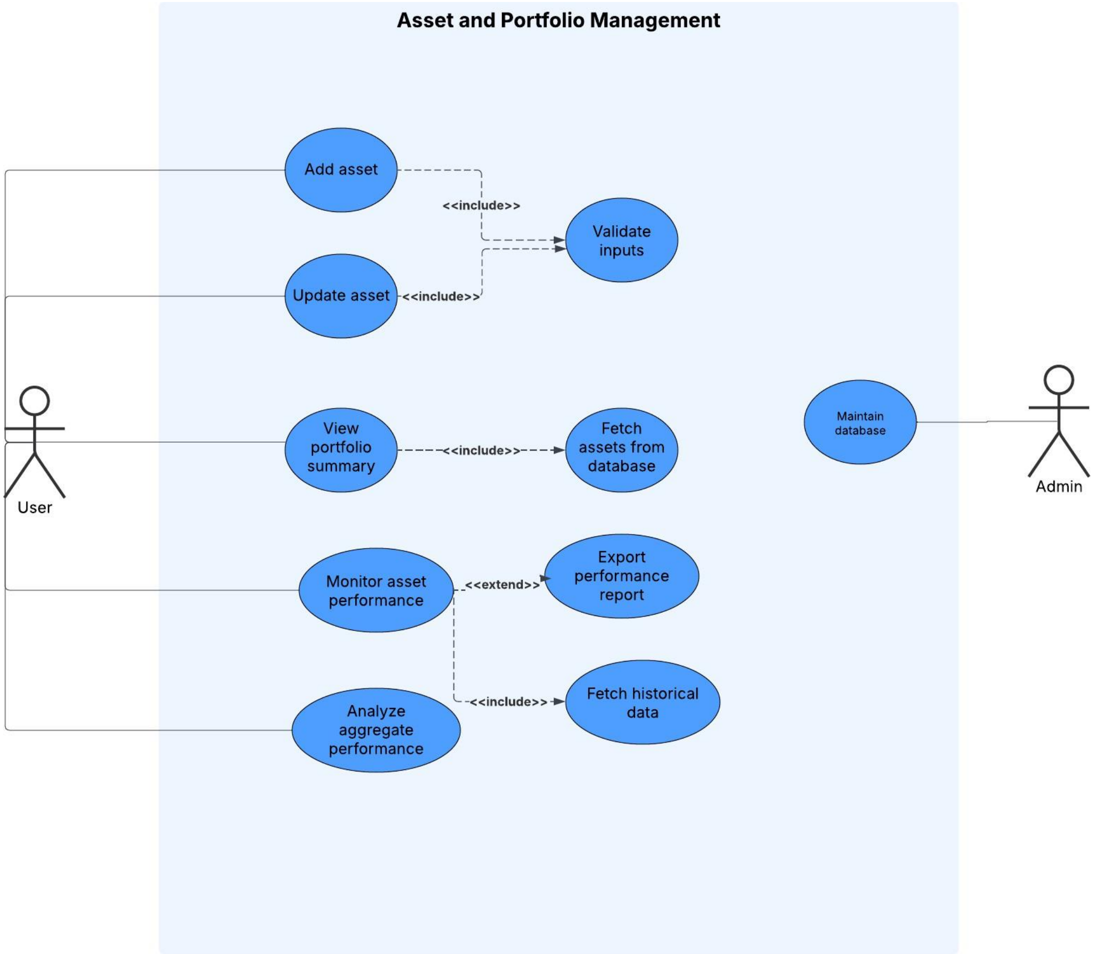


Figure 33 Use case diagram for asset and portfolio management

5.2.6 Wireframe - Asset and Portfolio Management

This figure showcases the wireframe for the asset and portfolio management interface. It outlines an architecture for components such as asset entry forms, performance graphs, summary tables. The wireframe helped when it came to designing an intuitive layout for the development. I focused on usability such that users can essentially manage their asset well. It acted as a visual representation of the process of building the frontend. The structured design greatly improved the overall experience for the user.

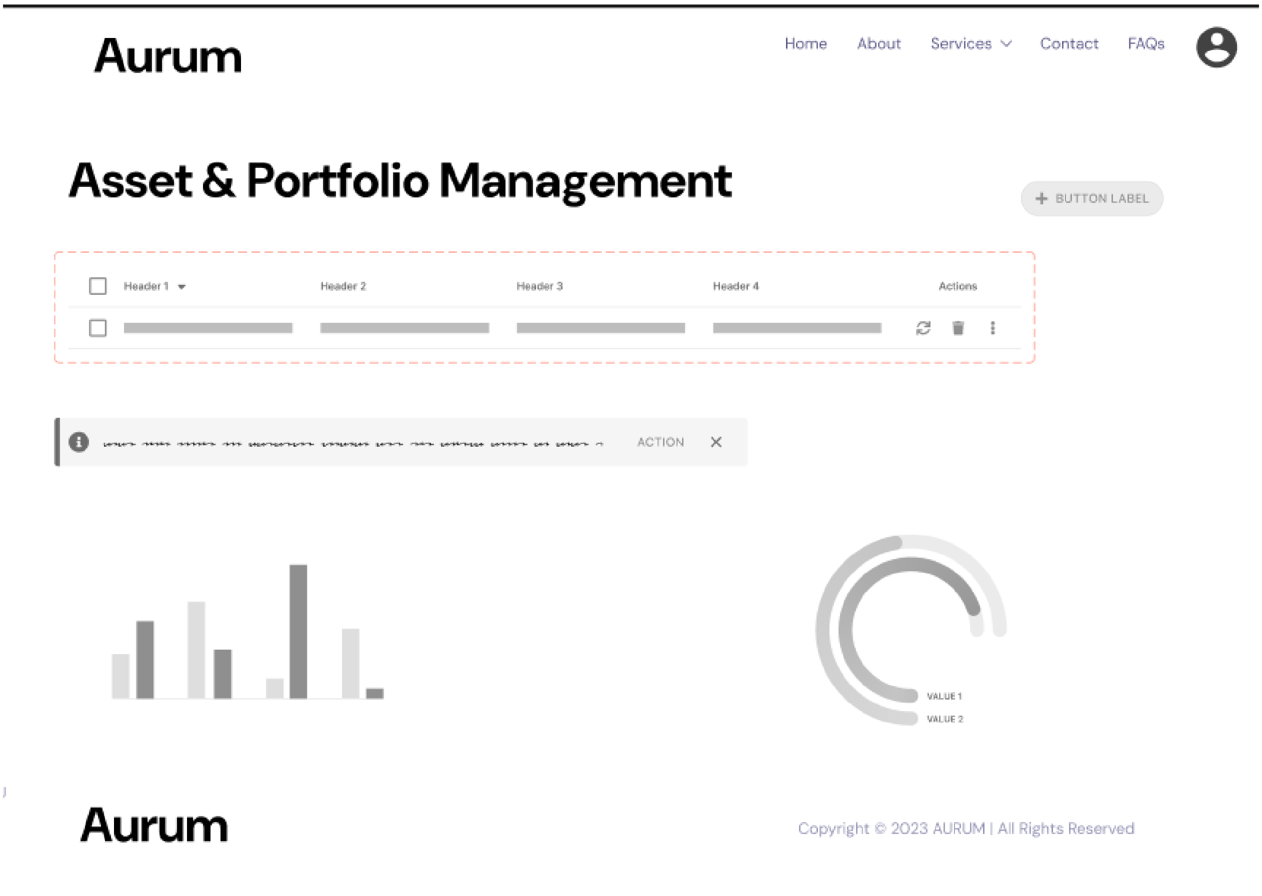


Figure 34 Wireframe - asset and portfolio management

5.2.7 Wireframe - Edit Assets

The Wireframe for the edit assets page where users can edit the existing assets' details have been given in this figure. It has a field of asset name, value and category to update the information. This makes the editing process easier for the users. During implementation this wireframe was used a reference to maintain the same design. By designing this interface ahead of time, I set the user up for an easy life. Validation of the accuracy, completeness of data is also considered in the design.

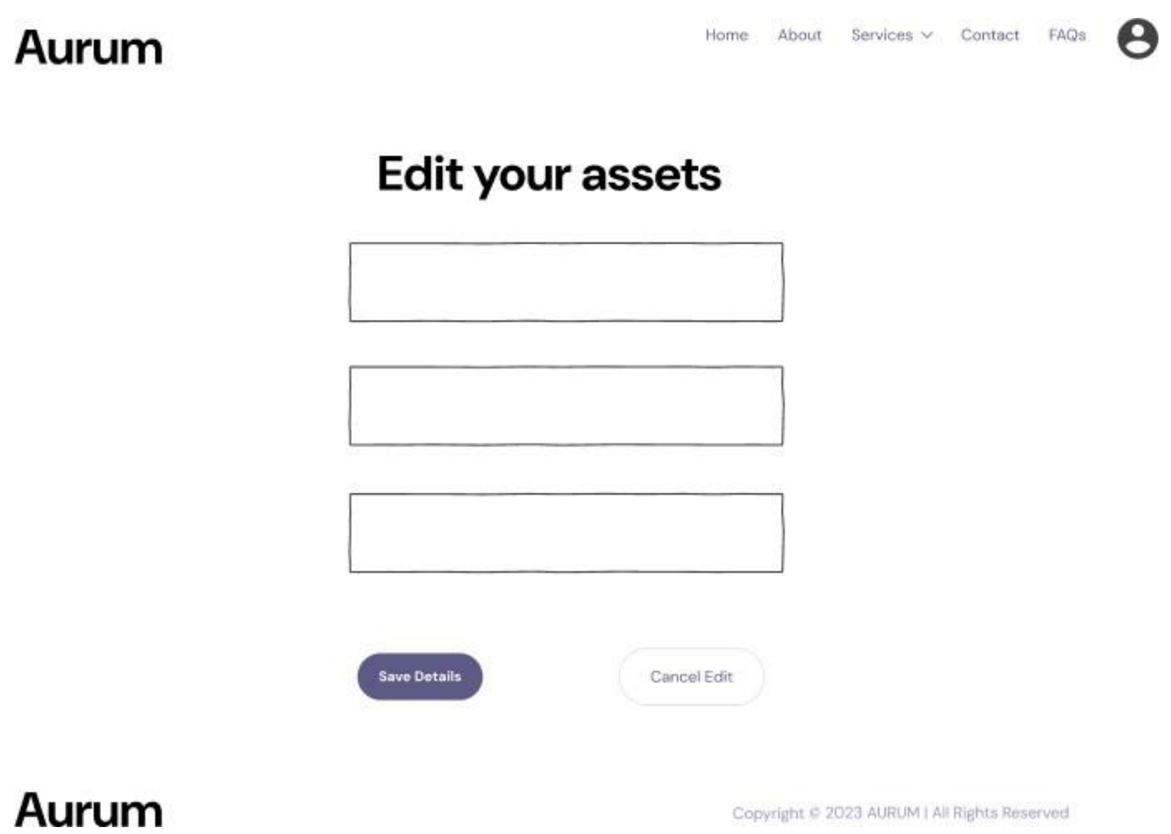


Figure 35 Wireframe - edit assets

5.2.8 ERD of Asset and Portfolio Management

Asset and Portfolio Management module ERD is shown in this figure. This defines relationship of assets, users and performance records. The ERD created a well-structured database schema to easily store the data centrally and progressively. I designed entity relationships to optimize database queries and storage. Moreover, this diagram also played a role in maintaining the data integrity and consistency. It helped the module to implement the backend structure of the module at its core.

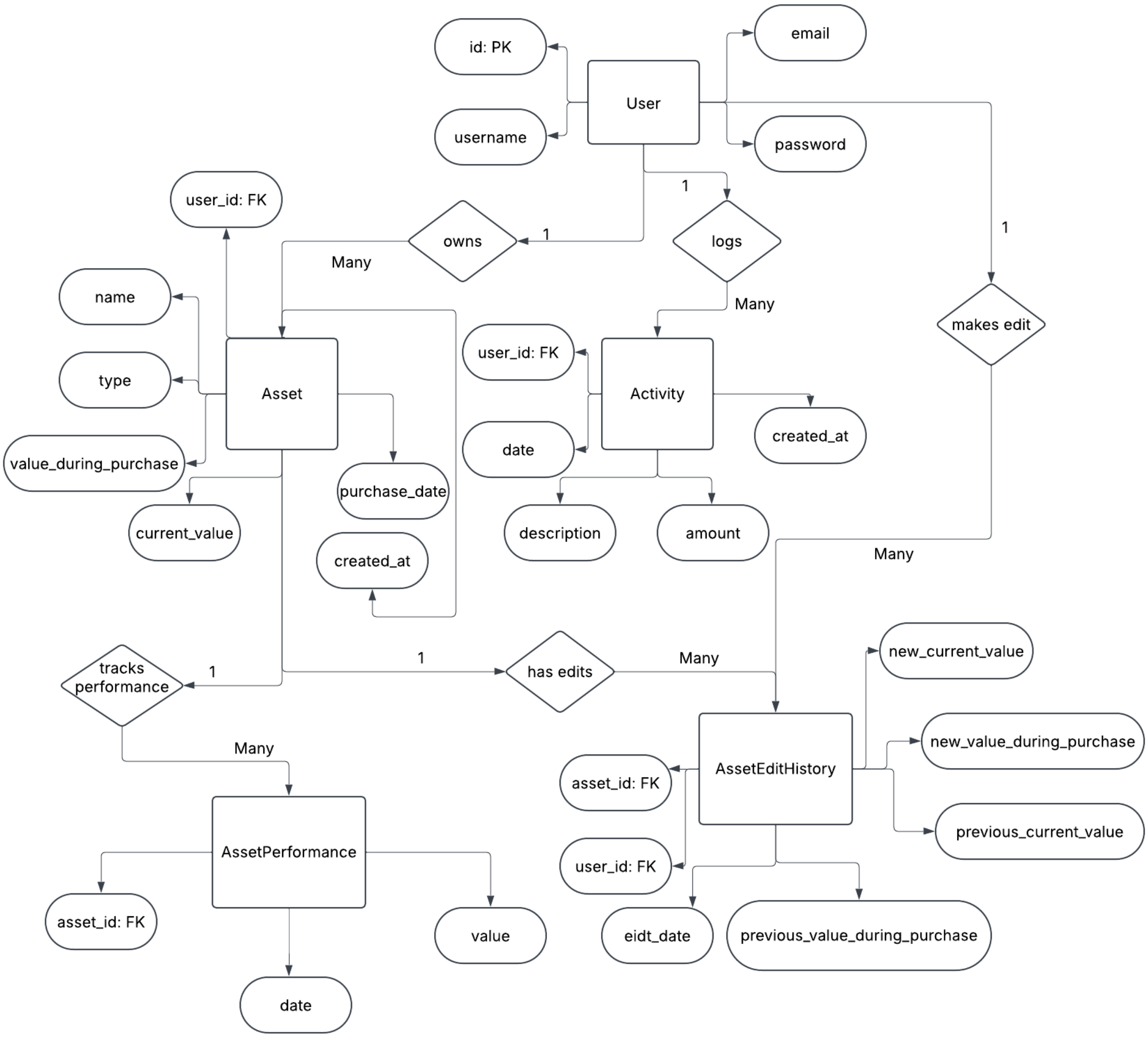


Figure 36 ERD of asset and portfolio management

5.2.9 Class Diagram of Asset and Portfolio Management

This figure shows the class diagram of the asset and portfolio management module showing overall structure of various object oriented components. Classes for asset management, performance tracking and user interactions are included. I defined class relationships to define a module and maintainable code structure. One of the benefits of the diagram was that it aided in implementing object oriented principles efficiently. Using such structured approach made code reusability and scalability easy. It was an essential reference when developing.

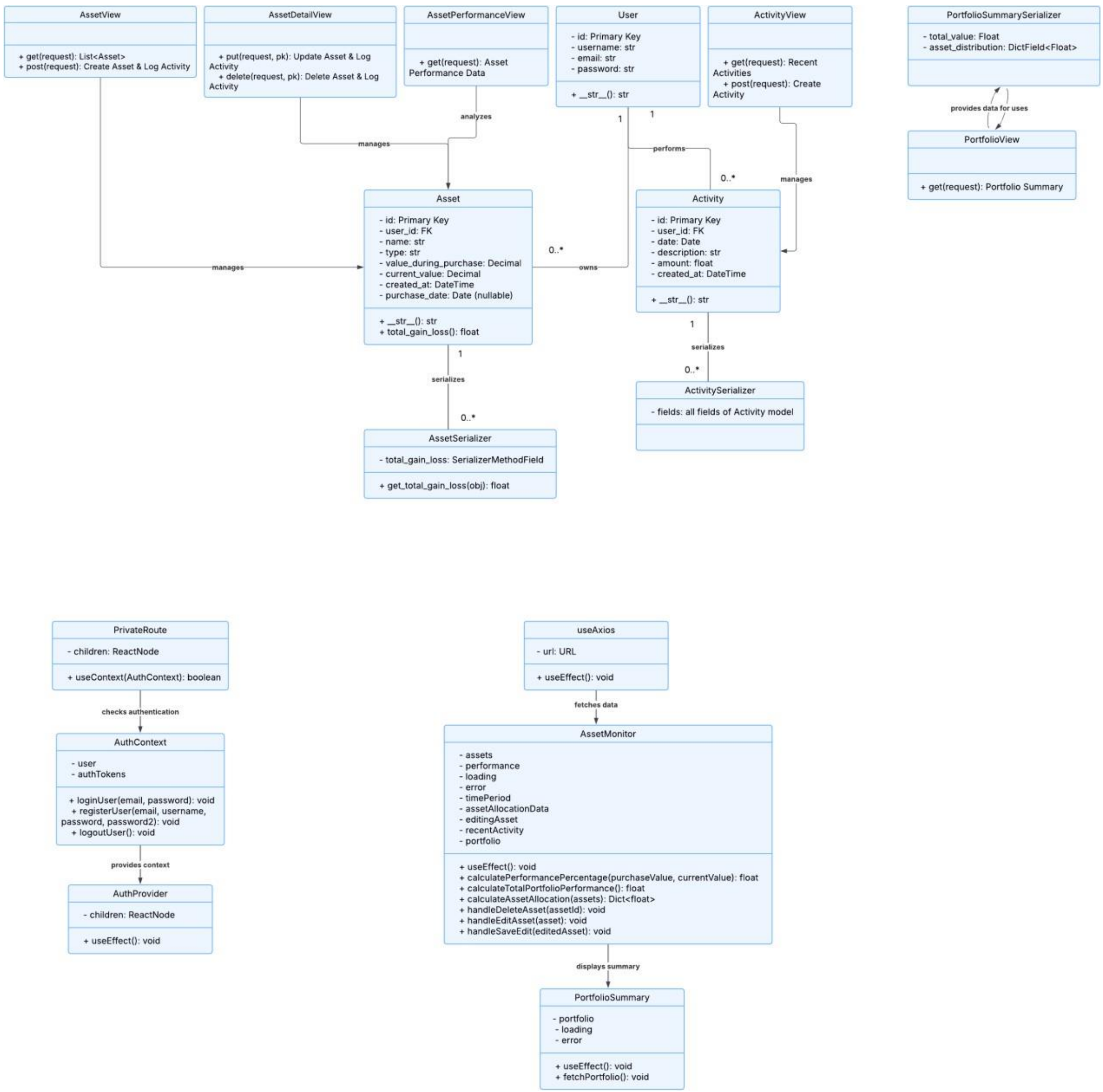


Figure 37 Class diagram of asset and portfolio management

5.2.10 Sequence Diagram of Asset and Portfolio Management

The sequence diagram for asset and portfolio management module is shown in this figure that represents the flow of interactions between the users and the system. In this section it discusses about how to add, edit and analyze assets, it gives step by step. I gave myself the logical and efficient workflow through structuring the sequence of operations. This diagram helped us identify what we might be doing inefficient or his response the response times. It also helped in building smooth user interactions with this structured approach. Backend implementation was key by taking reference from the sequence diagram.

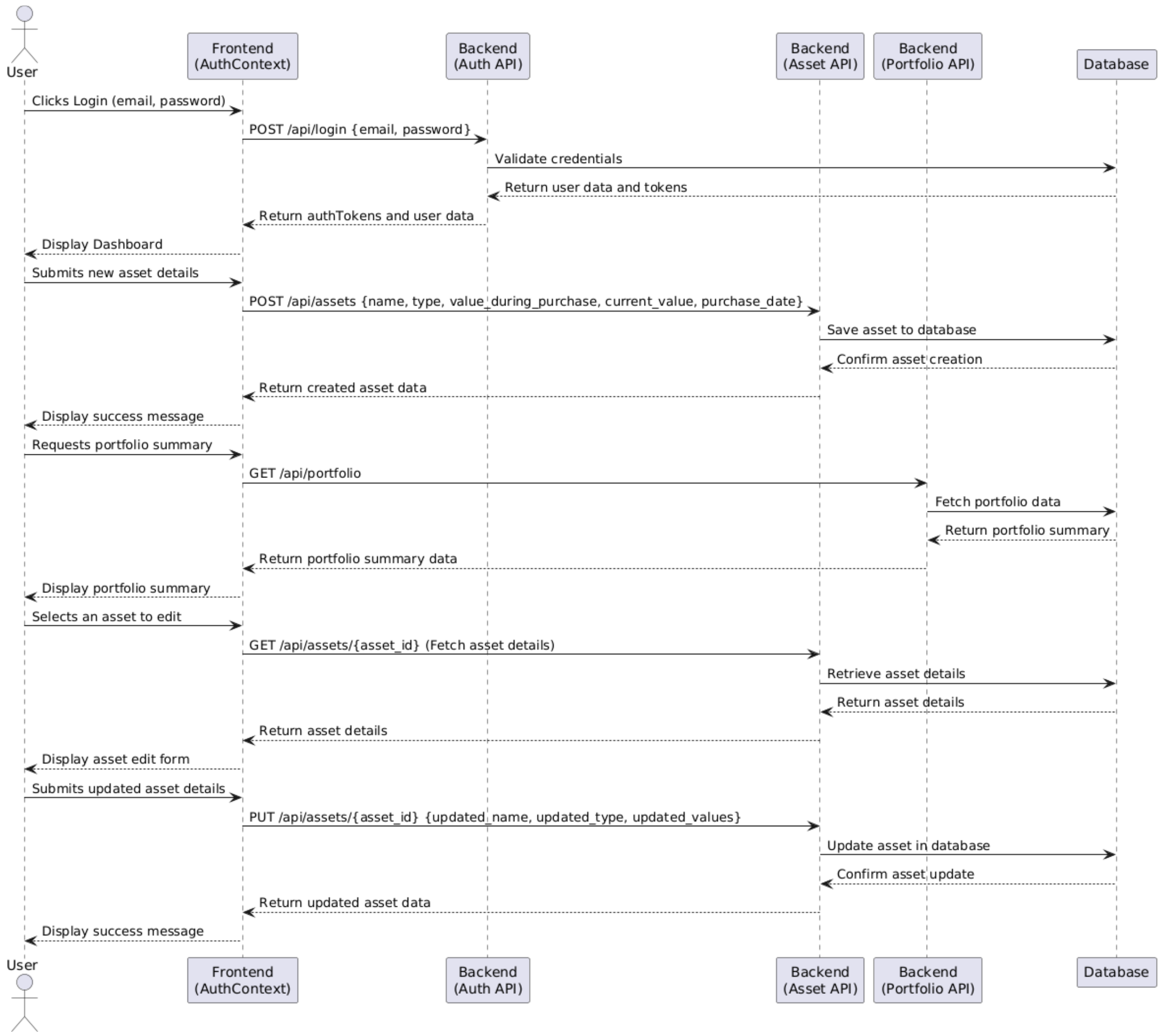


Figure 38 Sequence diagram for asset and portfolio management

5.2.11 Testing

This figure shows a testing process for the asset and portfolio management module and test cases results. To verify that users could successfully add, edit, and monitor their assets functional tests were carried out. To ensure that if input was incorrect or incomplete, then the system did not cause an error, validation tests were conducted. There was also work done on the performance testing to see how the module could work with large volume of asset data without lag. Iterative debugging was applied to any identified issues and they were logged and fixed. I made sure that the module would work by systematically running every feature so that it would work as expected and according to all the functional requirements. It also heightened the overall stability and reliability of the system.

Testing Document for Epic 2: Asset and Portfolio Management

User Story ID	Test Case	Priority	Steps	Expected Outcome	Status	Comments
APM-1	Add Asset	High	1. Log in to the application. 2. Navigate to the asset management page. 3. Input asset details (name, type, purchase value, current value). 4. Save the asset.	The asset is saved in the database. A success message is displayed: "Asset added successfully."	Passed	Asset added successfully.
APM-2	View Portfolio Summary	High	1. Log in to the application. 2. Navigate to the portfolio summary page.	The portfolio summary displays total asset count and total portfolio value.	Passed	Portfolio summary displayed.
APM-3	Update Asset Information	Medium	1. Log in to the application. 2. Navigate to the asset management page. 3. Select an existing asset and edit its details (e.g., name, type, value). 4. Save changes.	The updated asset details are reflected in the database. A success message is displayed: "Asset updated successfully."	Passed	Asset updated successfully.
APM-4	Monitor Asset Performance	High	1. Log in to the application. 2. Navigate to the asset performance monitoring page.	Graphs display individual asset performance trends (e.g., gain/loss over time).	Passed	Performance graphs displayed.
APM-5	View Aggregate Portfolio Analysis	Medium	1. Log in to the application. 2. Navigate to the portfolio analysis page.	Portfolio performance graphs and summaries (e.g., total gain/loss, overall growth) are displayed.	Passed	Aggregate analysis displayed.

Figure 39 Testing for asset and portfolio management

5.2.12 Example Test Cases:

Users all assets are displayed along with the portfolio summary.

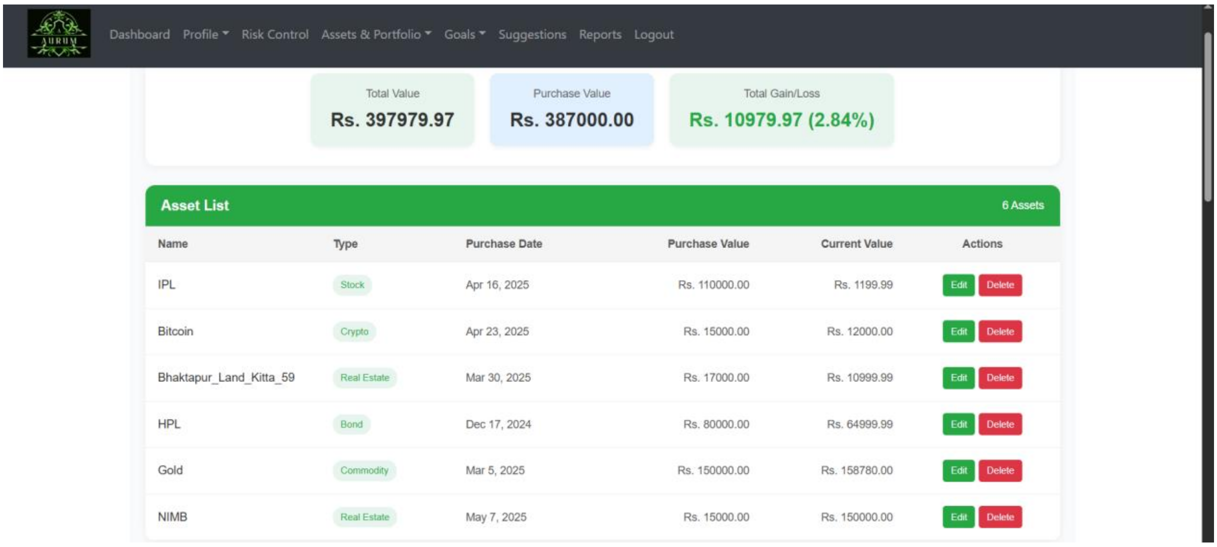


Figure 40 Example test case

CRUD operation is successful on the asset.

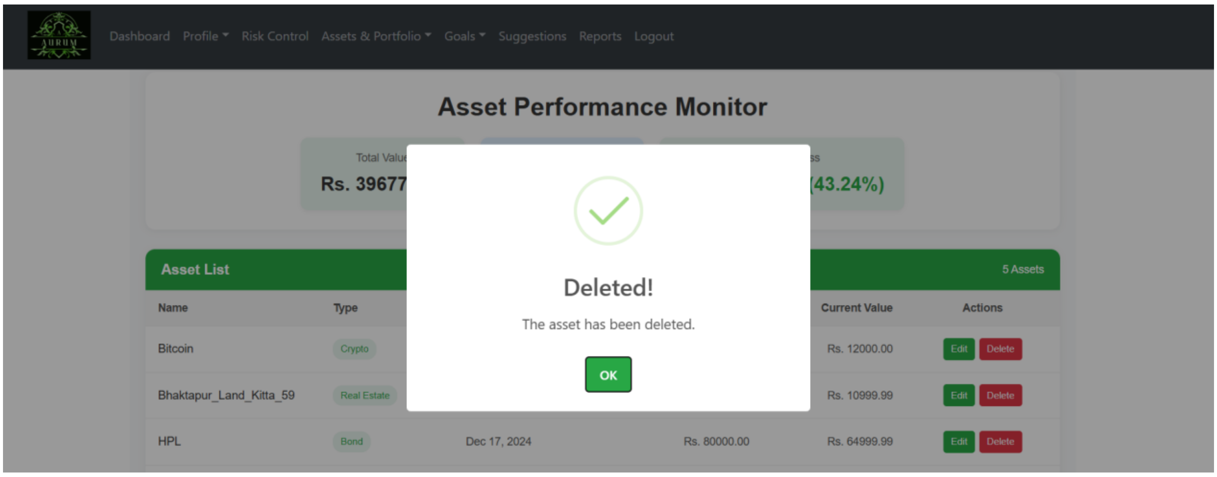


Figure 41 Example test case

5.3 Artefact – Financial Goal Management

5.3.1 FDD of Financial Goal Management

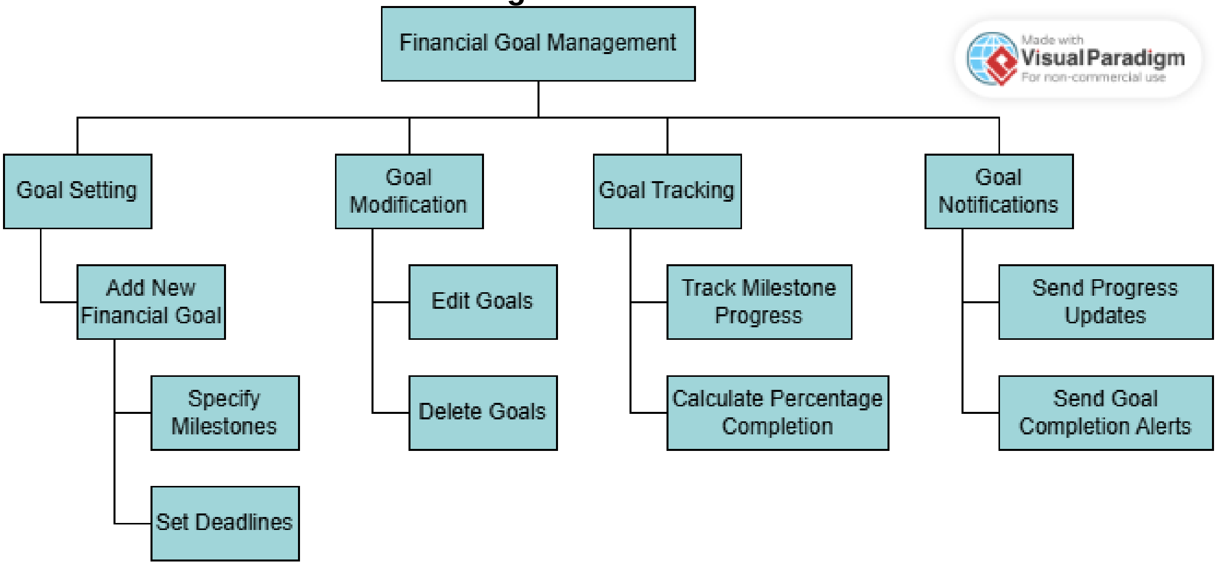


Figure 42 FDD of Financial Goal Management

5.3.2 Product Backlog for Financial Goal Management

This epic was made for the functionality of setting goals and its CRUD operation along with setting milestones and give the notification to the users whereby it is beneficiary for the users to keep progress of their financial goals.

Epic 3: Financial Goal Management

ID	User Story	Priority	Acceptance Criteria
FGM-1	As a user, I want to set financial goals with deadlines so that I can work toward specific targets.	High	Goals are saved with milestones and deadlines.
FGM-2	As a user, I want to modify or delete goals so that I can keep them relevant.	Medium	Goals can be updated or removed from the system.
FGM-3	As a user, I want to track the progress of my financial goals so that I know how close I am to them.	High	Progress bars or milestones reflect goal progress in percentages.
FGM-4	As a user, I want to receive notifications about my goals so that I stay informed of updates.	Medium	Notifications for progress and goal completion are sent via email and dashboard.

Figure 43 Product Backlog of Financial Goal Management

5.3.3 Sprint Planning for Financial Goal Management

With this system, users can easily keep track of their financial plans. It includes four primary features called functionalities.

Setting Objectives: Through this feature, users can define their money goals with specific dates for achieving them.

It includes tools to alter or get rid of your current goals.

Monitoring Progress: Provides visual markers and figures that show the status of tasks.

Alerts are sent to show how far a person has come in reaching their goal.

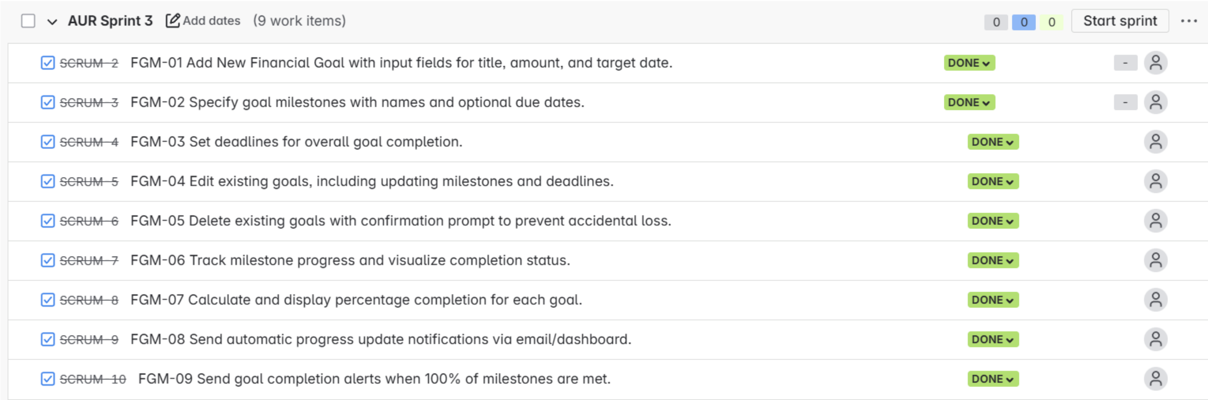


Figure 44 Sprint Planning for Financial Goal Management

5.3.4 Activity Diagram for Financial Goal Management

It displays the way in which users access the system to handle their financial goals. A new user starts by signing in, navigating the app and setting a goal that includes description, important milestones and the deadline. The program verifies if the goal has been saved properly. They can alter goals by selecting to edit or delete them, but for deletion, a confirmation will be required. Every milestone reached is tracked and the system determines the overall percentage. Once a goal completes, you get a notification. During the process, users get notifications and updates on how much progress they are making.

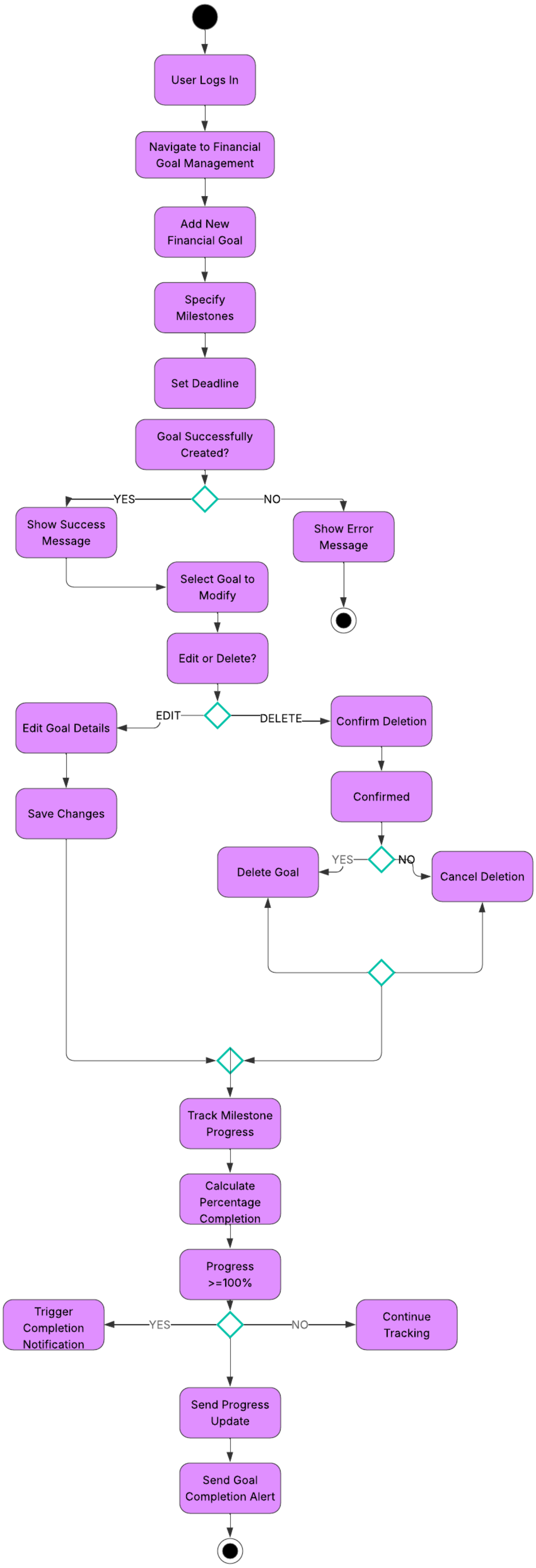

User Logs In
Navigate to Financial Goal Management
Add New Financial Goal
Specify Milestones
Set Deadline
Goal Successfully Created?
YES
NO
Show Success Message
Show Error Message
Select Goal to Modify
Edit or Delete?
EDIT
DELETE
Edit Goal Details
Confirm Deletion
Save Changes
Confirmed
YES
NO
Delete Goal
Cancel Deletion
Track Milestone Progress
Calculate Percentage Completion
Progress >=100%
YES
NO
Trigger Completion Notification
Continue Tracking
Send Progress Update
Send Goal Completion Alert

Figure 45 Activity Diagram for Financial Goal Management

5.3.5 Use Case Diagram for Financial Goal Management

It illustrates how a user can use the system to manage their financial goals. You can choose to enrich the main function, Manage Financial Goals, by adding, editing, deleting, viewing your progress and getting notifications. When a user creates a goal, the system reminds them to determine important milestones and set a deadline. Additionally, monitoring the work requires figuring out the current completion rate, whereas providing notifications covers both informing of successes and of finishing the task. With include and extend relations between processes the system can distinguish through core and extra processes.

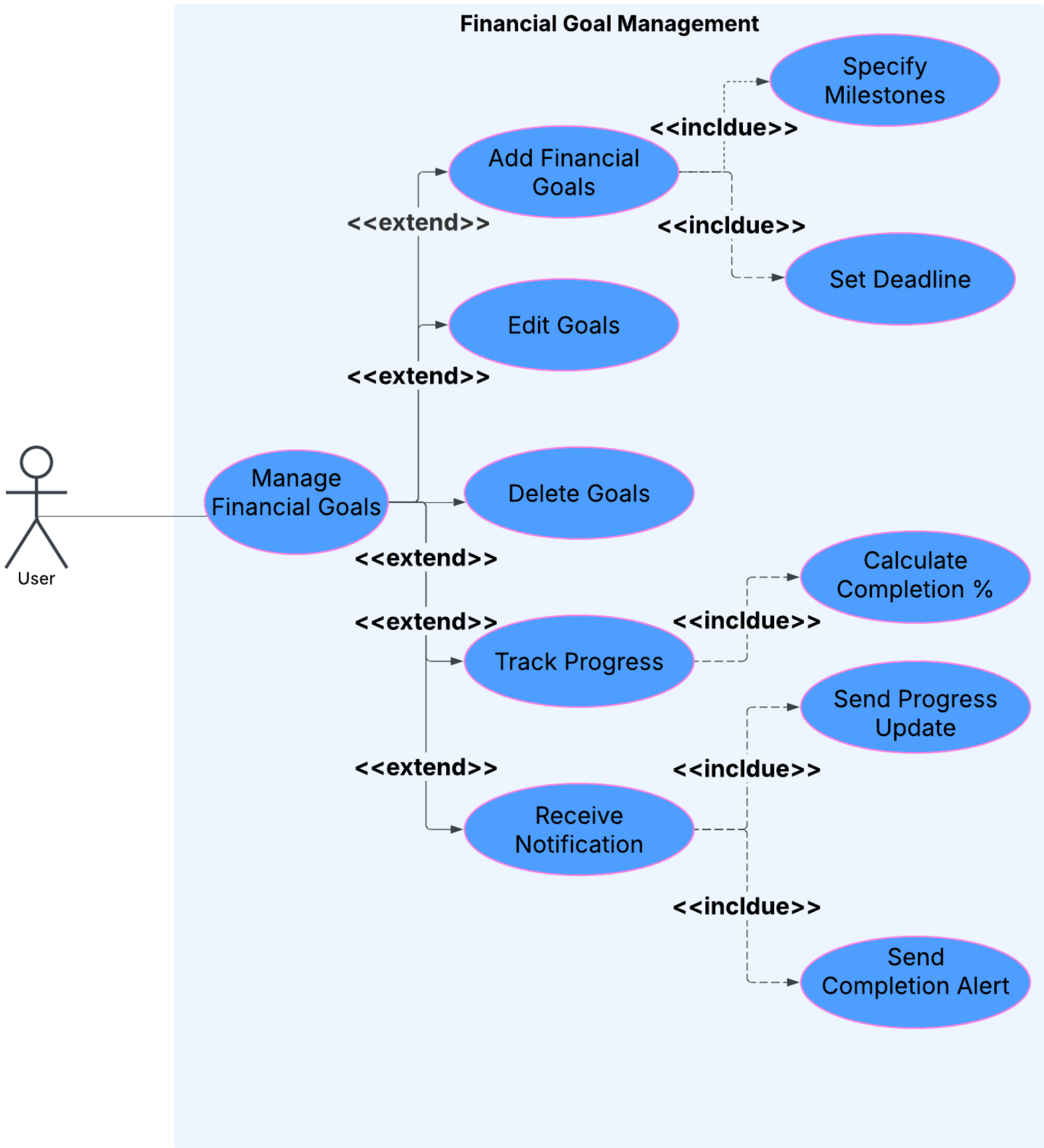


Figure 46 Use Case Diagram for Financial Goal Management

5.3.6 Wireframe

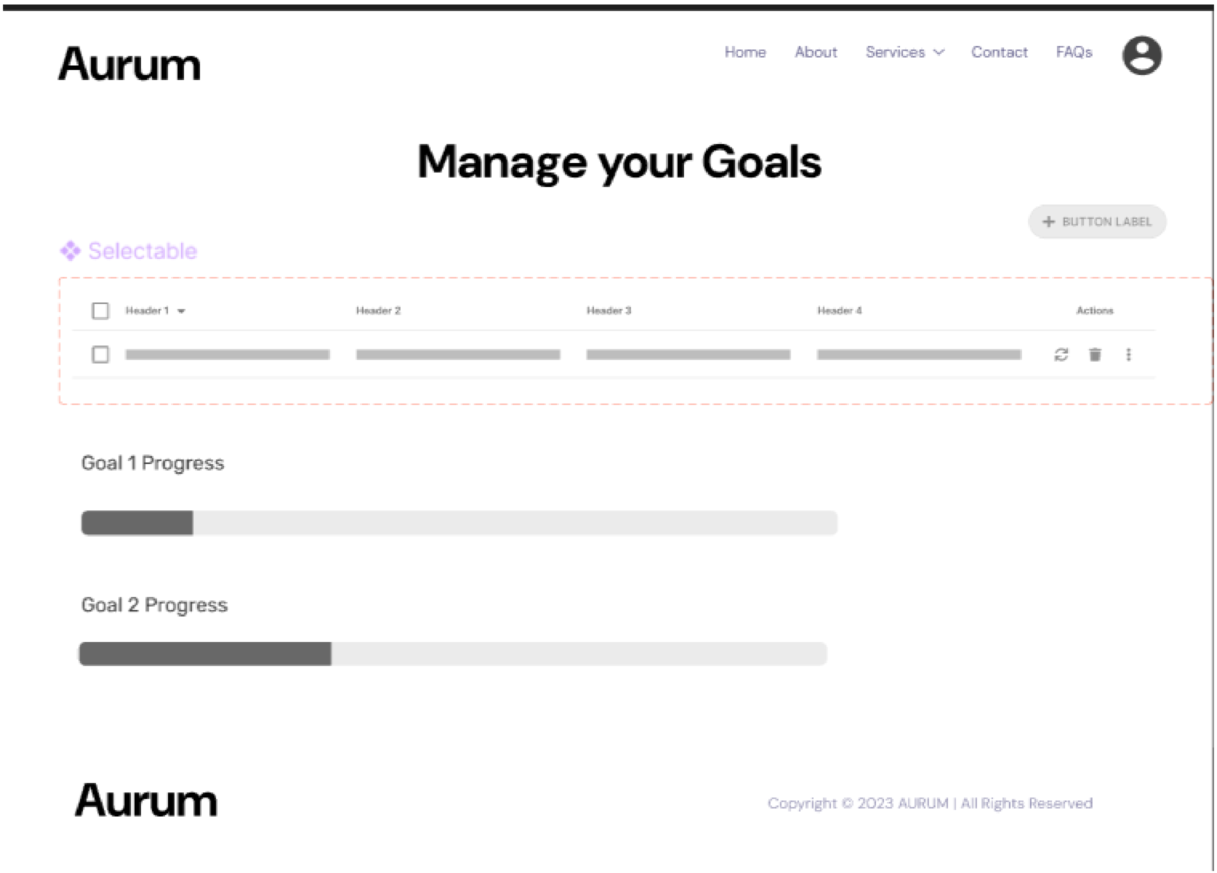


Figure 47 Wireframe – Goals

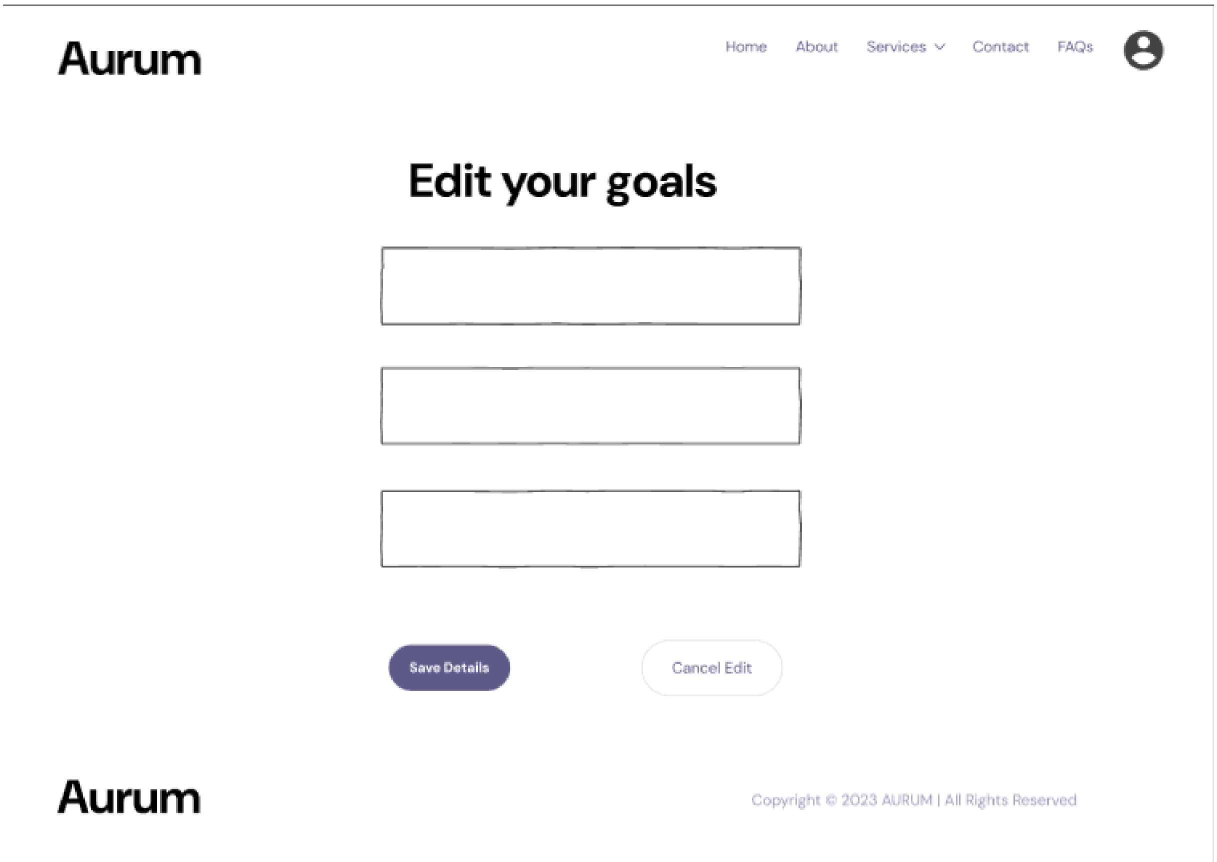


Figure 48 Wireframe 2 - Goals

5.3.7 ERD for Financial Goal Management

The ERD for the Financial Goal Management system shows the important data entities along with their connections. It consists of four main elements called User, Goal, Milestone and Notification. A user can set up as many financial goals as needed and each of these goals can be measured by various milestones. Any changes to the user's goals or final achievements trigger notifications which are attached to respective goals. With this ERD, it can be clearly seen how the data is organized and linked which aids in reaching the allocated and created goals and communicating with users.

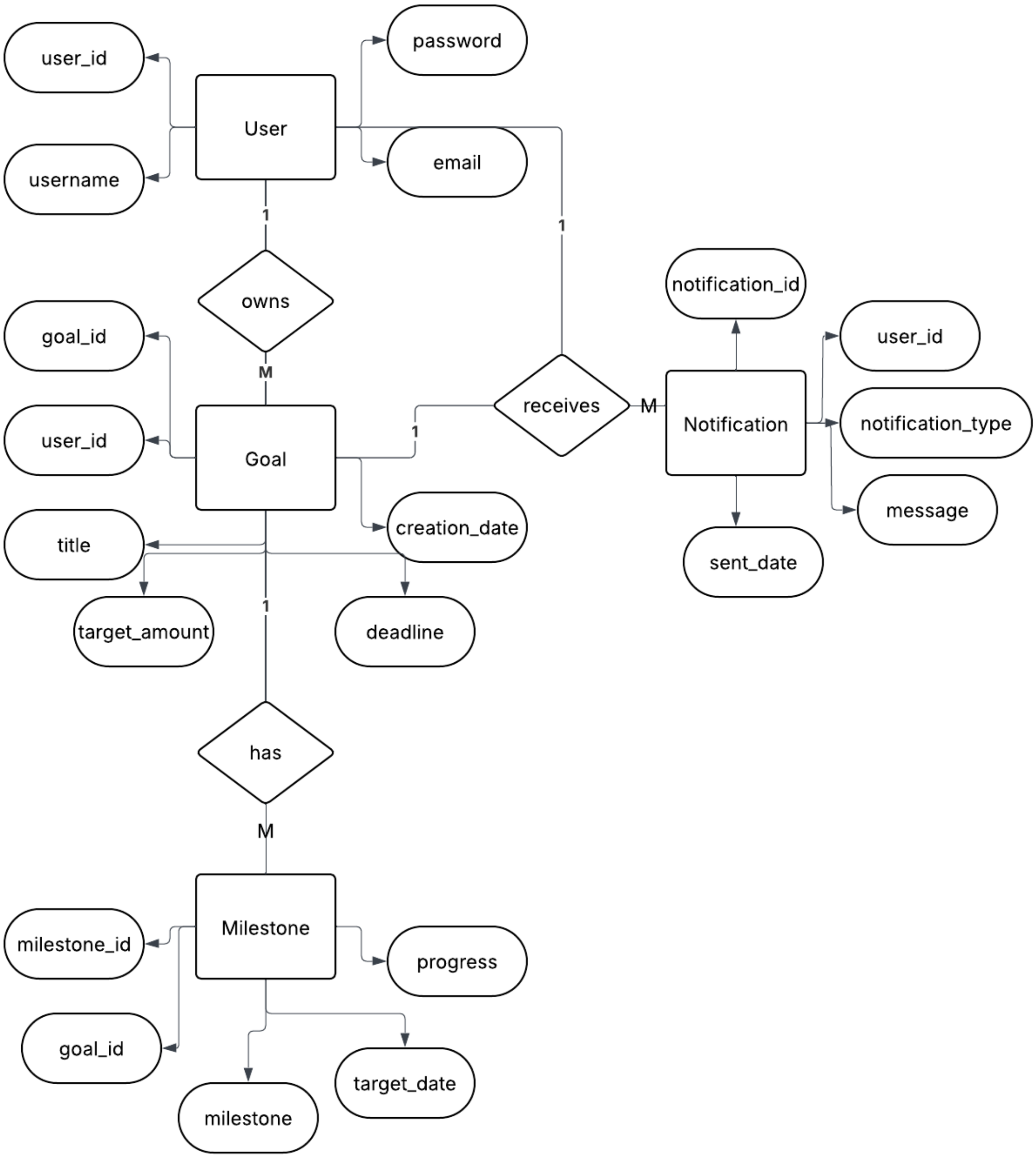


Figure 49 ERD for Financial Goal Management

5.3.8 Class Diagram for Financial Goal Management

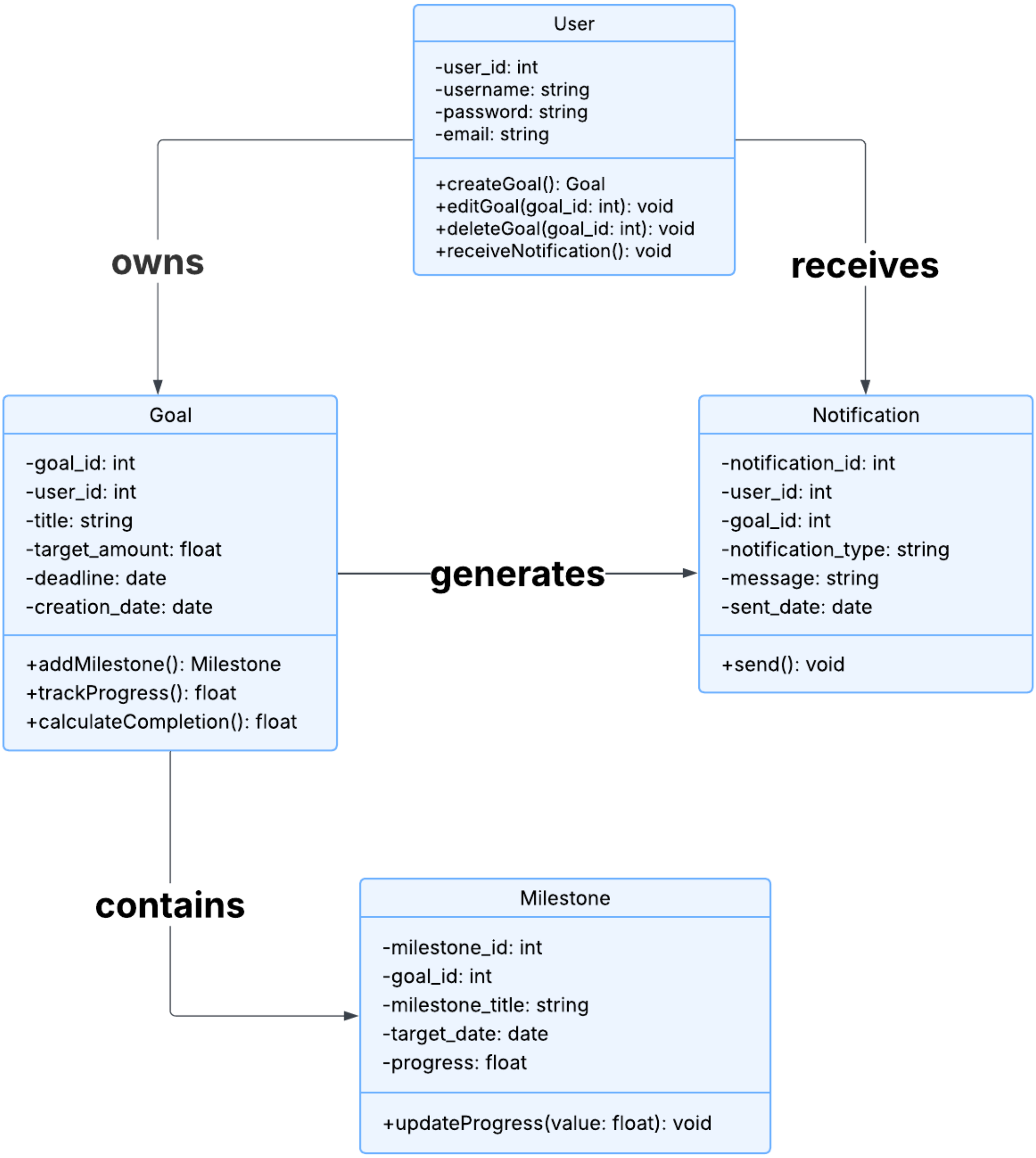


Figure 50 Class Diagram for Financial Goal Management

In the Financial Goal Management system, the class diagram entails the structure and actions of the important pieces: User, Goal, Milestone and Notification. A user has attributes like user ID, a username, password and email and has the ability to create, edit or delete goals. A Goal consists of details like the title, desired amount, target deadline and is connected to a user. You can add important goals, track the project's progress and see the percentage that is already finished. For every Milestone, there is a particular objective, as well as its specific status and date. This class manages sending alerts to users, connecting them to the user directly or to the specific goal they relate to. A one-to-many structure is utilized to indicate that users manage their goals, goals have several milestones and notifications are linked to users as well as goals. It clearly sets out how both the data and the management of goals are presented.

5.3.9 Sequence Diagram for Financial Goal Management

It outlines the process that takes a financial goal from the beginning to its conclusion. After a new goal is entered on the screen by the user, the GoalController handles it and sends it to the GoalService for storage in the database. After that, the user creates milestones and establishes a due date and these are recorded and arranged by each service. They may then review their progress by looking at the milestones they have achieved. If the progress is completed, the system sends an email to the user using the NotificationService. It illustrates the step-by-step interaction the user and all backend parts take.

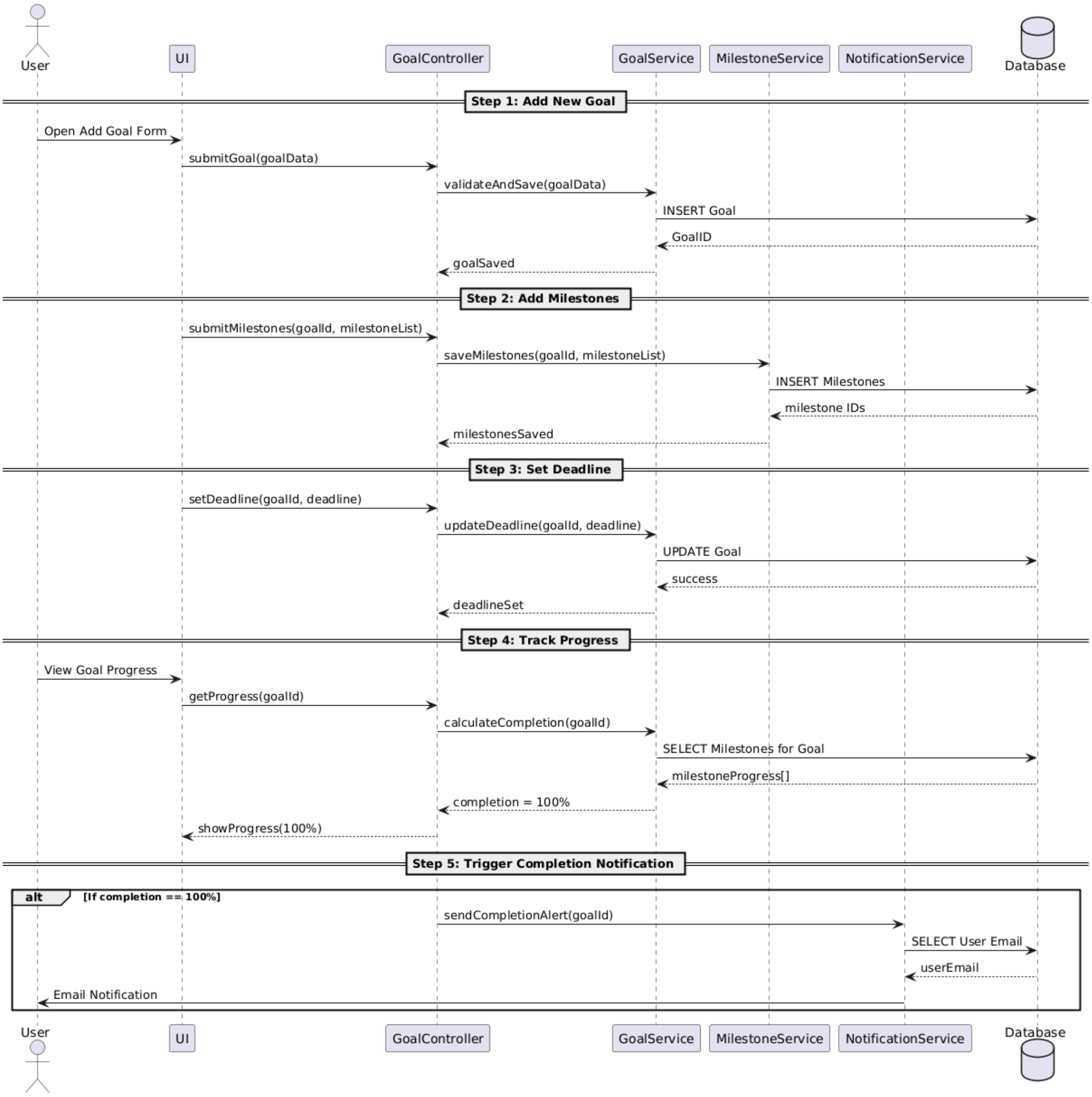


Figure 51 Sequence Diagram for Financial Goal Management

5.3.10 Testing

All parts of the Financial Goal Management artefact are tested to verify they function as they should and do not fail or misbehave. Adding, editing and removing goals from the system should be done properly with functional testing. The goal is measured through the proper milestone tracking as well as percentage completion. Alerts are tested to check that they become visible when an important step is reached in a project or when every step is done. The aim of usability testing is to verify that the interface is easy to use. Moreover, the system is also checked with cases such as empty inputs, unknown dates or missing milestones to see whether or not if it remains stable. With such thorough testing, the artefact is tested enough to be used in real situations.

User Story ID	Test Case	Priority	Steps	Expected Outcome	Status	Comments
FGM-1	Set Financial Goal	High	1. Navigate to goal form. 2. Enter goal title, milestones, and deadline. 3. Submit form.	Goal is created and saved with milestones and deadline.	Passed	Goal created successfully.
FGM-1	Duplicate Goal Title Check	Medium	1. Attempt to create a goal with the same title. 2. Submit the form.	System displays error: 'Goal title already exists.'	Passed	Duplicate title blocked.
FGM-2	Edit Goal	Medium	1. Open existing goal. 2. Edit goal details. 3. Save changes.	Goal details updated successfully.	Passed	Goal updated successfully.
FGM-2	Delete Goal	Medium	1. Select a goal to delete. 2. Confirm deletion.	Goal is removed from the system.	Passed	Goal deleted successfully.
FGM-3	Track Progress	High	1. View goal progress screen. 2. Review milestone completion status.	Progress bar shows correct percentage based on completed milestones.	Passed	Progress tracked correctly.
FGM-4	Send Progress Notification	Medium	1. Update a milestone. 2. Check notification log.	Notification sent to user about goal progress.	Passed	Notification received.
FGM-4	Send Completion Alert	Medium	1. Complete all milestones. 2. Check for completion alert.	User receives goal completion notification.	Passed	Completion alert sent.

Figure 52 Testing for Financial Goal Management

5.3.11 Example Test Cases

The goals along with the deadlines and milestones are shown.

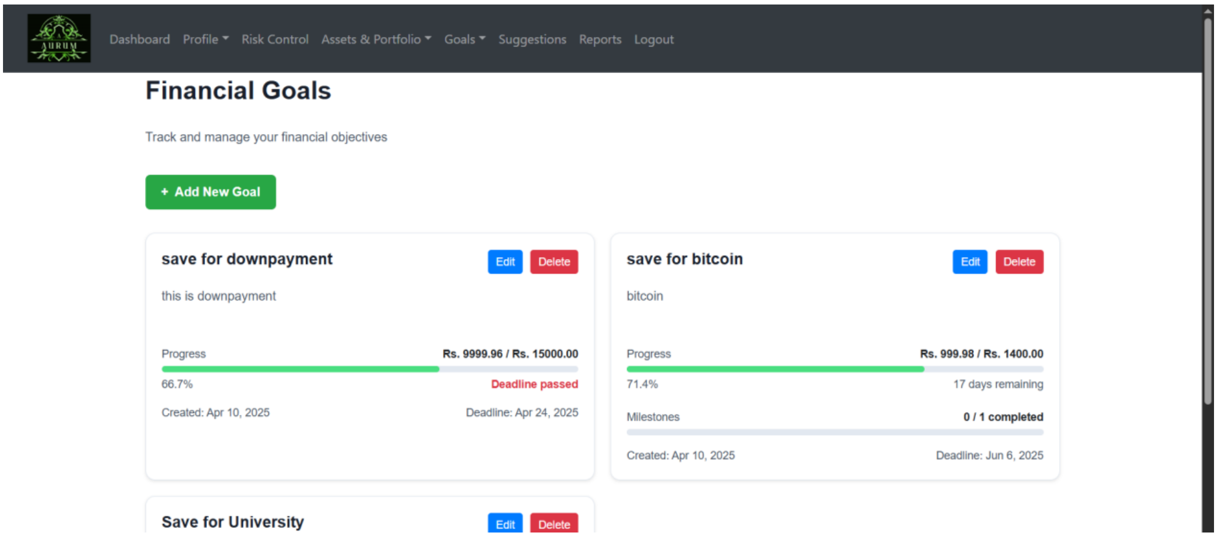


Figure 53 Example test case

CRUD operation in goals are successful.

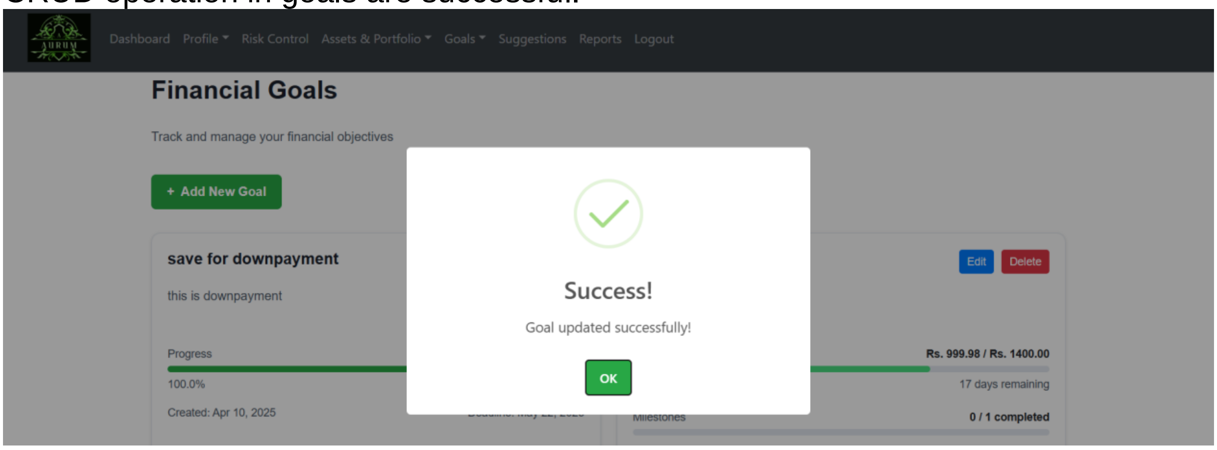


Figure 54 Example test case

5.4 Artefact – Notifications

5.4.1 FDD for Notifications

The FDD for notification is made such that both email and dashboard notifications can be sent. Users are to receive important alerts from the website like the updates on their portfolio, how they are achieving their goals, thorough email. Likewise, dashboard notifications of live alerts within the website, including actionable and general updates such that they can take action. Wherever the user are active, the user receive important messages.

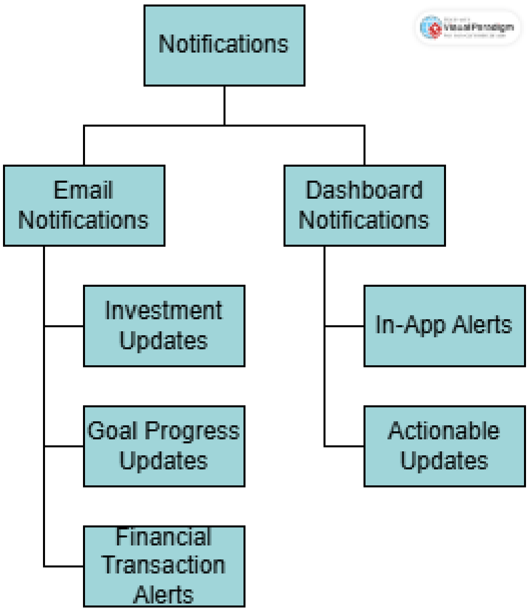


Figure 55 FDD of Notification

5.4.2 Product Backlog for Notification

Notification is made to keep users engaged and keep them up to date through email and in-app notifications whereby the ids ensures the users receive email notifications for urgent or critical events for examples, goal progress, helping them staying up to date. Also it is to provide users with a dedication section of the interface, such that it allows them for immediate awareness and action.

Epic 5: Notifications

ID	User Story	Priority	Acceptance Criteria
NTF-1	As a user, I want to receive email notifications about key updates so that I stay informed.	High	Email notifications are sent for significant events like investment changes or goals.
NTF-2	As a user, I want to see notifications on my dashboard so that I can take immediate action.	High	Notifications appear in a designated section of the dashboard.

Figure 56 Product Backlog for Notification

5.4.3 Sprint Planning for Notification

For this sprint, full notification system was added such that the user would be aware of what is happening and the user gets engaged. As part of this, I created email reminders and alerts that appear in the app for financial matters that actually matter such as deadline approaching, achieving goals and milestones reached etc. The task included creating the system's architecture, developing the frontend parts, creating API endpoints and including email features. Time when a notification was sent or opened was included in the structure along with read/unread information. At last phase, each unit and the interconnections between them were tested such that there was a smooth communication.

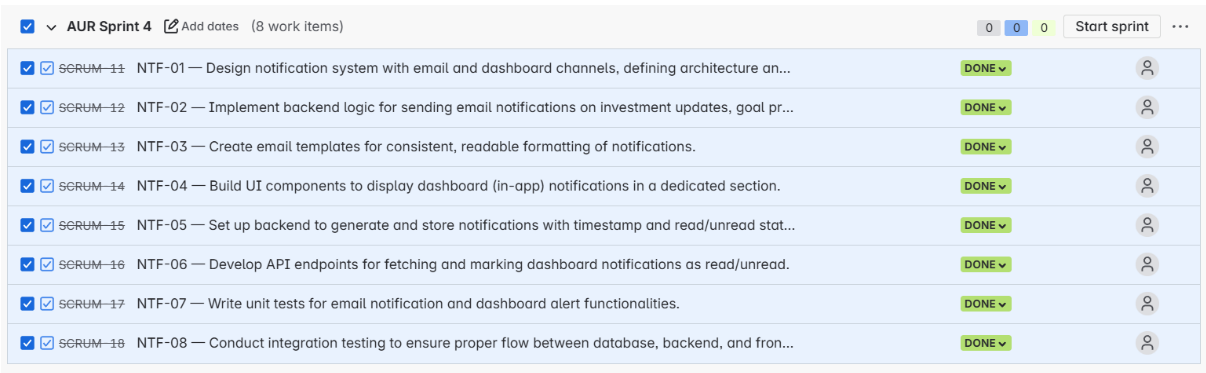


Figure 57 Sprint Planning for Notification

5.4.4 Activity Diagram of Notification

It outlines how activities are carried out, including the part that sends out email notifications and the one visible in the dashboard (in-app) alerts. A notification is sent when a user takes an action that could require attention such as any investment, progress towards a goal or a financial transaction. The process begins with checking if the event is significant. If the condition is met, the method goes to the notification module. Following the user's choice, the system produces and sends an email if the option for email notifications is set. No matter how emails are configured, the system automatically sends a notification which is displayed on the dashboard. When a user enters the system, it first looks for any new notifications. If anything needs attention, it is made visual for the user and actions can be taken easily. If there are no new notifications, you will see a message saying there is nothing new.

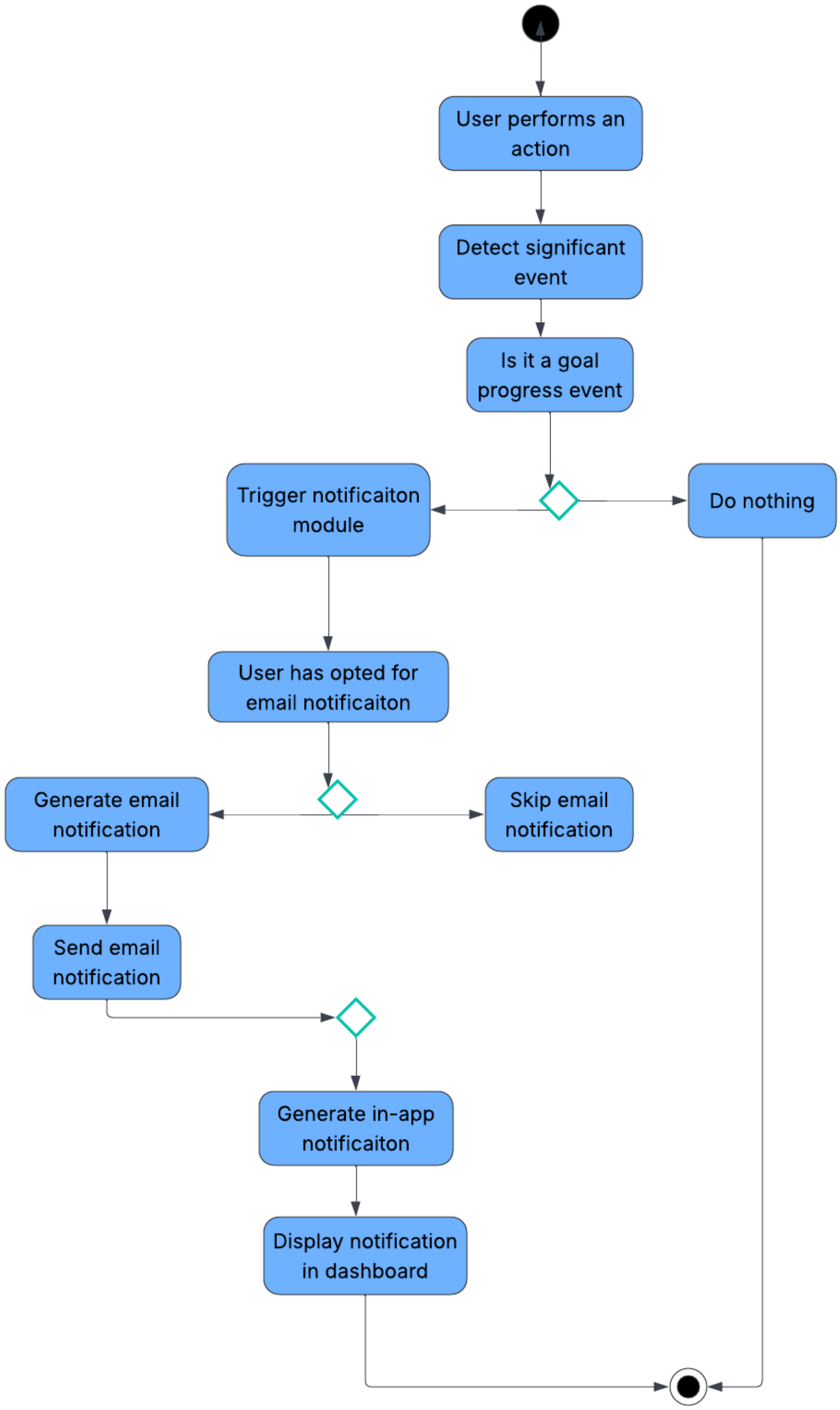


Figure 58 Activity Diagram of Notification

5.4.5 Use Case Diagram of Notification

The case diagram mainly focuses on how users and the software interact when dealing with notification features. The User is the main character of the story and is able to set up notifications and login to the dashboard. The systems work is to notify the users when important user activities are detected including the updates on goals and progress towards it.

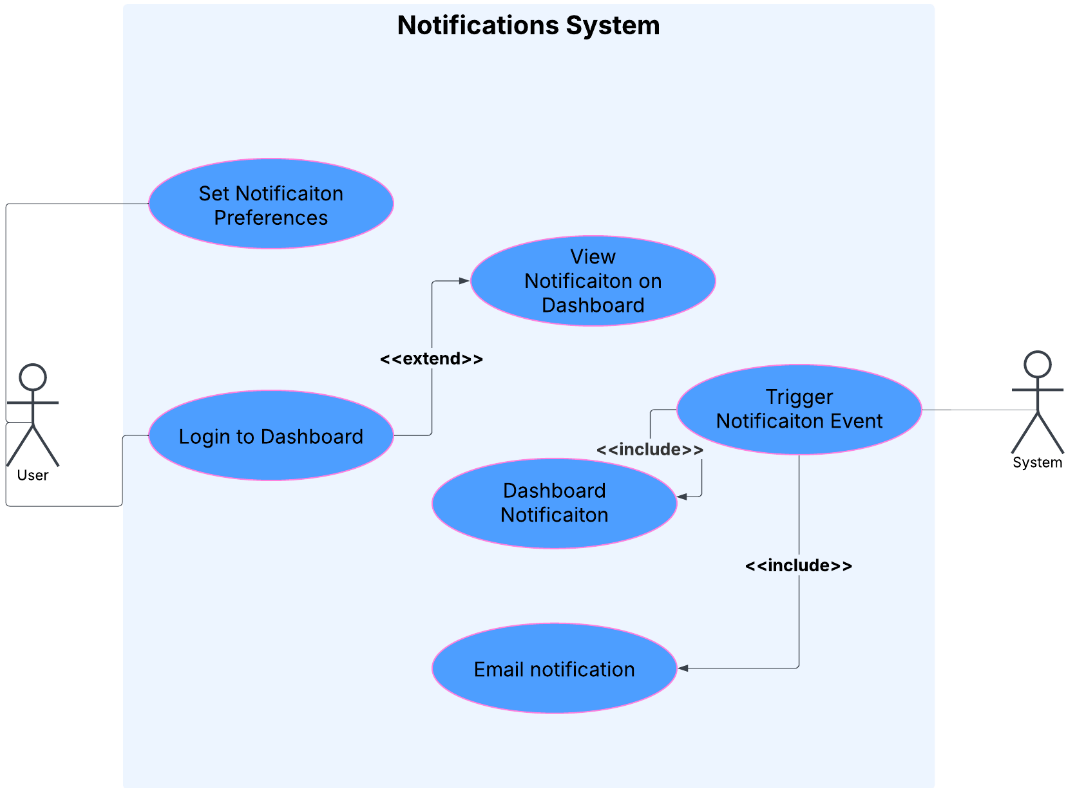


Figure 59 Use Case Diagram of Notification

5.4.6 ERD of Notification

It’s important because the system sends updates to the user about key aspects of their activities such as investment changes, progress on their goals or anything related to their financial accounts. To communicate, it uses either email messages or prompts displayed inside the application dashboard. Users can customize the system by changing the notifications they receive from email and in the app. Each time, you are provided the name of the notification, its text and whether you have checked it.

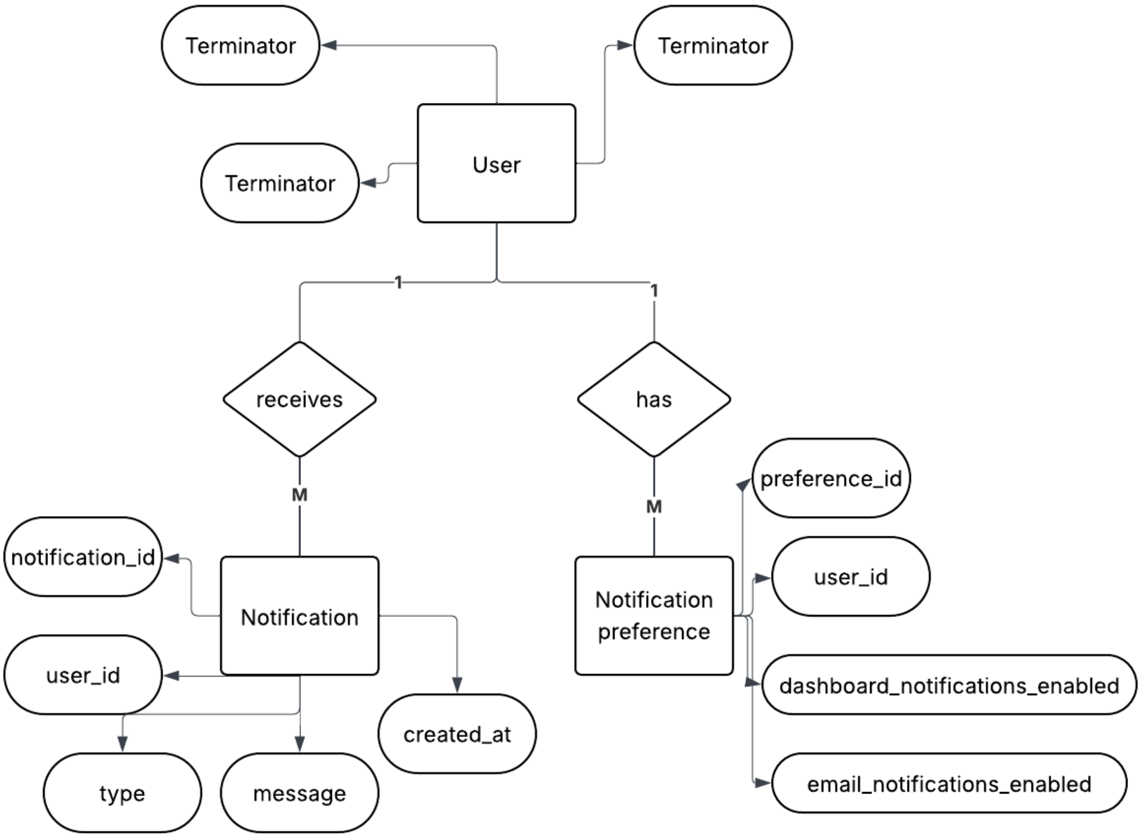


Figure 60 ERD of Notification

5.4.7 Class Diagram of Notification

It illustrates the primary classes in the Notification System and their interactions in the form of relationships. The main parts of the system include User, Notification and NotificationPreference. User information and functions for setting preferences, receiving messages and reviewing them are part of the User class. The Notification class stores data about notification type (Email or Dashboard), the message's content, whether it has been read or not and the time when it was created. It involves techniques to deliver the notification and letting your device know it has been read. NotificationPreference records how you would like to be notified and offers a solution to modify your preferences. It means a user may get different notifications and can manage their preferences by linking one choice for each notification. It shows how notifications are organized and handled using an object oriented approach.

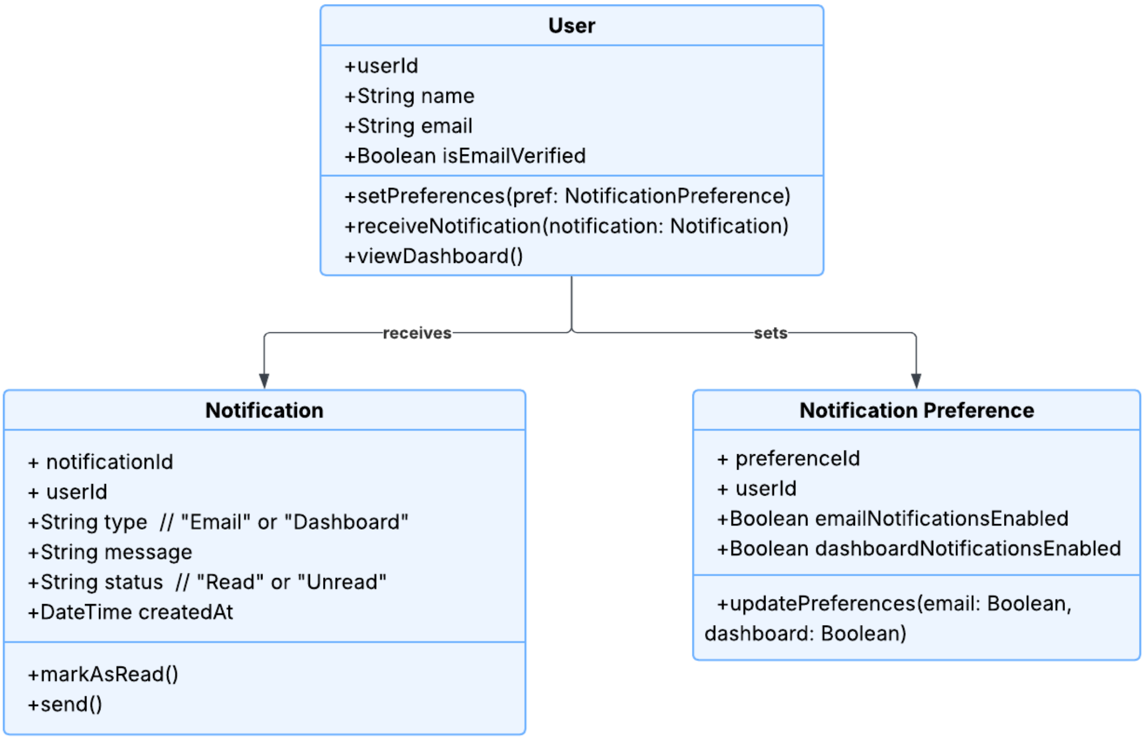


Figure 61 Class diagram of notification

5.4.8 Sequence Diagram of Notification

The diagram demonstrates each step in how notifications are delivered to users using email and the dashboard alert system. First, when a User takes an action, the Frontend records it and delivers the data to NotificationService on the backend. The service finds the display preferences for notifications in the Database for the user. If email is set up as a notification, the EmailService sends an email to the user. When dashboard notifications are activated, an in-app notification gets saved in the database. As soon as the user logs in or refreshes the dashboard on their mobile, the frontend grabs unread messages from the database and displays them.

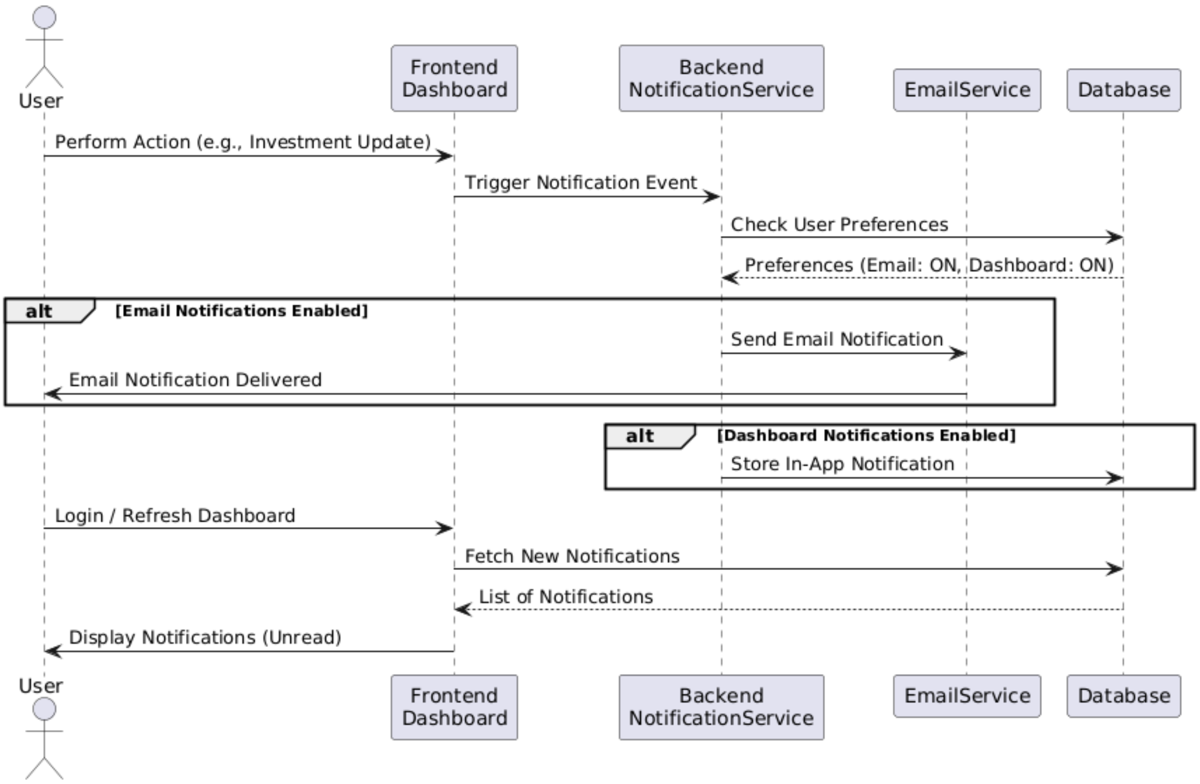


Figure 62 Sequence Diagram of Notification

5.4.9 Testing

The system's ability to update a user is tested by sending alerts to both the user's email and dashboard in various ways. Users are confirmed to receive prompt updates on their email after large events happen. Make sure that notifications are shown on the dashboard when the user accesses it either by login or refreshing it. Every test instance passed, indicating that things work as they should with the notification system.

User Story ID	Test Case	Priority	Steps	Expected Outcome	Status	Comments
NTF-1	Email Notification	High	1. Trigger a significant event (e.g., investment update). 2. Check if the user has email notifications enabled. 3. Confirm email is received.	The user receives an email with relevant update information. Email includes subject, message, and timestamp.	Passed	Email received successfully.
NTF-2	Dashboard Notification	High	1. Log in to the application. 2. Navigate to the dashboard. 3. Check for new in-app notifications.	Dashboard displays the new notification with message, type, and timestamp.	Passed	Notification displayed corre

Figure 63 Testing for Notification

5.4.10 Example test case

Notifications are being shown for important events.

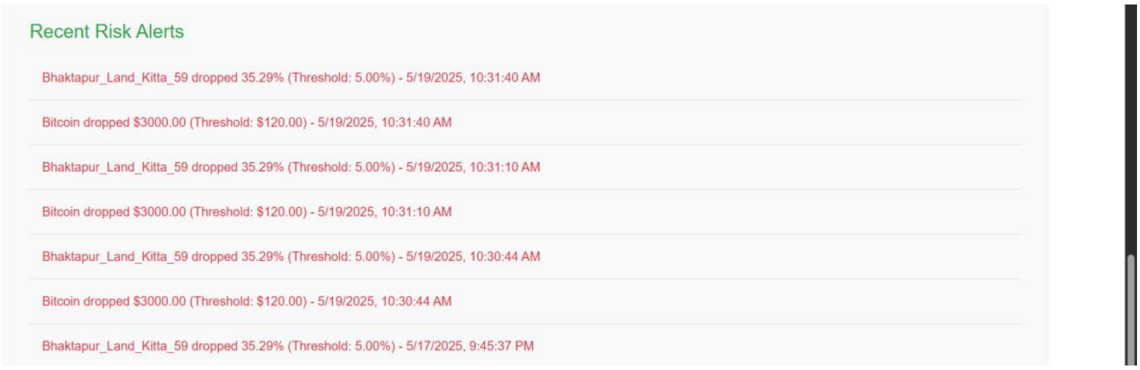


Figure 64 Example test case

5.5 Artefact - Reporting and Visualization

5.5.1 FDD of Reporting and Visualization

The image shows how the Reporting and Visualization module is put together in a financial application. The main sections are Data Visualizations and Basic Reports. The charts as well as graphs in the section of data visualizations outline the closeness of the users to meeting their financial objectives and how the investments are being developed over the years. Similarly, Basic Reports cover the user's main targets in the Summary of Financial Goals and report on all financial assets in the Asset Portfolio section. The way this software is built allows it to present key visual graphs and detailed finance reports.

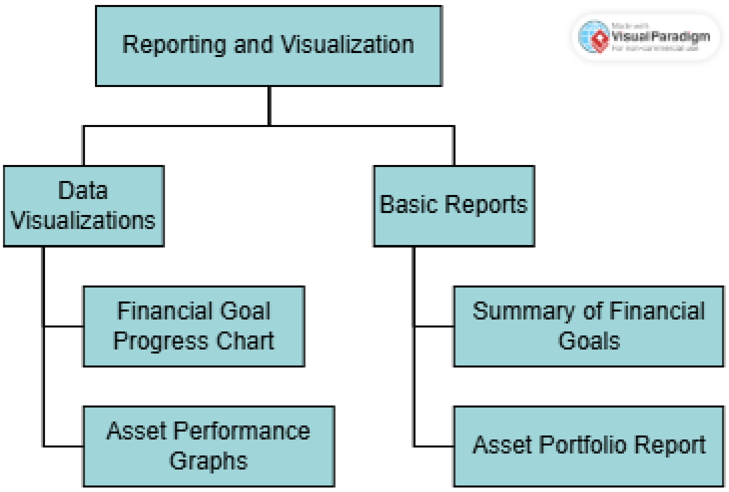


Figure 65 FDD of Reporting and Visualization

5.5.2 Product Backlog of Reporting and Visualization

Reporting and Visualization is meant to help individuals manage their finances by presenting their information in both beautiful graphs and printable reports. The key aspects are the two user stories. To help a user who wants to use their financial reports when offline, RV-2 makes it easy to download PDF or Excel copies.

Epic 7: Reporting and Visualization

ID	User Story	Priority	Acceptance Criteria
RV-1	As a user, I want to see visualizations of my portfolio and goals so that I can track performance.	High	Charts and graphs display financial goal progress and asset performance.
RV-2	As a user, I want to download basic reports so that I can analyze my financial data offline.	Medium	Reports can be downloaded in PDF or Excel formats.

Figure 66 Product Backlog of Reporting and Visualization

5.5.3 Sprint Planning of Reporting and Visualization

Real-time data for the visualizations was provided by implementing backend APIs. As well, two downloadable reports called “Summary of Financial Goals” and “Asset Portfolio Report” were developed so users can save their data in PDF or Excel formats and assess it without needing the internet.

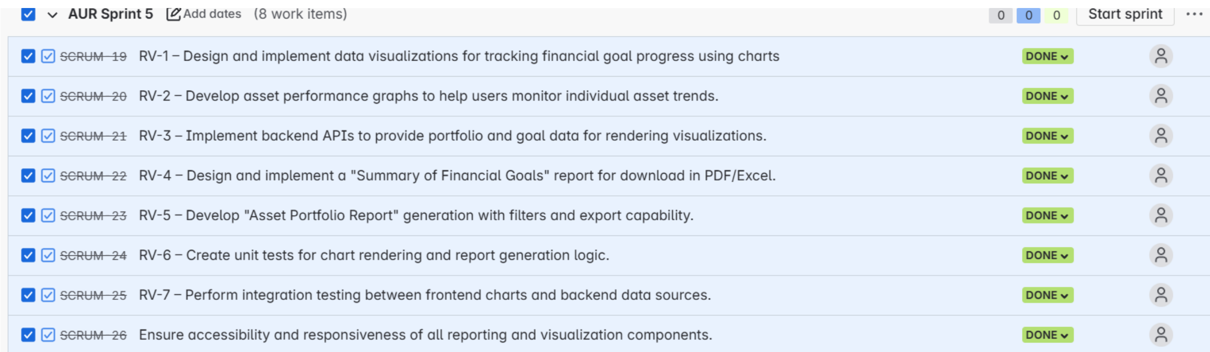


Figure 67 Sprint planning of reporting and visualization

5.5.4 Activity Diagram of Reporting and Visualization

It explains the steps a user takes while using the module to get financial insights. The reporting section is where the user reaches the first step: whether to view their portfolios or goals. If the user selects the visualization option, they get to compare a chart of their financial targets with an asset performance graph. If the user wishes to get reports, they can download either a report on their financial goals or an asset portfolio breakdown. When the report is selected, you are offered the option to download it as a PDF or an Excel file. If the user does not decide between options, they leave the reporting module. With this design, users can access and look at their finances, either online or by downloading the data.

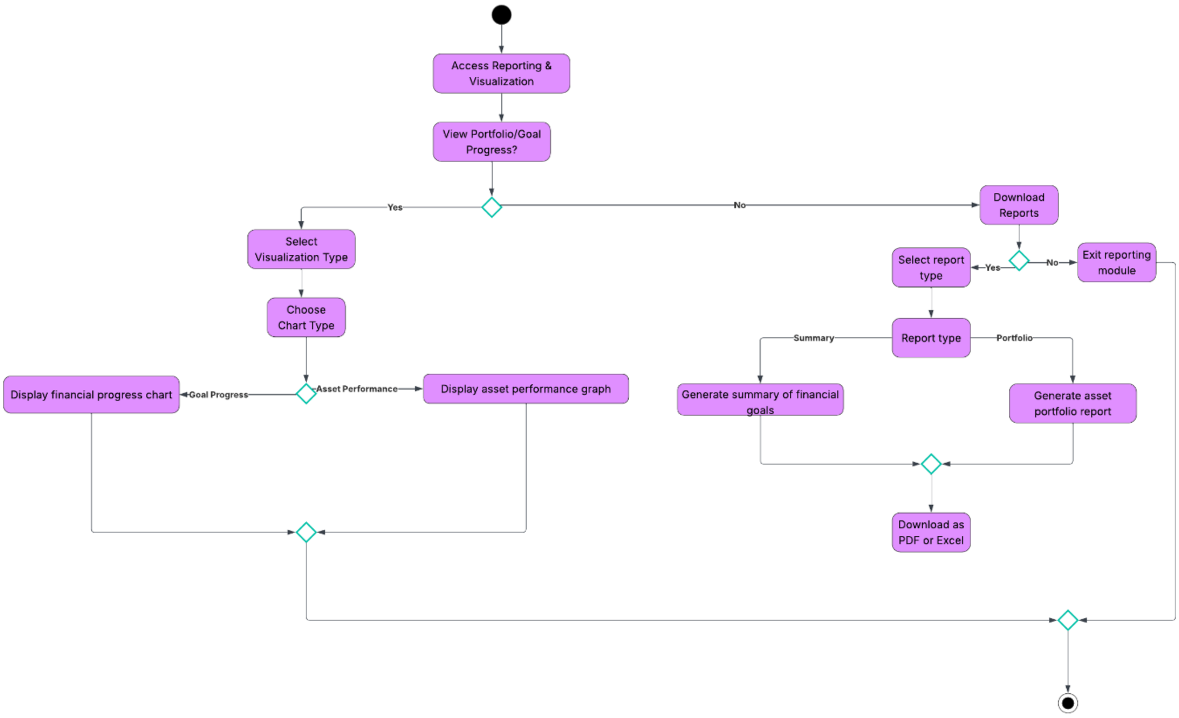


Figure 68 Activity Diagram of Reporting and Visualization

5.5.5 Use Case of Reporting and Visualization

It reveals how a user can use the Reporting and Visualization module to gain access to financial information. Access Reporting & Visualization allows users to mainly perform two tasks: checking charts and graphs, as well as downloading reports. These features are offered to all users.

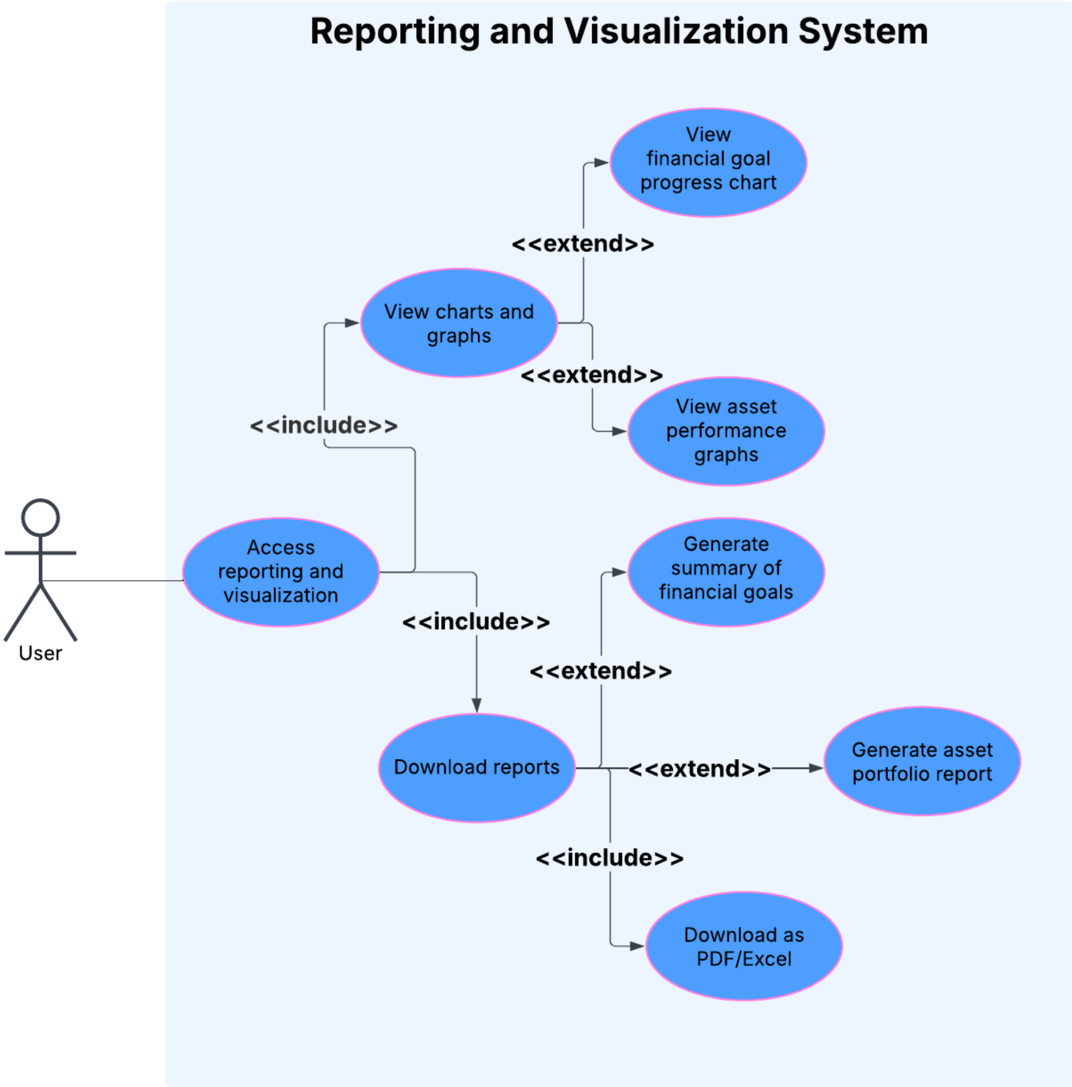


Figure 69 Use case of reporting and visualization

9.5.6 Wireframe

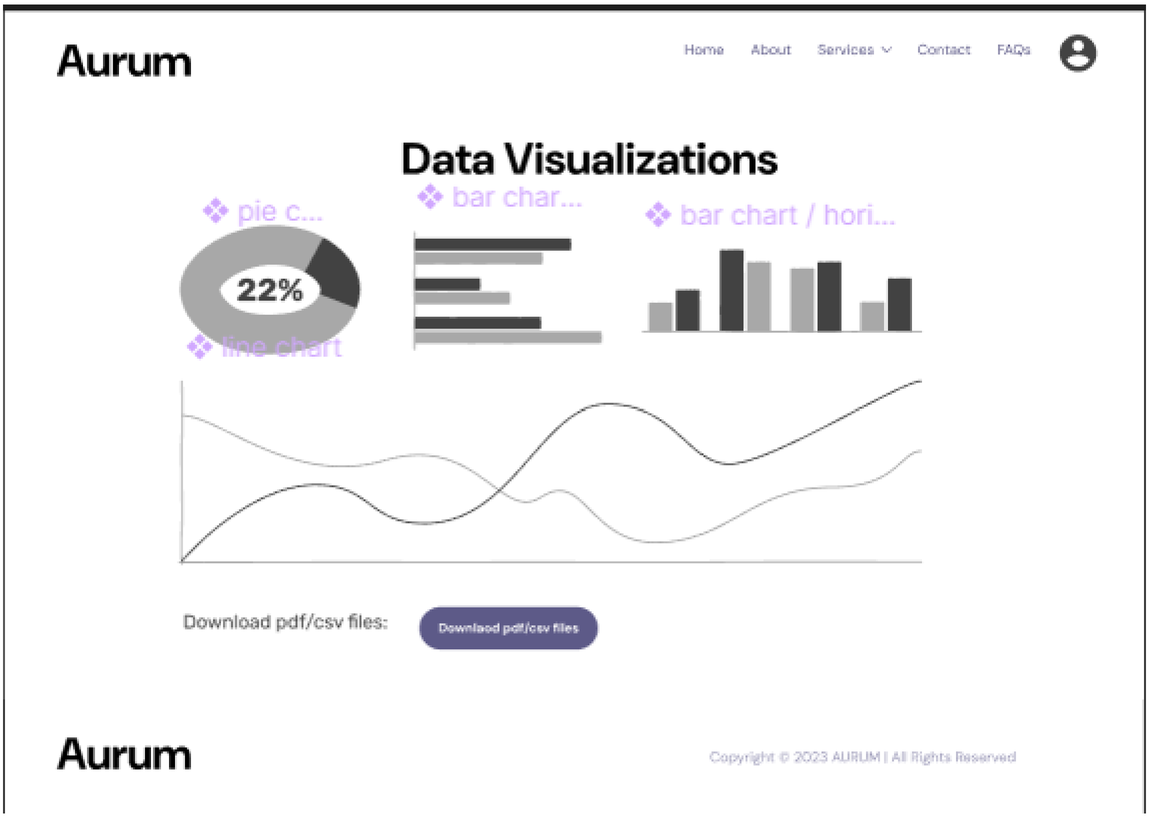


Figure 70 Wireframe - reporting and visualization

5.5.7 ERD of Reporting and Visualization

How all the data components of the Reporting and Visualization module are connected is described in the ERD. The most important thing is the User which has connections to many Portfolios, Financial Goals and Reports. Information stored in each portfolio allows the creation of financial goal or asset graphs. After setting a FinancialGoal, the system keeps track of it and the Report stores any metadata linked to downloaded reports, including their format and location. The architecture helps organize user financial data in a way that makes it accessible for analysis, backups and use later without an internet connection.

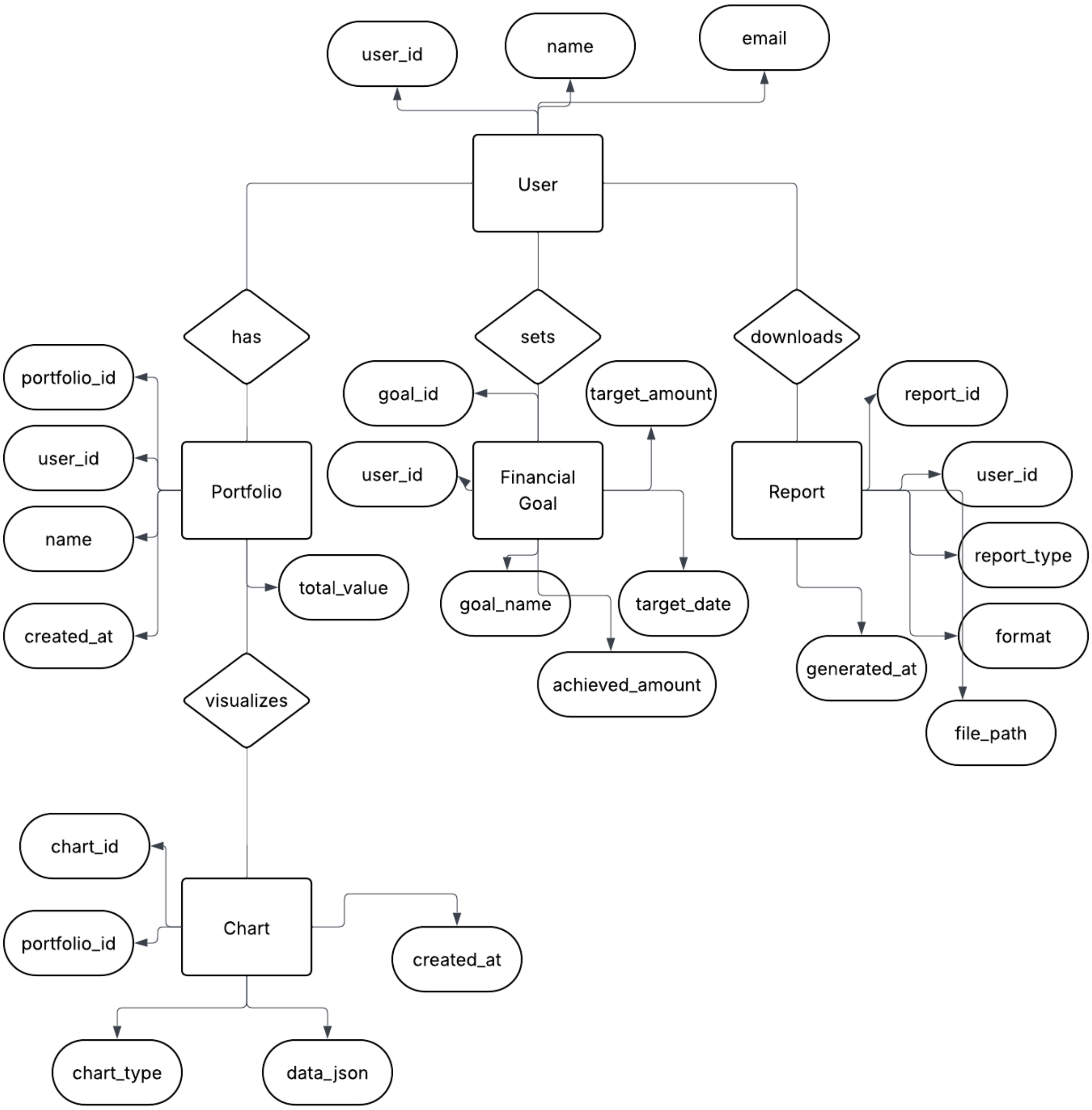


Figure 71 ERD of Reporting and Visualization

5.5.8 Class Diagram of Reporting and Visualization

At the top is User which has connections to Portfolios, Financial Goals and Reports. Each Portfolio carries investment information and is paired with one or more Charts that show how it is performing. With this class, the user's saving or investing targets can be managed and progress evaluated. Reports store downloadable documents, listing what format they are in and where they are stored, whereas Charts help create visual elements, for example, various graphs.

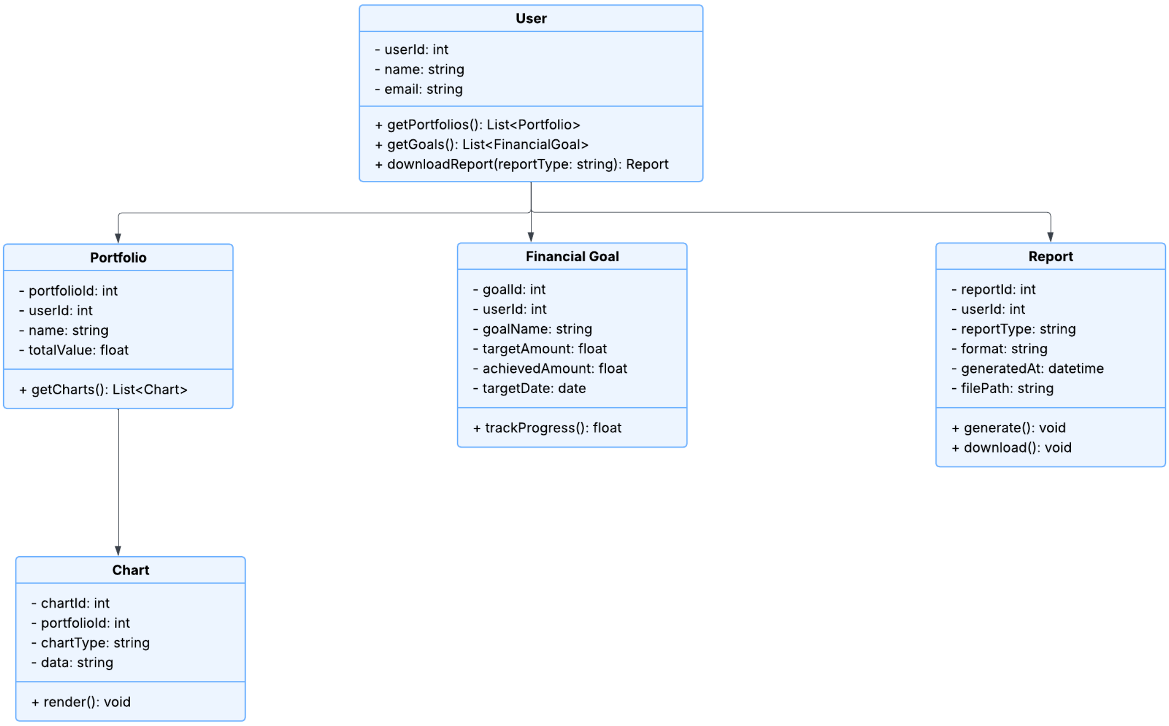


Figure 72 Class Diagram of Reporting and Visualization

5.5.9 Sequence Diagram of Reporting and Visualization

The sequence diagram explains the connections between the user and the systems involved in the Reporting and Visualization module. This means that people can view charts or download reports as the main actions they can take. While viewing a chart, the first step is for the user to make a request via the reporting module which talks to the chart service to get relevant data from the proper database. The chart is made and displayed on the user’s screen. When a person wants to download a report, they indicate the report type and format. After that, the system runs the reporting module and connects to the report service. It takes the data from the database, prepares the report and offers a link to download it. The user can view the report in PDF or Excel format. It marks out the main services that work together to present useful visuals and content for downloading.

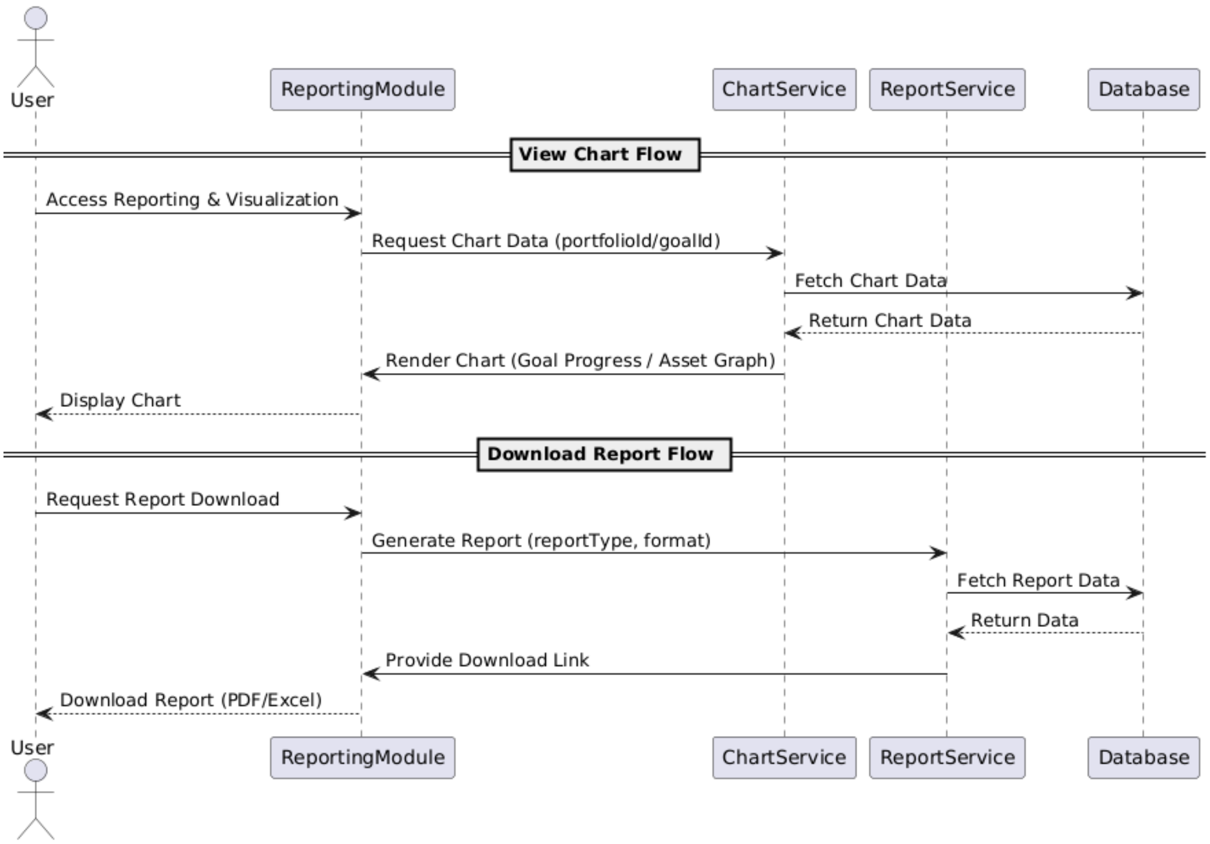


Figure 73 Sequence Diagram of Reporting and Visualization

5.5.10 Testing

For RV-1, the key to the test was to confirm that they could observe their achievements and portfolio returns using interactive graphs. The trial of RV-2 found that users can effortlessly download reports in the formats of their choice.

User Story ID	Test Case	Priority	Steps	Expected Outcome	Status	Comments
RV-1	View Portfolio & Goal Charts	High	1. Log in to the application. 2. Navigate to the reporting/visualization page. 3. Select chart type (Goal Progress or Asset Performance).	Charts and graphs display financial goal progress and asset performance accurately.	Passed	Charts rendered successfully.
RV-2	Download Basic Reports	Medium	1. Log in to the application. 2. Navigate to the reporting/download section. 3. Choose report type (Summary or Portfolio). 4. Select format (PDF/Excel). 5. Download the file.	Report is generated and successfully downloaded in the selected format.	Passed	Report downloads verified.

Figure 74 Testing of Reporting and Visualization

5.6 Artefact - Customized Suggestions for Investment

5.6.1 FDD of Customized Suggestions for Investment

Here is a clear representation of a personalized investment recommendation system. First, information about the user's age, irritation with risk and investment goals is gathered and stored under investment parameters. The system of recommendation uses the inputs provided by the users to recommend items tailored to each person. According to the user's information, the system's goal is to recommend suitable investments which match the requirements and goals. This allows the recommendation system to provide accurate and tailored tips to manage investments.

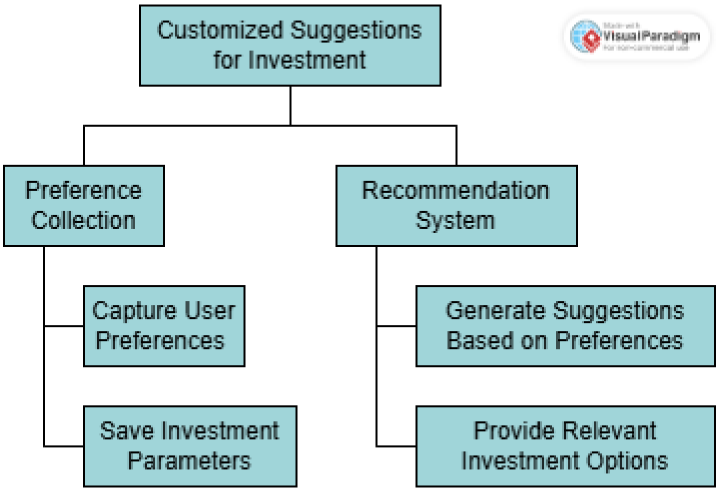


Figure 75 FDD of Customized Suggestions for Investment

5.6.2 Product Backlog of Customized Suggestions for Investment

Epic 4 provides investment advice that matches the needs and wishes of the user. Three main user stories are included in it:

Users of CSI-1 can specify their preferences which will be securely saved in their account for further reference.

The important aspects of CSI-2 include providing users with accurate and practical suggestions, since these depend on what they have set in their profile.

Users can look into every suggestion in detail using CSI-3 to ensure better assessments and comparisons before investing.

With this app, each user can explore personalized and interactive ways to invest based on their data.

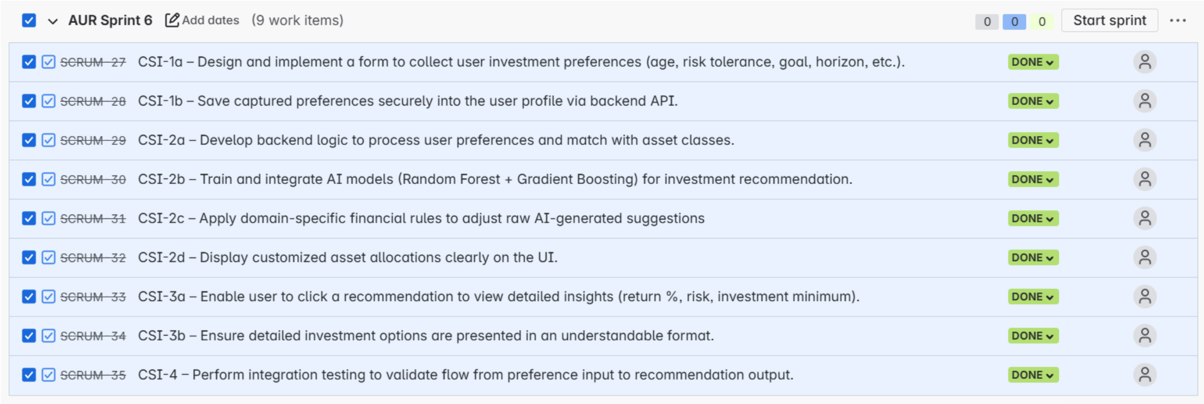
Epic 4: Customized Suggestions for Investment

ID	User Story	Priority	Acceptance Criteria
CSI-1	As a user, I want to input my preferences for investments so that I get personalized suggestions.	High	Preferences are captured and saved in the user profile.
CSI-2	As a user, I want the system to recommend investments based on my preferences so that I can decide.	High	Recommendations are accurate and relevant to user preferences.
CSI-3	As a user, I want to view detailed investment options so that I can analyze them further.	Medium	Clicking a recommendation provides more detailed insights about the investment.

Figure 76 Product Backlog of Customized Suggestions for Investment

5.6.3 Sprint Backlog of Customized Suggestions for Investment

The objective of the Customized Suggestions for Investment sprint was to make personalized recommendations for users. Initially, the team developed simple forms for the website so that users could provide their age, level of risk, goals for their savings and the number of years they wished to invest. User profiles were safely secured by storing the information through backend APIs. I made sure to use Random Forest and Gradient Boosting AI models to review user information and recommend the best way to spread assets. Domain-related guidelines were used to correct and improve the recommendations based on the user’s and company’s guidelines. During the sprint, users were able to look at detailed options for investments mentioned in the recommendations and finally, the system was tested to ensure everything functioned as expected. The sprint helped attract users because it offered smart personalized recommendations on investments.



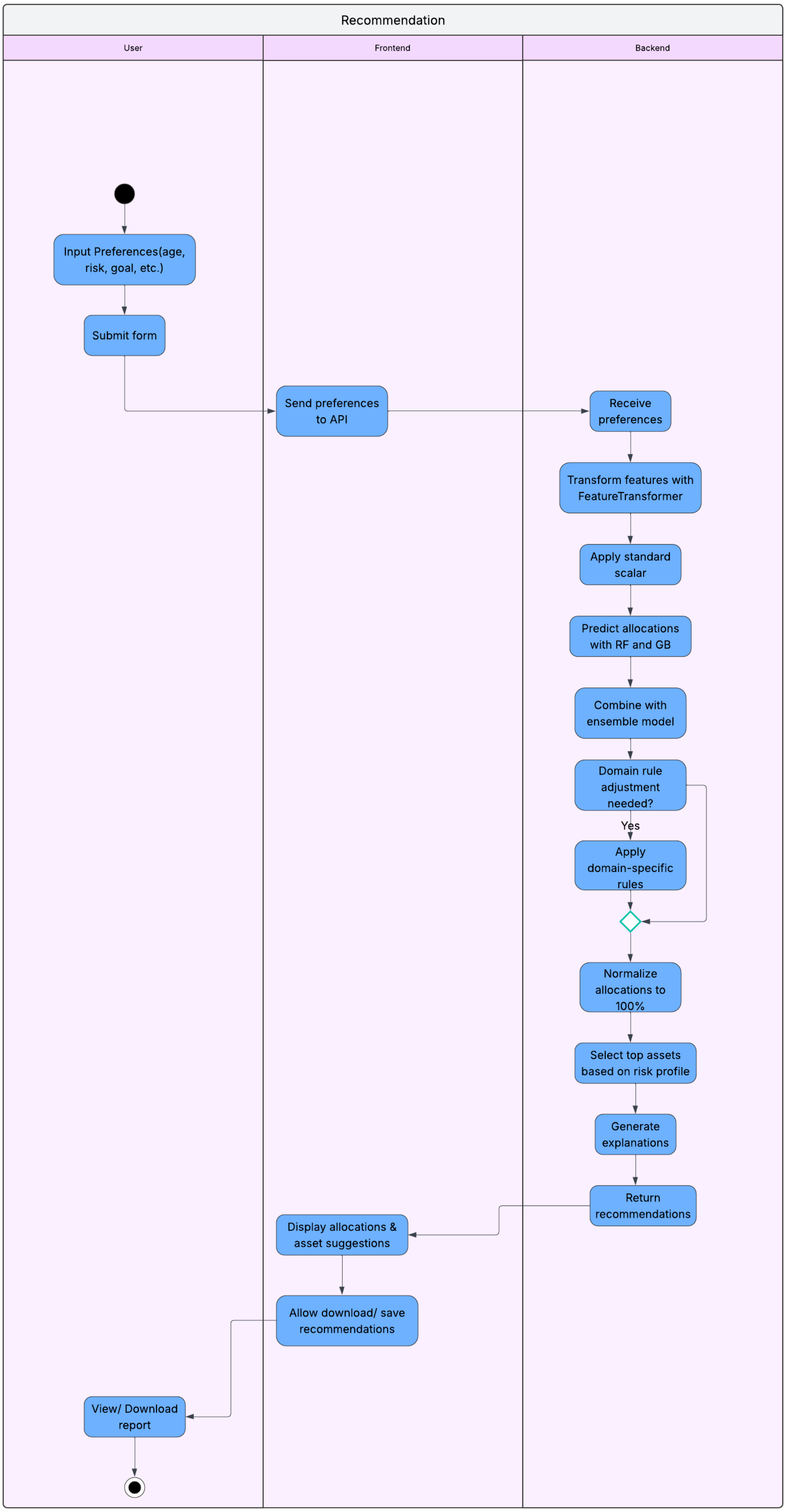
AUR Sprint 6 Add dates (9 work items) 0 0 0 Start sprint

~~SCRUM-27~~	CSI-1a – Design and implement a form to collect user investment preferences (age, risk tolerance, goal, horizon, etc.).	DONE	
~~SCRUM-28~~	CSI-1b – Save captured preferences securely into the user profile via backend API.	DONE	
~~SCRUM-29~~	CSI-2a – Develop backend logic to process user preferences and match with asset classes.	DONE	
~~SCRUM-30~~	CSI-2b – Train and integrate AI models (Random Forest + Gradient Boosting) for investment recommendation.	DONE	
~~SCRUM-31~~	CSI-2c – Apply domain-specific financial rules to adjust raw AI-generated suggestions	DONE	
~~SCRUM-32~~	CSI-2d – Display customized asset allocations clearly on the UI.	DONE	
~~SCRUM-33~~	CSI-3a – Enable user to click a recommendation to view detailed insights (return %, risk, investment minimum).	DONE	
~~SCRUM-34~~	CSI-3b – Ensure detailed investment options are presented in an understandable format.	DONE	
~~SCRUM-35~~	CSI-4 – Perform integration testing to validate flow from preference input to recommendation output.	DONE	

Figure 77 Sprint Backlog of Customized Suggestions for Investment

5.6.4 Activity Diagram of Customized Suggestions for Investment

The activity diagram reveals the entire process from a user's input to suggested investments. After logging in, a user enters their age, how much risk they want, the amount of money they have to invest and what their goal is. In the backend, these inputs are turned into new forms and their scales are adjusted. After that, the data is given to both the Random Forest and Gradient Boosting models to predict the right allocation for assets. Allocations are improved by applying particular rules that suit the user and their investment plan. The results are standardized and used to provide users with recommendations and clear explanations, alongside charts and diagrams. The process helps provide personal, data-based and understandable guidance on investments.



Recommendation
User
Frontend
Backend
Input Preferences(age, risk, goal, etc.)
Submit form
Send preferences to API
Receive preferences
Transform features with FeatureTransformer
Apply standard scalar
Predict allocations with RF and GB
Combine with ensemble model
Domain rule adjustment needed?
Yes
Apply domain-specific rules
Normalize allocations to 100%
Select top assets based on risk profile
Generate explanations
Return recommendations
Display allocations & asset suggestions
Allow download/ save recommendations
View/ Download report

Figure 78 Activity Diagram of Customized Suggestions for Investment

5.6.5 Use Case Diagram of Customized Suggestions for Investment

The diagram illustrates the interactions between the investment recommendation system and the users for the feature. Basically, the user first indicates what they want to invest in, then stores the choices and creates an asset allocation from them. Moreover, applying domain-specific rules to the recommendation ensures it is surprisingly accurate for each user. After having decided on the proper portfolio, the service offers advice centered on the user's willingness to take risks and their goals. The users may also decide to explore more about their investments or download a report if that is needed for their situation. Authentication is necessary in this use case to keep personal data secure. It shows the sequence of the app's blocks and points out which capabilities are necessary and which are optional.

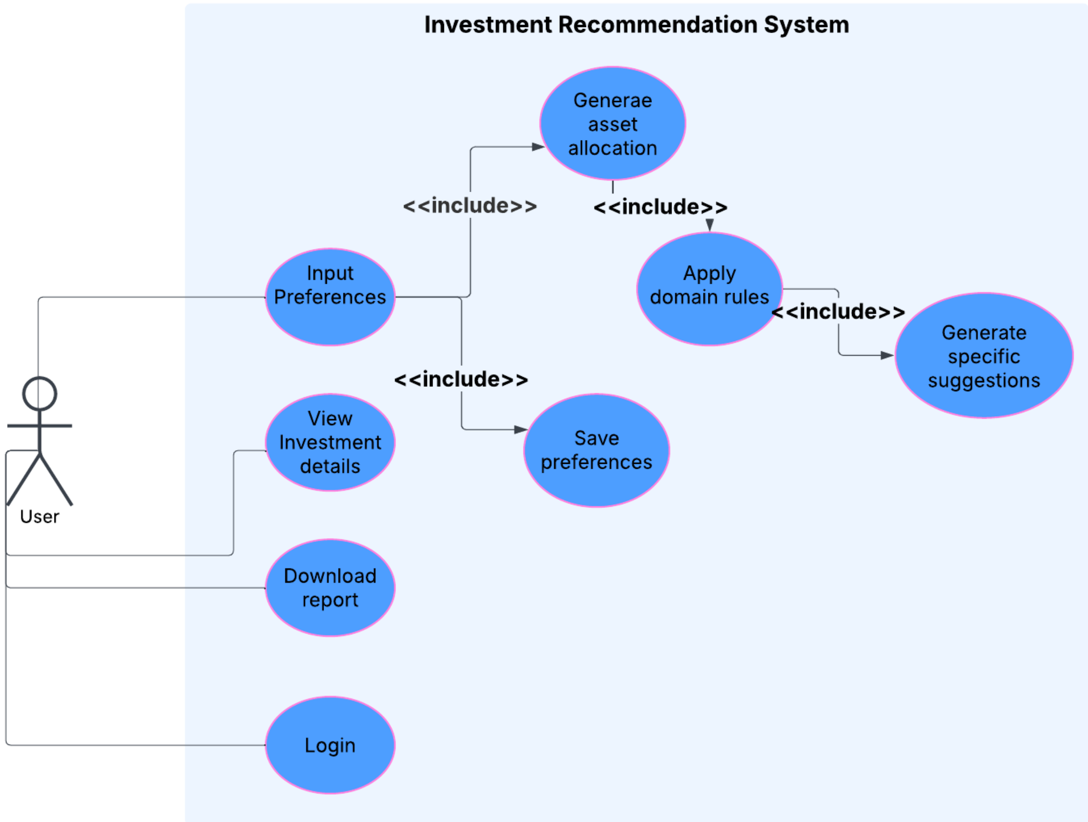


Figure 79 Use Case Diagram of Customized Suggestions for Investment

5.6.6 Wireframe

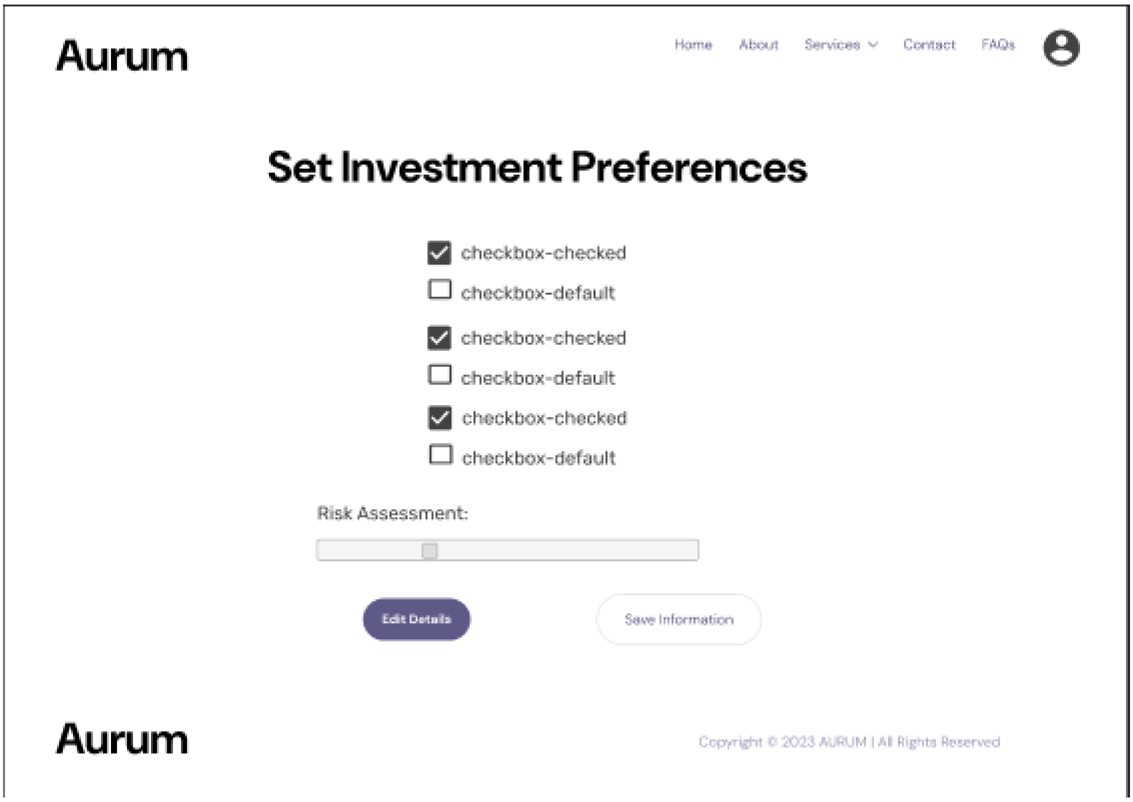


Figure 80 Wireframe - customized suggestions for investment

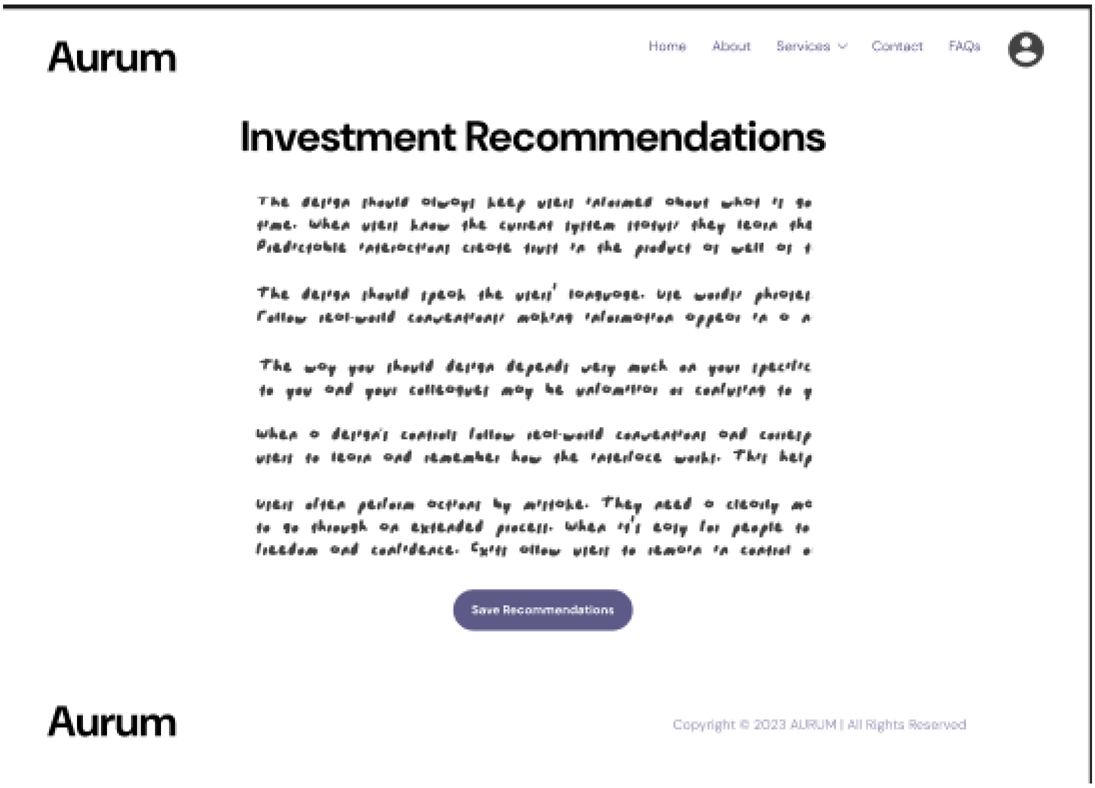


Figure 81 Wireframe - investment recommendations

5.6.7 ERD of Customized Suggestions for Investment

The diagram shows the main structure of the data involved in the Nepal Investment Recommendation System. The system is focused on the User, who possesses many

Goals (such as saving for retirement and a house) and their present Holdings (investments). Each person can be given numerous Recommendations based on the system's asset allocation suggestions. Every recommendation offers AssetAllocations and may have SpecificRecommendations that identify exact investments for specific assets. Assets are often defined by their return, how volatile they are and how liquid they remain. It is clear that a user can provide recommendations and has aims and belongings; these recommendations specify which assets should be purchased. Because of this structure, advice given to clients is personalized and considers all the necessary information.

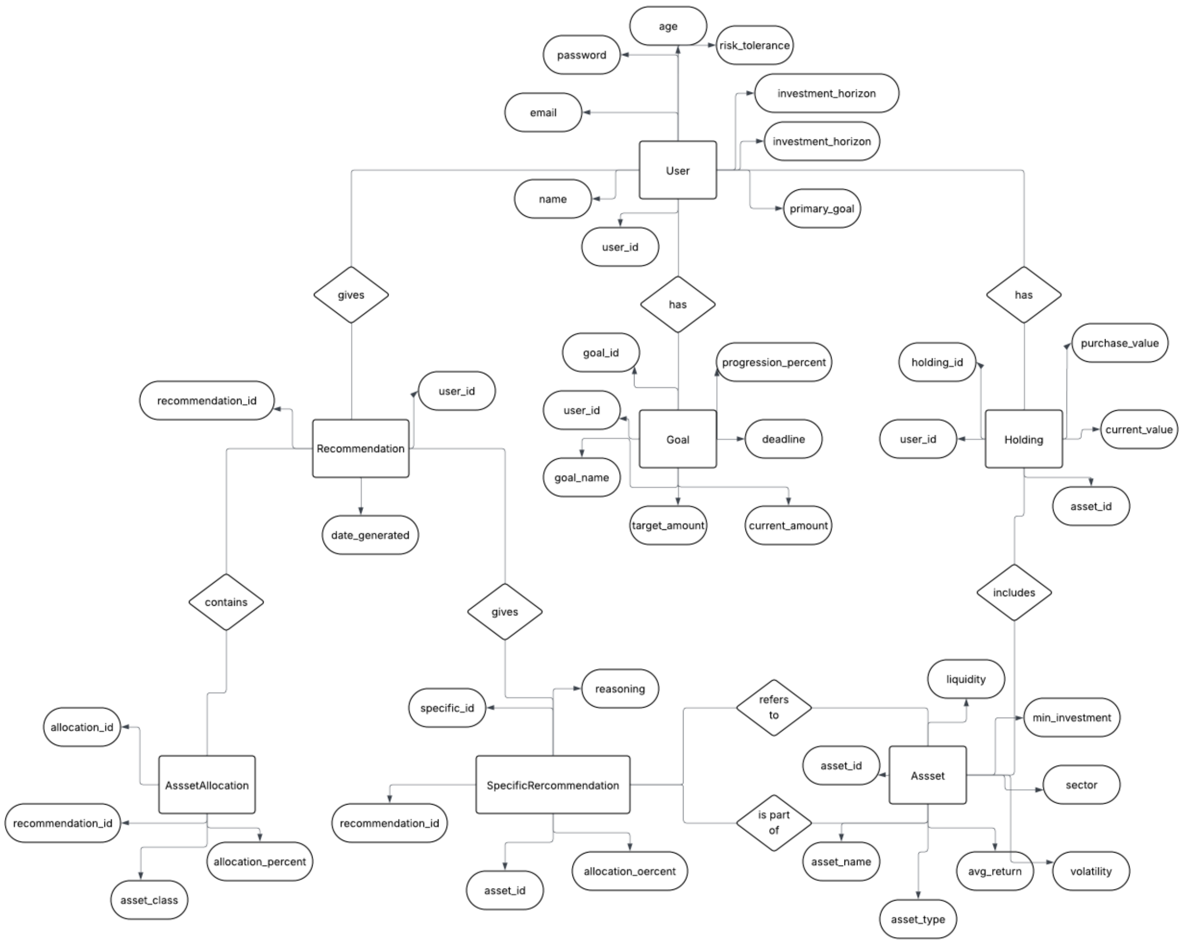


Figure 82 ERD of Customized Suggestions for Investment

5.6.8 Class Diagram of Customized Suggestions for Investment

The diagrams describe the structure and operation of all parts in the Nepal Investment Recommendation System. The User class focuses on a user's unique details and finances and it is linked to several Goal, Holding and Recommendation objects. Goal is used to track financial objectives and their progress, whereas Holding handles a user's investment portfolio. An AssetAllocation is provided for each

Recommendation and each includes several SpecificRecommendation instances that detail specific assets and why they should be chosen. The Asset class is home to data about the market such as wins, risks and the methods used to invest. The logic behind it is powered by the Recommender class that runs algorithms to create a personalized list of ideas for investments and how to distribute them. It is clear from the relationships that a User owns Holdings, has Goals and is recommended Assets by way of SpecificRecommendations. In every class, I added methods that demonstrate real-world behaviors such as making recommendations, analyzing assets and updating data.

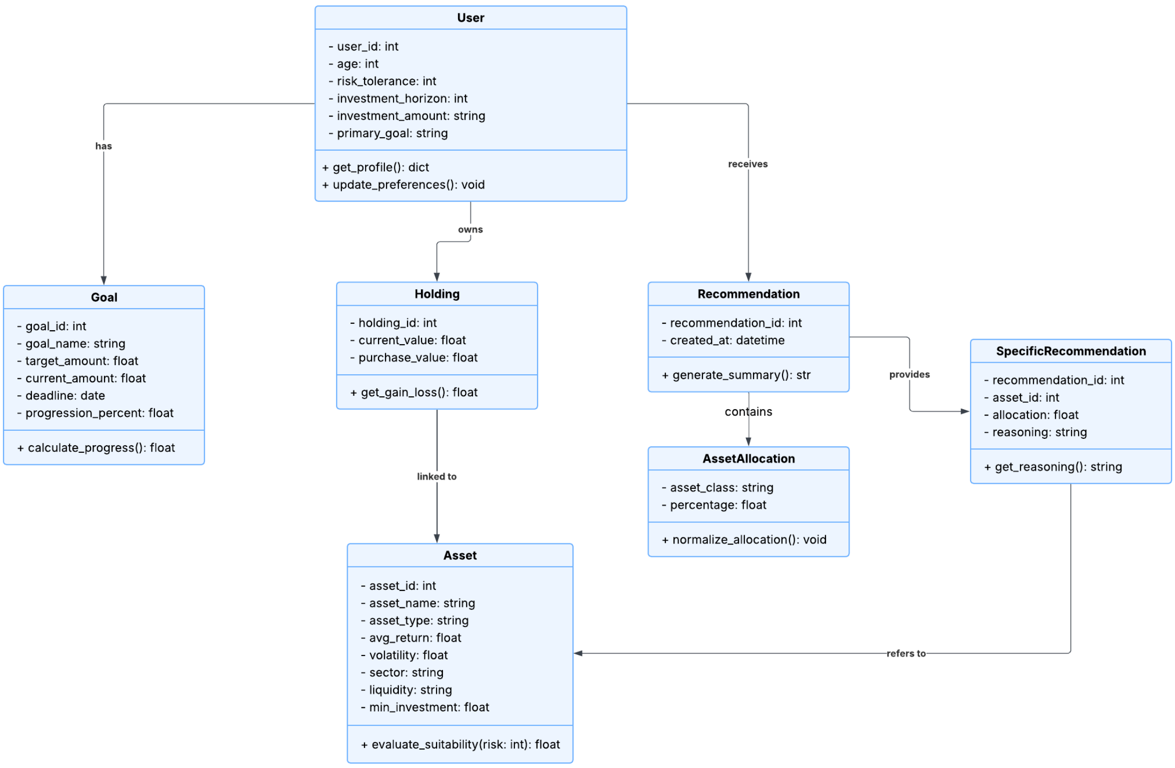


Figure 83 Class Diagram of Customized Suggestions for Investment

5.6.9 Sequence Diagram of Customized Suggestions for Investment

The sequence diagram shows how the user and Nepal Investment Recommendation System interact with each other. To get started, the user enters their preferences (including age, how they handle risks and what they want to do with their money) in the UI of the web application. As a result, the Django API sends the data to our Recommender System for use. The recommender prepares and updates the data, scales it and passes it to the ensemble which merges the forecasts from the Random Forest and Gradient Boosting models.

Having determined the percentages to invest, the system uses rules relevant to finance to ensure the allocations are practical. Then, the recommender asks the assets database for investment choices in the asset classes and assigns a score to each based on the user's amount of risk. The top assets and personalized data are combined, sent to the API and the API sends this data back as the final recommendation to the frontend. After that, the user gets a detailed suggestion of various assets and investments best suited to them. Recommender is in charge of computing recommendations for investors based on their accounts, what they hope to achieve and tolerance to risk. Example: A User is mapped to Goals, Holdings and Recommendations and a SpecificRecommendation represents an Asset that each specific recommendation relates to. In every class, the methods show behaviors such as making recommendations, evaluating assets and updating data.

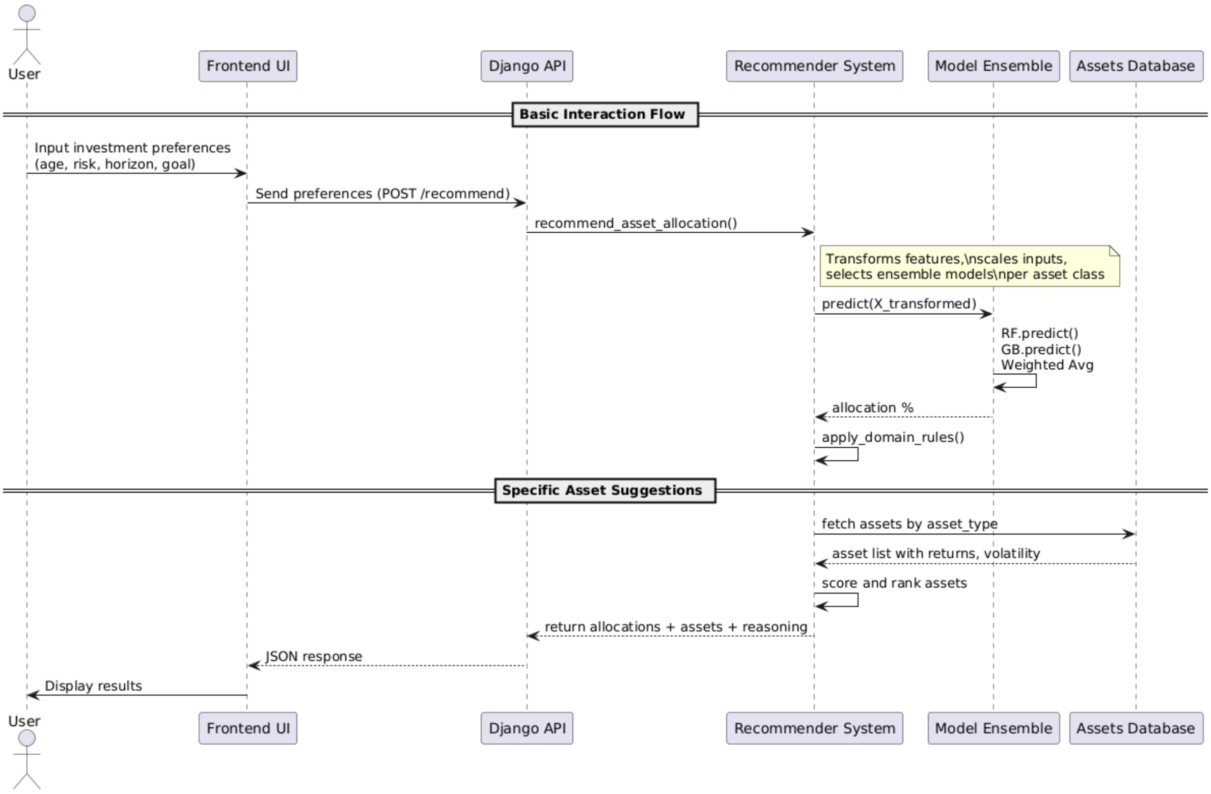


Figure 84 Sequence Diagram of Customized Suggestions for Investment

5.6.10 Testing of Customized Suggestions for Investment

Epic 4 focused the testing on making sure the recommendation system was functioning properly and suggesting the right investments based on what a user likes. Various situations were listed in the test cases, like taking user preferences, assigning asset allocations based on them and listing many more investment possibilities for analysis. When making each test, I included assertions that show the success message, confirm the data being saved and prove the logic used for recommendations. The study revealed that the system met all necessary expectations in important areas and allowed for a reliable experience for users.

User Story ID	Test Case	Priority	Steps	Expected Outcome	Status	Comments
CSI-1	Capture Investment Preferences	High	1. Log in to the application. 2. Navigate to the preference input page. 3. Enter investment preferences (e.g., age, risk tolerance, goals). 4. Save preferences.	Preferences are saved in the user profile. A confirmation message is displayed.	Passed	Preferences saved s
CSI-2	Generate Investment Recommendations	High	1. Log in to the application. 2. Ensure preferences are saved. 3. Navigate to the recommendation page. 4. View personalized investment suggestions.	System displays personalized investment recommendations relevant to the preferences.	Passed	Recommendations
CSI-3	View Detailed Investment Options	Medium	1. Log in to the application. 2. Navigate to the recommendation page. 3. Click on a recommendation to view more details.	Detailed view of selected investment is displayed, including reasoning and metrics.	Passed	Detailed informatio

Figure 85 Testing for Customized Suggestions for Investment

5.6.11 Example Test cases

The preferences collection and the recommendations given are saved.

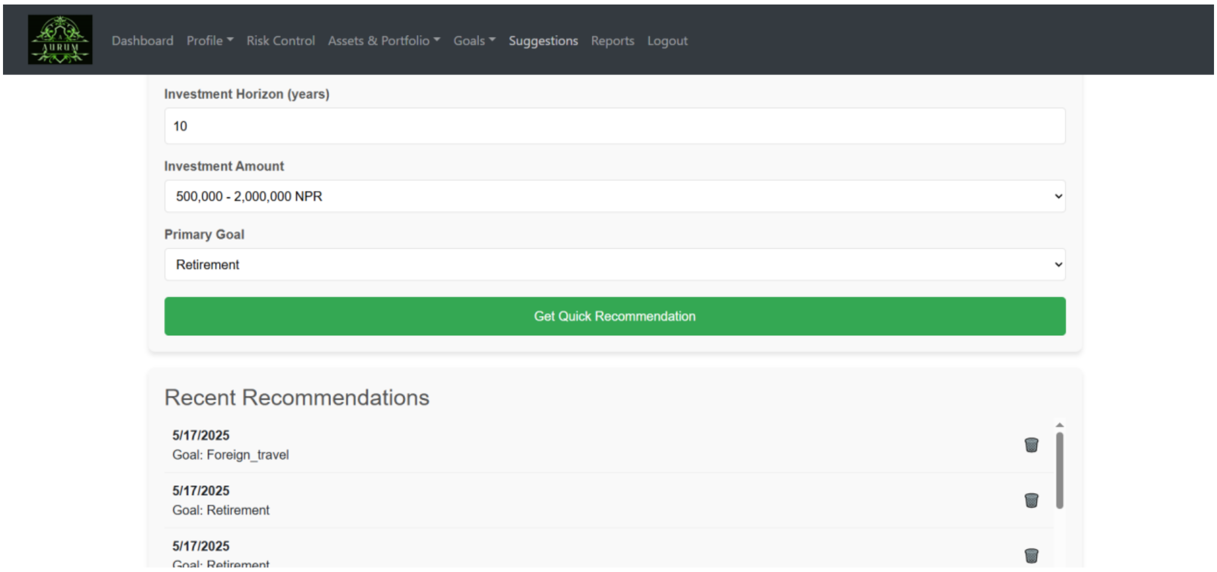


Figure 86 Example test case

The recommendations are shown as intended:

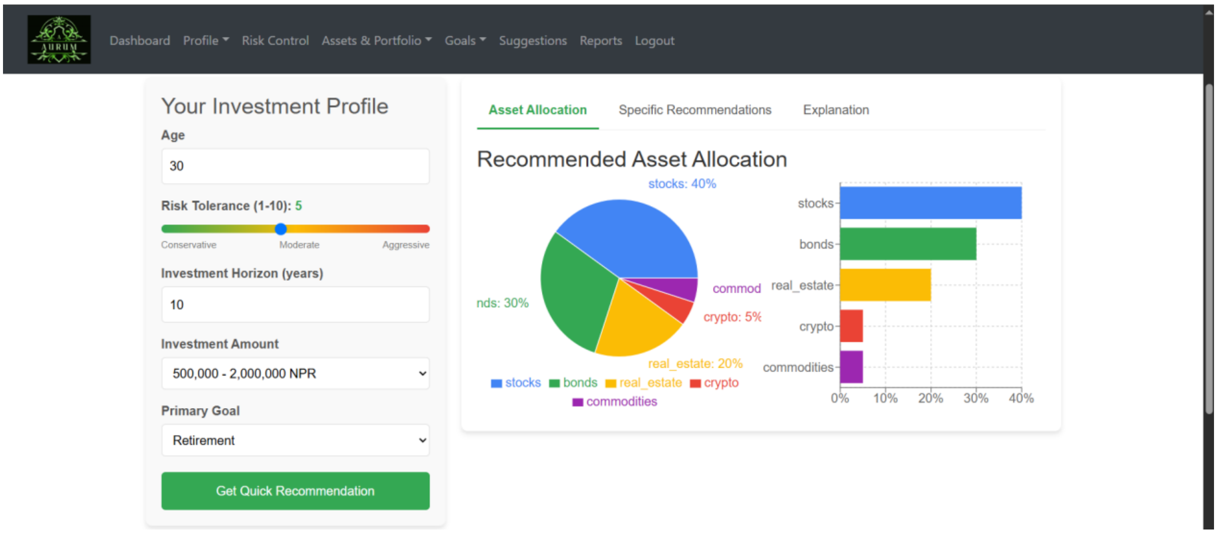


Figure 87 Example test case

5.6.12 Data Collection

- Dataset size: **3,000** recommendation samples.
- Source files: users.csv, assets.csv, quick_recommendation.csv

 Features: age_range, risk_tolerance, investment_horizon

 Target: allocation percentage for 5 asset classes.

5.6.13 Model Development

 Used RandomForestRegressor and GradientBoostingRegressor for each asset class.

 Applied advanced feature engineering:

- Age normalization
- One-hot encoding of goals
- Domain-specific interaction features (e.g., risk_x_horizon)

5.6.14 Optimization Evaluation

Model	**R^2 Score**	**RMSE**	**MAE**
Stocks	0.904	3.72	2.94
Bonds	0.928	3.77	3.11
Real Estate	0.968	1.16	0.84
Crypto	0.921	1.53	1.20
Commodities	0.929	0.54	0.44

Ensemble approach (60% RF + 40% GB) used for final output.

5.6.15 API Integration

 Exposed predictions via **Django API**.

 Example endpoints:

- /quick_recommendation: for age, risk, horizon, goal-based output
- /comprehensive_recommendation: uses holdings, goals, risk controls

 Model loaded using pickle and returned JSON predictions.

5.6.16 Comparing Algorithm Performance

Asset Class	RF R^2	GB R^2	Ensemble R^2
Stocks	0.897	0.909	0.904
Bonds	0.923	0.931	0.928
Real Estate	0.965	0.969	0.968
Crypto	0.917	0.924	0.921
Commodities	0.926	0.925	0.929

 Gradient Boosting generally performs best alone.

 Ensemble is used for balanced performance.

5.6.17 AI Testing

A/B Testing Style Comparison

Bar chart saved as allocation_comparison.png:

- Compared asset allocation % for:
  - Conservative (60yo, Risk 2)
  - Moderate (35yo, Risk 5, Home Purchase)
  - Aggressive (25yo, Risk 9)

Purpose:

It is to compare how the model uses AI to recommend vivid asset allocations based on the varying and different investor profiles using a bar chart.

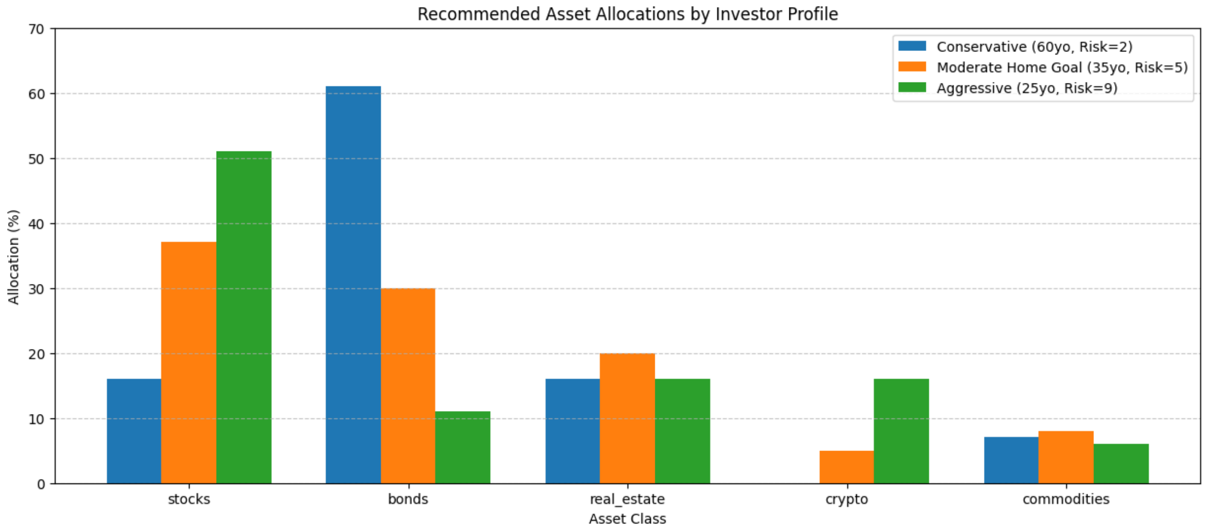


Figure 88 Recommended Asset Allocations by Investor Profile

5.6.18 Investor Profiles Compared

Profile	**Age**	**Risk Tolerance**	**Investment Horizon**	**Primary Goal**
Conservative Investor	60	2	5 years	Retirement
Moderate Investor (Home Goal)	35	5	7 years	Home Purchase
Aggressive Young Investor	25	9	30 years	Wealth Preservation

Here's what I learned from the chart:

Conservative Investor considers bonds (61%) as the most important investment and has no investments in cryptocurrencies.

Moderate Investor favors stocks (37%) in the portfolio and also has a large stake in real estate (20%).

Aggressive Investor has most of its funds in stocks (51%) and holds 16% in crypto for fast growth.

6. Conclusion

It has reached its goal of developing an AI software that handles wealth and makes daily life tasks easier. People could sign up, report on how risky they want their investments to be, set personal goals, view assets, benefit from AI recommendations and remain updated with reports on their progress. Thanks to this, users can get recent information and decide how to manage their finances. Every objective for the project was accomplished properly. It allows users to easily track their finances with data and grouped information. You can manage your finances by using various tools, whether for small or large decisions such as checking what you own, observing your portfolio or preparing your money goals ahead. Clearly, the system demonstrates how having technology helps in planning finances and provides guidelines for a better way of life.

With the system reaching its main goal, people can now use an AI platform to assist them with money management and planning their life. A customer can simply take care of their accounts, keep an eye on their finances, establish financial goals and receive investment advice, all to become more independent and understand finances better. All of these features were included so that users could use the system as they wished and set the main financial targets from the beginning. A system is safer and simpler to use when user accounts cannot be shared and email addresses are verified. Furthermore, it makes it possible for users to track and oversee the activities of all their investments on the platform. Users have the option of setting their goals and selecting dates for achieving them which are simple to monitor using graphs and stats. By using the system, users learn about their finances and receive personal guidance, suggestions and recommendations on how to manage their investments better. This supports the central goal of giving people a means to increase their wealth and plan their activities wisely.

Overall, the system is complete in its design, tailored to users and mixes financial knowledge with daily trends. It helps to prove that AI converts old finance tools into flexible ones offering both useful advice and easy-to-use decision support. Thanks to this system, users can spend and invest more wisely, pursue their major goals and plan for the future. The project achieved its technical goals and also significantly contributes to linking technology, finance and helping people become more empowered.

7. Critical Evaluation of the Project

7.1 Final Report Evaluation

The final report covered the entire project process, from setting the initial aims and objectives to putting the system into action and reviewing it. I liked that the report's structure was clear, as it made it easy to see the progression of the project. All modules, including user management, handling assets, goal tracking, giving investment suggestions and visualization, were further discussed by focusing on how they contributed to the system's objectives. Because the document included diagrams, user stories, sprints and technical specifications, it looked like a document created by professionals. Moreover, users were able to understand how every artifact helped with their financial planning and optimal investment choices.
Project Findings and Development Process

Before completing the development of Wealth Health, it was important to have a well-defined plan and to repeated design the platform to include AI suggestions, managing your investments and setting goals. The findings demonstrated that using AI to support financial decision-making leads to useful, accurate and helpful tips for each user. Due to using modules, it was possible to test, enhance or extend any part of the code apart from the rest. It was difficult at times to guarantee that the data between goals and assets moved steadily and remained the same. However, after thoroughly testing and improving the software, the system became stable and simple for users. Choosing these development tools and frameworks allowed us to finish the work satisfactorily within the given time.

7.2 System Evaluation

All scenarios were handled competently by the last system. You could sign up, manage your profile, set goals for your money, keep an eye on all investments and receive useful tips all in the same place. The information I saw and the alerts I received made it easier to choose appropriate actions. The module for risk control improved matters by prompting users whenever their investments were not performing well. Even though the system included all the necessary functions, there is still room to make it better, for example, by including live market information, chatbot advice and updating the AI model using RandomForestRegressor and GradientBoostingRegressor for every type of asset. Advanced feature engineering was applied like Age normalization, One-hot encoding means that every goal is shown separately. Specialized features for communication. In general, the system is solid in its approach and important for practical reasons.

7.3 Planning, Management and Sources

The development of projects was led by agile strategies using sprint routes and user stories. As a result, it was possible to monitor steady improvement and choose features based on their significance. Despite effective time planning and regular goals, a few phases, mainly the one where the AI model was combined, took more effort due to matching user interests with the AI's results. I included reliable sources, including research reports on AI in finance as well as plans and guides meant for developing software and tools. They helped guide all technical operations and fundamental ideas, leading to a satisfactory conclusion for the project.

7.4 Self-Reflection

Taking part in this project made a significant difference for me in many ways. While working on the project, I developed my ability to design systems, make them modular, plan user experience and build AI models. I was able to add machine learning to different applications and learn more about finances and how users use the platforms. I believe the project made me better at solving problems, managing my schedule well and adjusting when facing challenges I wasn't prepared for. Additionally, it gave me more confidence in handling big projects from the start to when they are used by customers. All in all, creating this system taught me to become a better and wiser developer, not only write software.

8. Evidence of Project Management

8.1 Log Sheet

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: KARAN	Surname: SAH
Student Number: 2329860	Supervisor: NIKUNJA LAMSAL
Project Title: AURUM	Month: NOV.

What have you done since the last meeting

- Brief Literature Review Research
- Project Proposal
- Presentation making
- Initial Research brief finished

What do you aim to complete before the next meeting

- Complete Initial Research
- Work on the flow of the project

Supervisor comments

- Remove the conclusion section.
- Proposal is accepted.
- Recorrect the mistakes and submit the proposal.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______________ Date: 24th Nov, 2024

Supervisor Signature: ______________ Date: ~~24th Nov, 2024~~ 1st Dec 2024.

Figure 89 Log Sheet 1

UNIVERSITY OF WOLVERHAMPTON

Faculty of Science and Engineering
School of Mathematics and Computer Science

PROJECT MANAGEMENT LOG	
First Name: KARAN	Surname: SAH
Student Number: 2329860	Supervisor: NIKUNSA LAMSAL
Project Title: AURUM	Month: Dec

What have you done since the last meeting

- Completed Initial Research
- Worked on the flow of the project
- Worked on timeline of the project

What do you aim to complete before the next meeting

- Set up React.js Environment
- Create a Basic Folder Structure for the project
- Set up Django with PostgreSQL
- Install Django REST framework & configure project settings
- Set up Version Control

Supervisor comments

- Start the project.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______________ Date: Dec 1, 2024

Supervisor Signature: ______________ Date: Dec 8, 2024.

Figure 90 Log Sheet 2

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: KARAN	Surname: SAH
Student Number: 2329860	Supervisor: NIKUNJA LAMSAL
Project Title: AURUM	Month: Dec

What have you done since the last meeting

- Set up Environments, version control, Django
- Create a Basic Folder structure for the project
- Made Product Backlog

What do you aim to complete before the next meeting

- Make UMLs diagrams
- Complete designs of the project
- Get datasets

Supervisor comments

- Start the wireframe design and UML diagrams.
- Start the development and take the front end/backend parallely.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______ Date: Dec 15, 2024

Supervisor Signature: ______ Date: Dec 22, 2024

Figure 91 Log Sheet 3

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG			
First Name:	KARAN	**Surname:**	SAH
Student Number:	2329860	**Supervisor:**	NIKUNJA LAMSAL
Project Title:	AURUM	**Month:**	

What have you done since the last meeting

- Completed wireframes of the project of admin and user side.
- Made some progress in the website.
- Visited Journal sites for literature review.

What do you aim to complete before the next meeting

- Literature Review
- Frontend of 2/3 artifacts

Supervisor comments

- Work on the literature review part and go through the papers reviewed.
- Work on the UI part and development.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______________ Date: 12/22/2024

Supervisor Signature: ______________ Date: 2025/01/05

Figure 92 Log Sheet 4

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: KARAN	Surname: SAH
Student Number: 2329810	Supervisor: NIKUNJA LAMSAL
Project Title: AURUM	Month: Jan

What have you done since the last meeting

- Completed Literature Review
- few parts of admin

What do you aim to complete before the next meeting

- User Management and verification (Frontend & Backend)

Supervisor comments

- Work on front end part.
- Prepare at least 2-3 diagrams.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ____________ Date: 2025/01/05

Supervisor Signature: ____________ Date: 2025/01/19

Figure 93 Log Sheet 5

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: KARAN	Surname: SAH
Student Number: 2329860	Supervisor: NIKUNJA LAMSAL
Project Title: AURUM	Month: Jan

What have you done since the last meeting

- User management and verification (Frontend and Backend)
-

What do you aim to complete before the next meeting

- Dashboard - User
- Asset add, monitor.
- Asset artifact.

Supervisor comments

- Simplify the UI.
- Add the features of your project.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______ Date: 2025/01/19

Supervisor Signature: ______ Date: 2025/01/26

Figure 94 Log Sheet 6

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: KARAN	**Surname:** SAM
Student Number: 2329860	**Supervisor:** NIKUNJA LAMSAL
Project Title: AURUM	**Month:** Jan

What have you done since the last meeting

- Asset
- Few errors of User Registration Handled.

What do you aim to complete before the next meeting

- Finish Asset artefact
- Finish Financial Goals artefact

Supervisor comments

- Complete the artifact designs.
- ~~Go~~ Generate a single FDD diagram.
- Complete at least 2 module with testing.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______________ Date: 2025/01/26

Supervisor Signature: ______________ Date: 2025/02/02

Figure 95 Log Sheet 7

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG			
First Name: KARAN		**Surname:** SAH	
Student Number: 2329860		**Supervisor:** NIKUNJA LAMSAL	
Project Title: AURUM		**Month:** Feb	

What have you done since the last meeting

- Completed artifact designs
- Generated a single FDD diagrams
- Completed 2 modules with testing.
- Quarter of work of financial goals artefact completed.

What do you aim to complete before the next meeting

. Complete financial goals artefact with testing.

Supervisor comments

- Start working on other modules.
- Prepare for the demo session,

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______________ Date: 2025/02/02

Supervisor Signature: ______________ Date: 2025/02/09

Figure 96 Log Sheet 8

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: KARAN	Surname: SAH
Student Number: 2329860	Supervisor: NIKUNJA LAMSAL
Project Title: AURUM	Month: Feb

What have you done since the last meeting

- Financial goals artifact.

What do you aim to complete before the next meeting

- Complete financial goals artefact & risk assesment

Supervisor comments

- Work on the remaining modules.
- Prepare for the demo session.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ____________ Date: 2025/02/09

Supervisor Signature: ____________ Date: 2025/02/16

Figure 97 Log Sheet 9

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: KARAN	Surname: SAH
Student Number: 2329860	Supervisor: NIKUNJA LAMSAL
Project Title: AURUM	Month: Feb

What have you done since the last meeting

- Frontend of goals.
- Fews backend of goals.

What do you aim to complete before the next meeting

- Complete goal artefact.
- Complete backend of risk assessment.

Supervisor comments

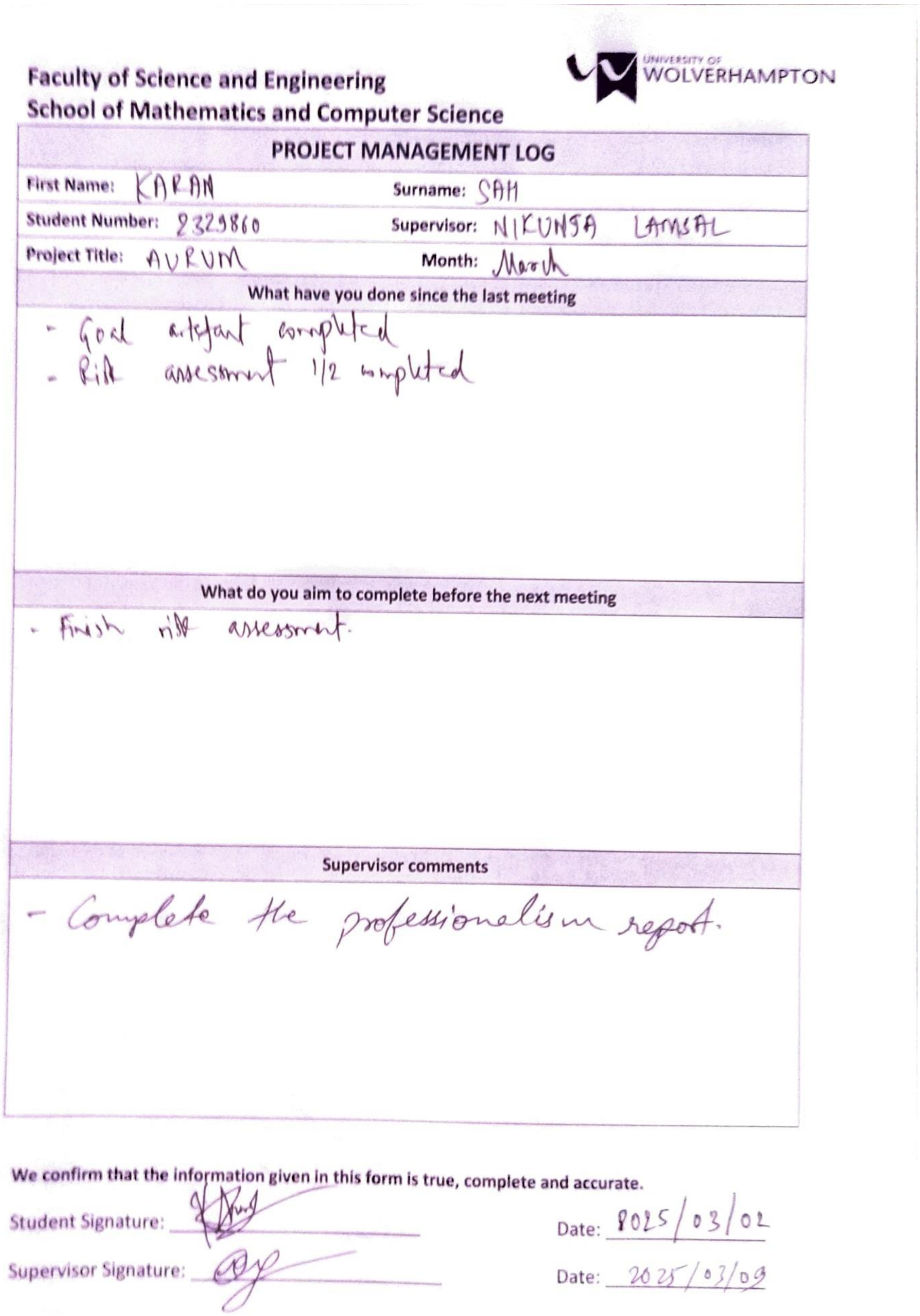
- make the correction as per the comment.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ____________ Date: 2025/02/18

Supervisor Signature: ____________ Date: 2025/03/02

Figure 98 Log Sheet 10



Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: KARAN	**Surname:** SAH
Student Number: 2329860	**Supervisor:** NIKUNJA LAMSAL
Project Title: AURUM	**Month:** March

What have you done since the last meeting

- Goal artifact completed
- Risk assessment 1/2 completed

What do you aim to complete before the next meeting

- Finish risk assessment.

Supervisor comments

- Complete the professionalism report.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______ Date: 2025/03/02

Supervisor Signature: ______ Date: 2025/03/09

Figure 99 Log Sheet 11

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: KARAN	**Surname:** SAH
Student Number: 2329860	**Supervisor:** NIKUNJA LAMSAL
Project Title: AURUM	**Month:** March

What have you done since the last meeting

- Risk Assessment finished

What do you aim to complete before the next meeting

- Finish all except recommendation system

Supervisor comments

- Complete the recommendation system.
- Integrate it with the reports.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______________ Date: 2025/03/09

Supervisor Signature: ______________ Date: 2025/4/6

Figure 100 Log Sheet 12

Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG

First Name: KARAN	Surname: SAH
Student Number: 2329860	Supervisor: NIKUNJA LAMSAL
Project Title: AURUM	Month: ~~March~~ April.

What have you done since the last meeting

- Research Completed for datasets
- Research completed for recommendation system.

What do you aim to complete before the next meeting

- Make recommendation system.
- Preprocess and clean datasets.

Supervisor comments

- Complete the remaining part by next week.

We confirm that the information given in this form is true, complete and accurate.

Student Signature: ______________ Date: 2025/04/06

Supervisor Signature: ______________ Date: 2025/04/13

Figure 101 Log Sheet 13

8.2 Gantt Chart

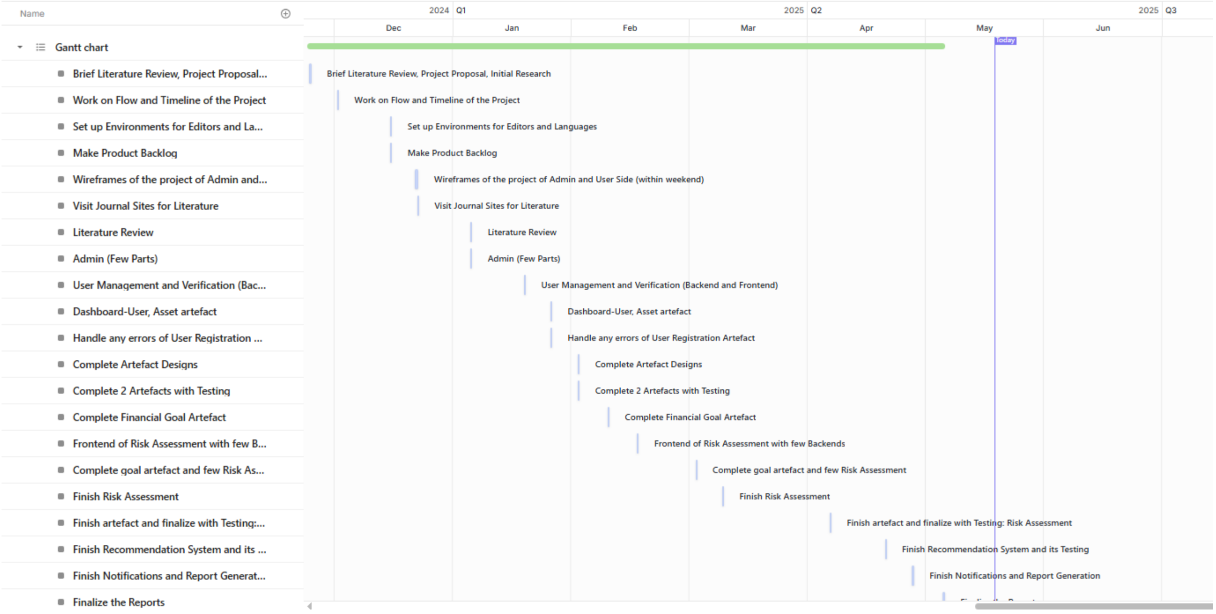


Figure 102 Gantt Chart

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Appendices

Questionnaire

Please fill up the form. It would be helpful.

Full Name *

Short answer text

Have you used any investment or goal-tracking platforms before?

- ○ Yes
- ○ No

Which features are most important to you in a financial platform?

- ☐ Email verification
- ☐ Profile management (edit/view details)
- ☐ Risk tolerance settings
- ☐ Setting financial goals
- ☐ Asset tracking
- ☐ Portfolio summary
- ☐ Performance monitoring

How would you prefer to set your risk tolerance level?

- ○ Manually
- ○ Through a guided questionnaire
- ○ Not sure

What types of assets do you currently manage or plan to track?

- [ ] Stocks
- [ ] Bonds
- [ ] Mutual Funds
- [ ] Real Estate
- [ ] Other...

How frequently do you review your portfolio summary?

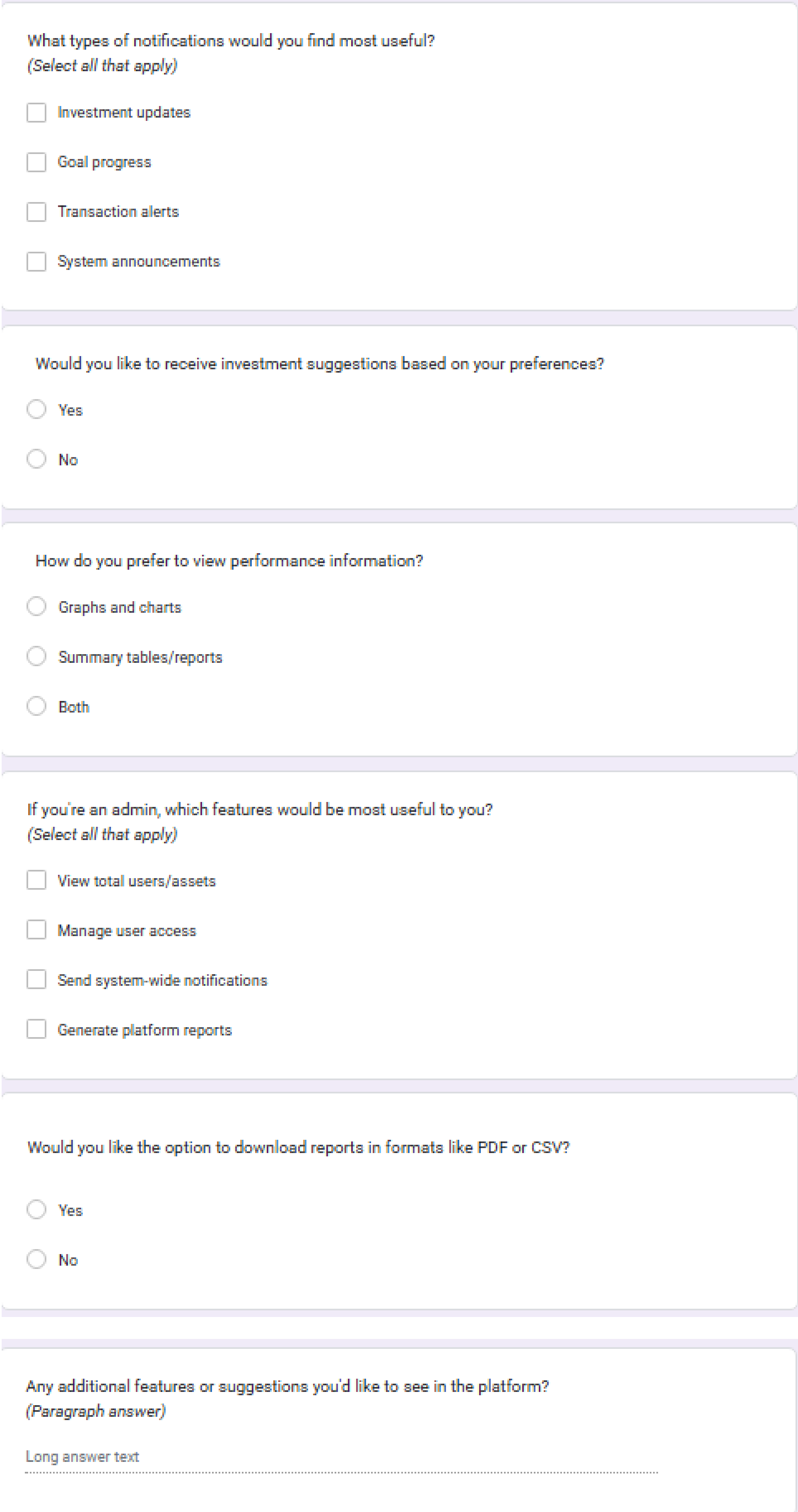
- ○ Daily
- ○ Weekly
- ○ Monthly
- ○ Only when significant changes occur

Would setting financial goals with milestones and deadlines be helpful to you?

- ○ Yes
- ○ No
- ○ Maybe

How would you like to receive notifications about your financial goals and investments?

- [ ] Email
- [ ] In-app dashboard
- [ ] SMS
- [ ] Push notifications



What types of notifications would you find most useful?
(Select all that apply)

- [ ] Investment updates
- [ ] Goal progress
- [ ] Transaction alerts
- [ ] System announcements

Would you like to receive investment suggestions based on your preferences?

- ( ) Yes
- ( ) No

How do you prefer to view performance information?

- ( ) Graphs and charts
- ( ) Summary tables/reports
- ( ) Both

If you're an admin, which features would be most useful to you?
(Select all that apply)

- [ ] View total users/assets
- [ ] Manage user access
- [ ] Send system-wide notifications
- [ ] Generate platform reports

Would you like the option to download reports in formats like PDF or CSV?

- ( ) Yes
- ( ) No

Any additional features or suggestions you'd like to see in the platform?
(Paragraph answer)

Long answer text

Figure 103 Form for data and preferences collections